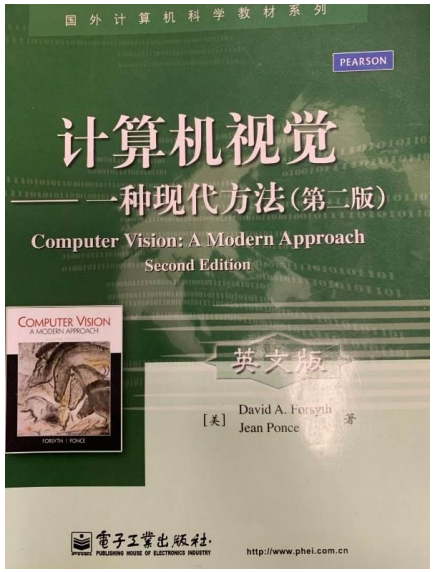
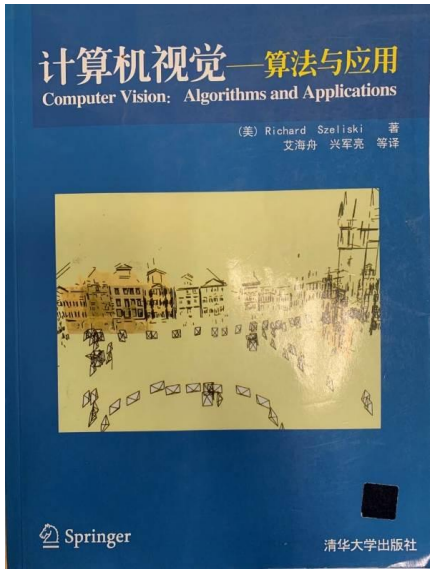
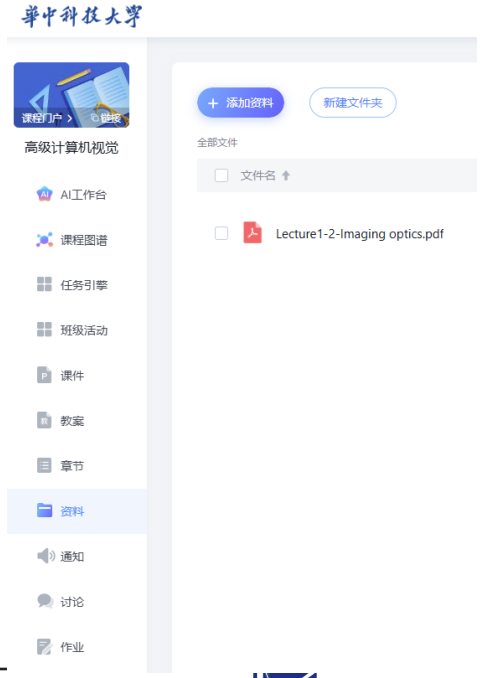
Computer Vision
Lecture 1
Light and Color

About the course

- 教师：桑农，王岳环，左峥嵘
- 教材及参考资料：Computer Vision，selected papers
- 成绩构成: 闭卷考试（课程结束后2周）70%+平时成绩30%（按照讲授内容写一份心得，包括对课程内容的理解，存在的问题和难点，对讲授内容的需求等。）
- 日常通知及课件: 课程平台
- 课程选定后不得无故修改，必须参加课程学习考核，如因特殊原因需要退课，必须在课程进度的前50%内申请，逾期不予处理；



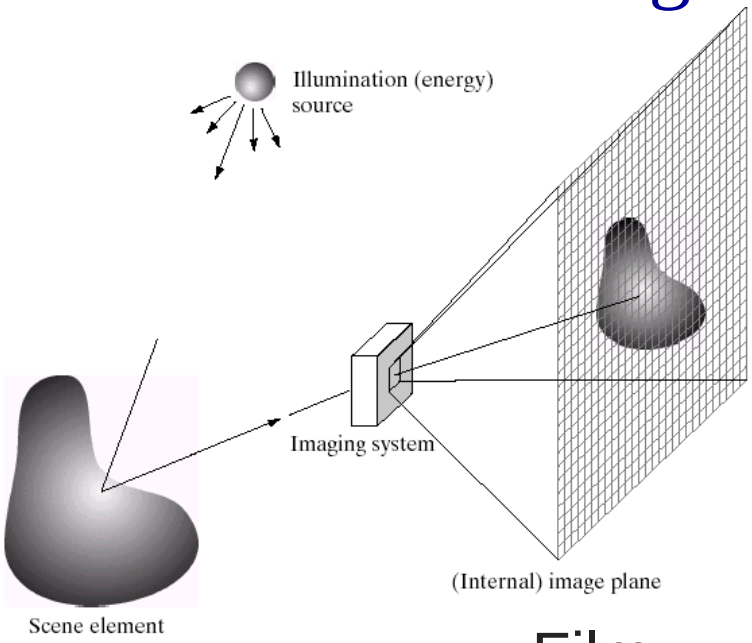




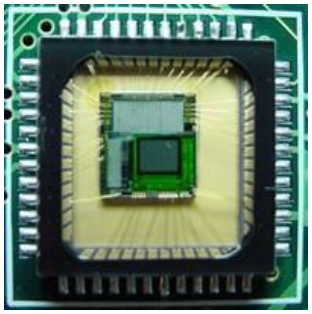
About the course

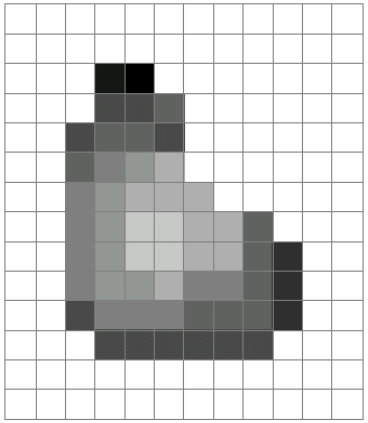
- What is computer vision?
- Where the ideas come from?

Image Formation



Film

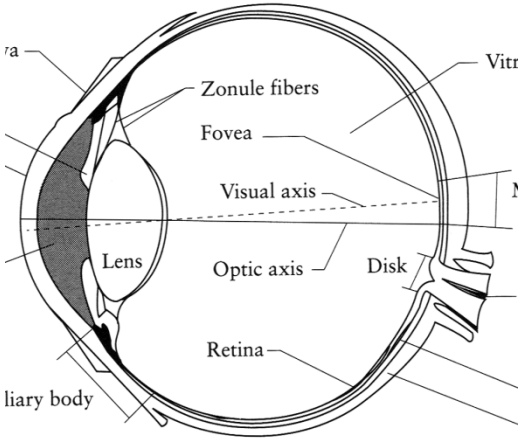




Digital Camera

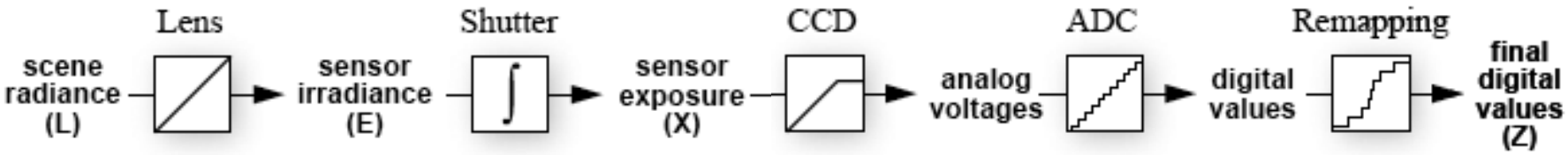
Major factors

- Illumination strength and direction
- Surface geometry
- Surface material
- Nearby surfaces
- Camera gain/exposure



The Eye

From light rays to pixel values

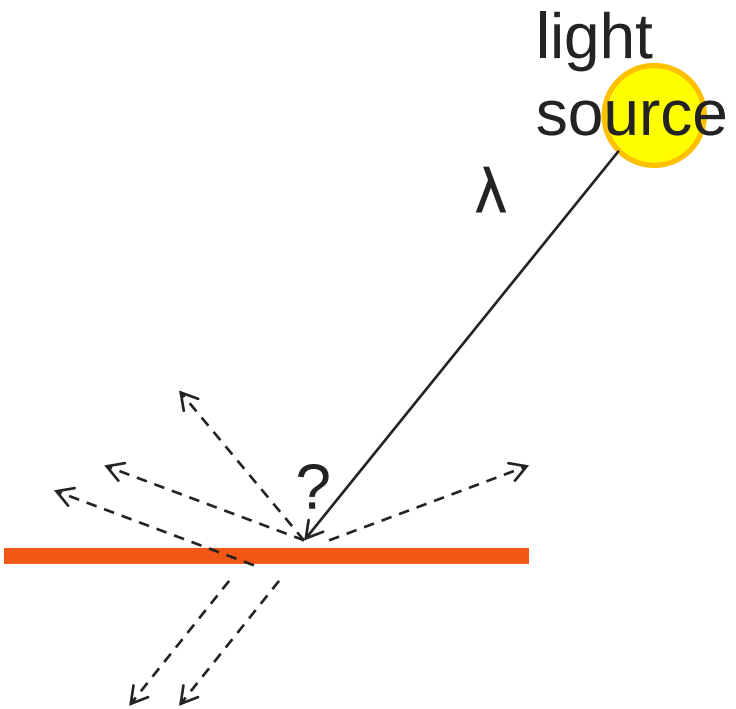


For more information:

- M. Brown, Understanding the In-Camera Image Processing Pipeline for Computer Vision, CVPR 2016 Tutorial

A photon's life choices

Absorption
Diffusion
Reflection
Transparency
Refraction
Fluorescence
Subsurface scattering
Phosphorescence
Interreflection



A photon's life choices

Absorption

Diffusion

Reflection

Transparency

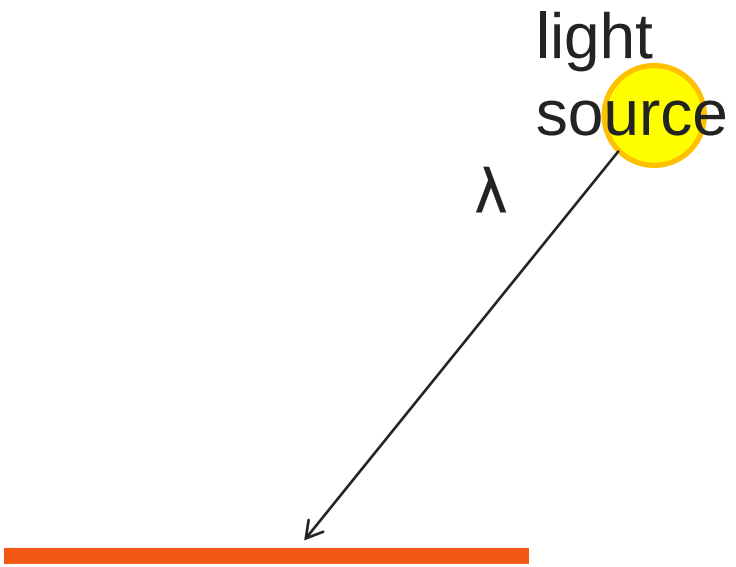
Refraction

Fluorescence

Subsurface scattering

Phosphorescence

Interreflection



A photon's life choices

Absorption

Diffuse Reflection

Reflection

Transparency

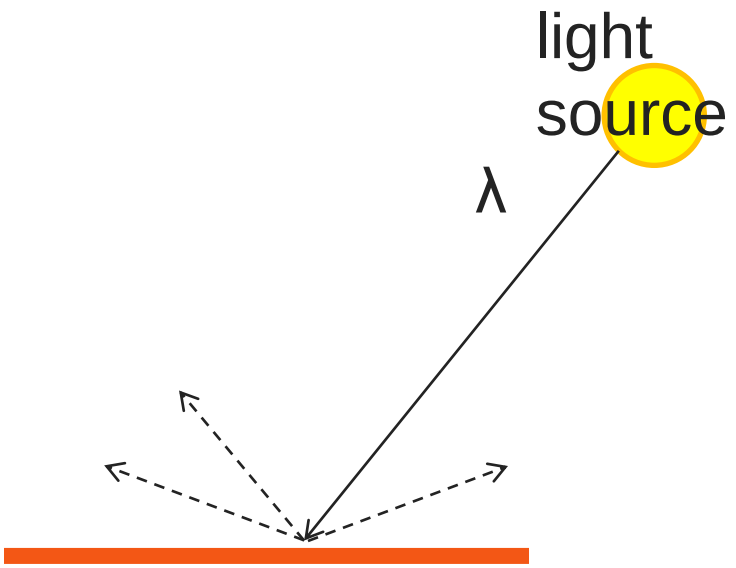
Refraction

Fluorescence

Subsurface scattering

Phosphorescence

Interreflection



A photon's life choices

Absorption

Diffusion

Specular Reflection

Transparency

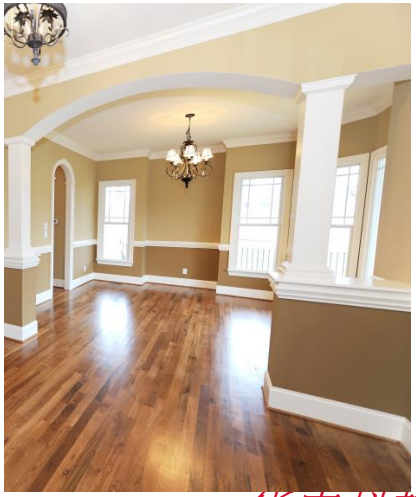
Refraction

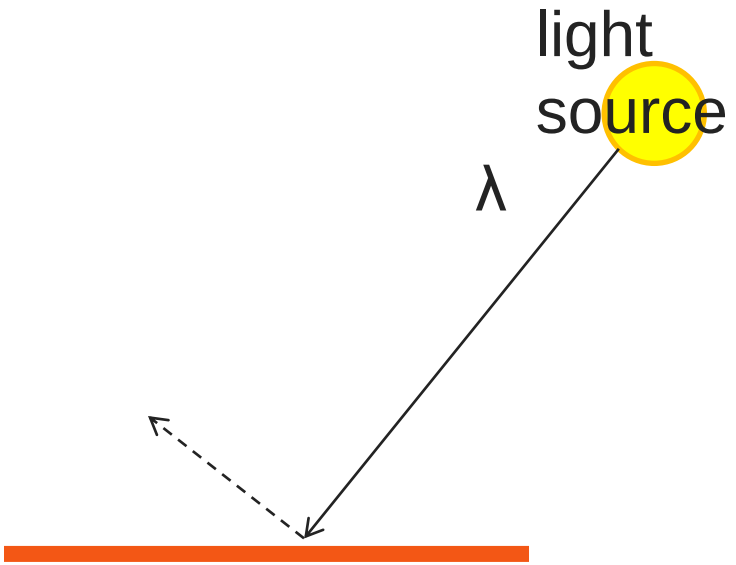
Fluorescence

Subsurface scattering

Phosphorescence

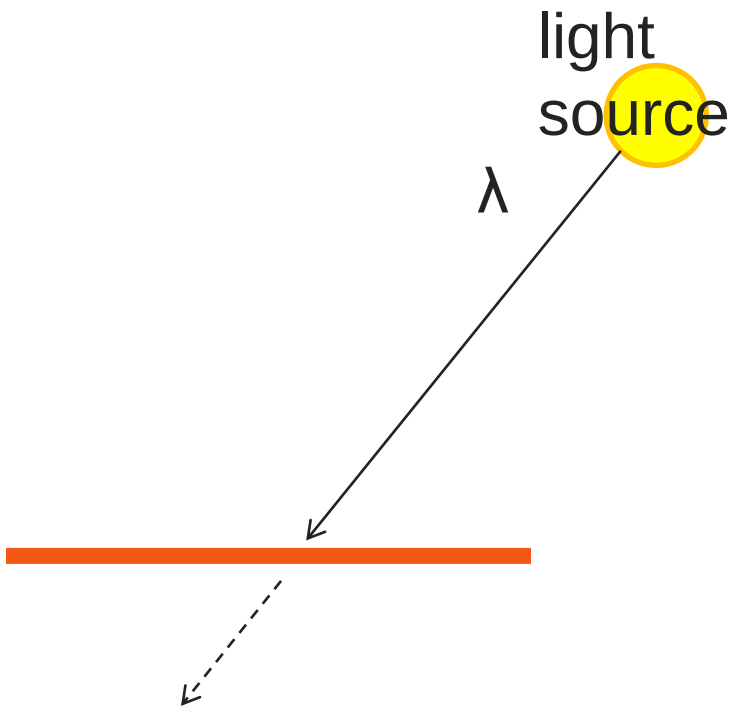
Interreflection





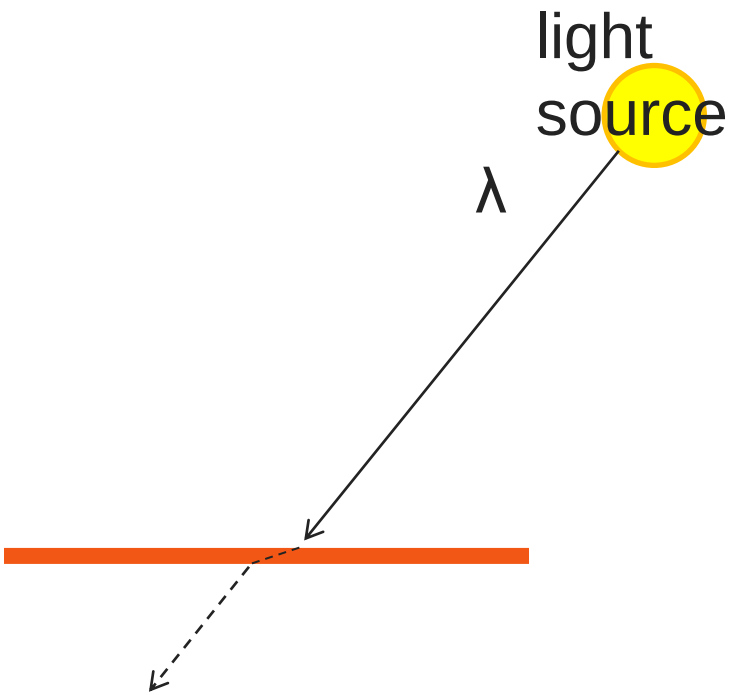
A photon's life choices

Absorption
Diffusion
Reflection
Transparency
Refraction
Fluorescence
Subsurface scattering
Phosphorescence
Interreflection



A photon's life choices

Absorption
Diffusion
Reflection
Transparency
Refraction
Fluorescence
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Phosphorescence
Interreflection

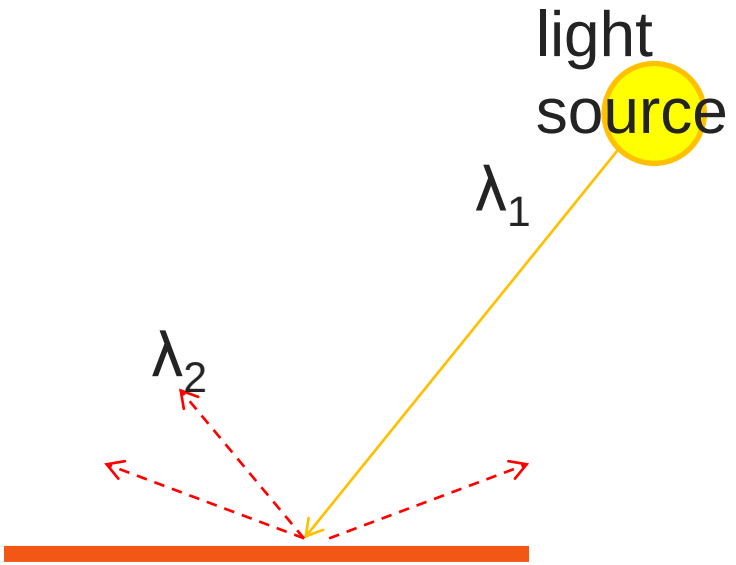




A photon's life choices

Absorption
Diffusion
Reflection
Transparency
Refraction
Fluorescence
Subsurface scattering
Phosphorescence
Interreflection

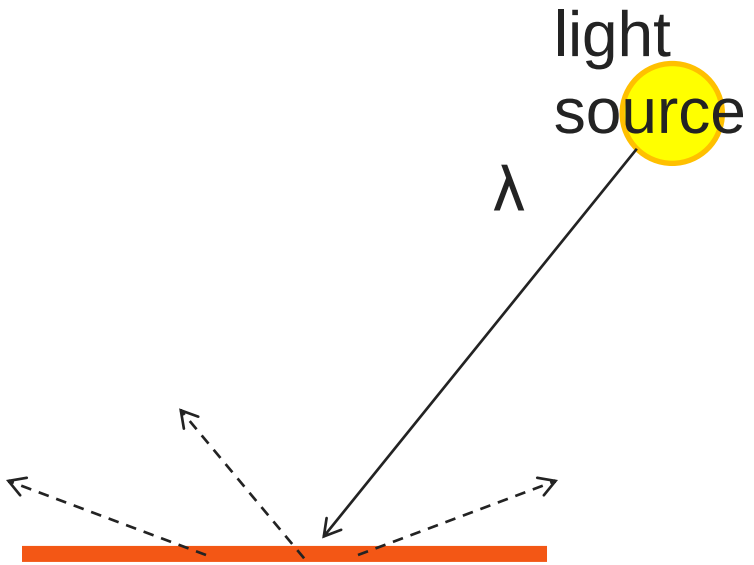




A photon's life choices

Absorption
Diffusion
Reflection
Transparency
Refraction
Fluorescence
Subsurface scattering
Phosphorescence
Interreflection

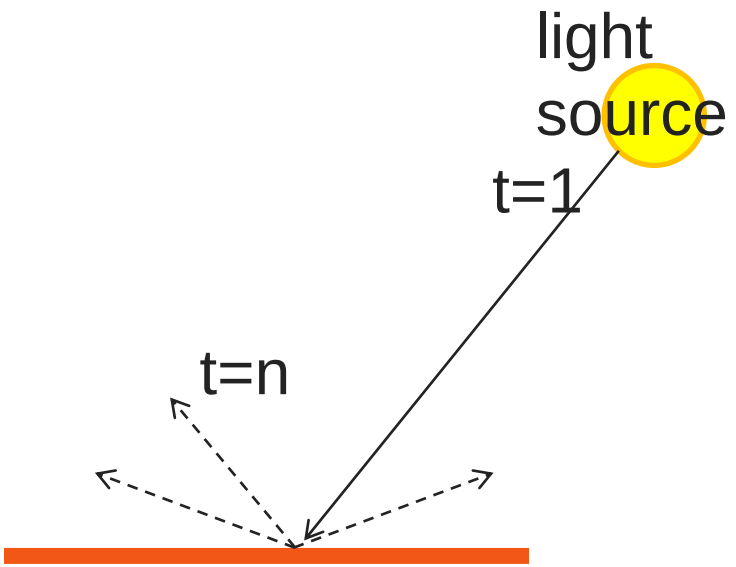




A photon's life choices

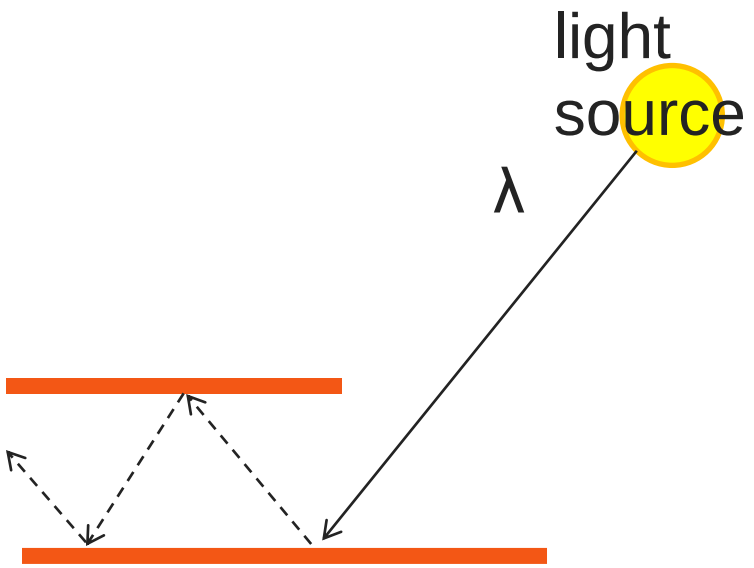
Absorption
Diffusion
Reflection
Transparency
Refraction
Fluorescence
Subsurface scattering
Phosphorescence
Interreflection





A photon's life choices

Absorption
Diffusion
Reflection
Transparency
Refraction
Fluorescence
Subsurface scattering
Phosphorescence
Interreflection



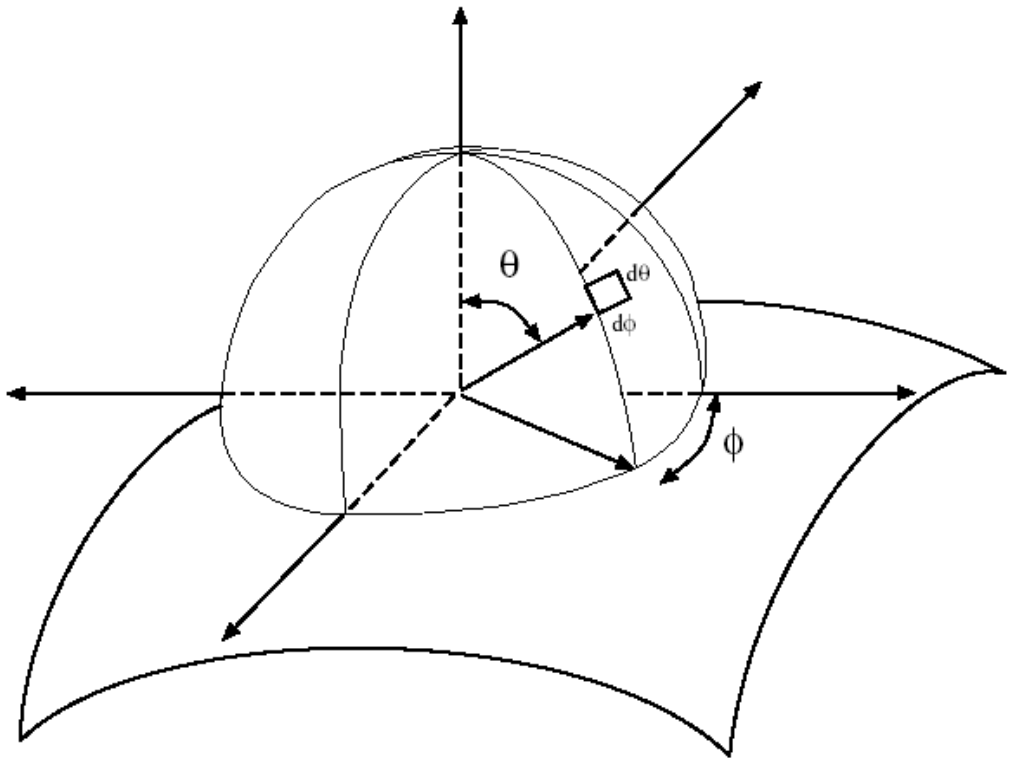
(Specular Interreflection)

Radiometry

- *Radiometry* is the part of image formation concerned with the relation among the amounts of
  - light energy emitted from light sources,
  - reflected from surfaces,
  - and registered by sensors.

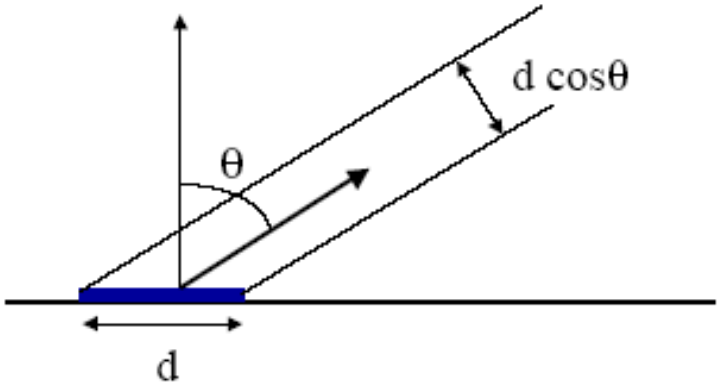
Foreshortening

- A point on a surface sees the world along a hemisphere of directions centered at the point.



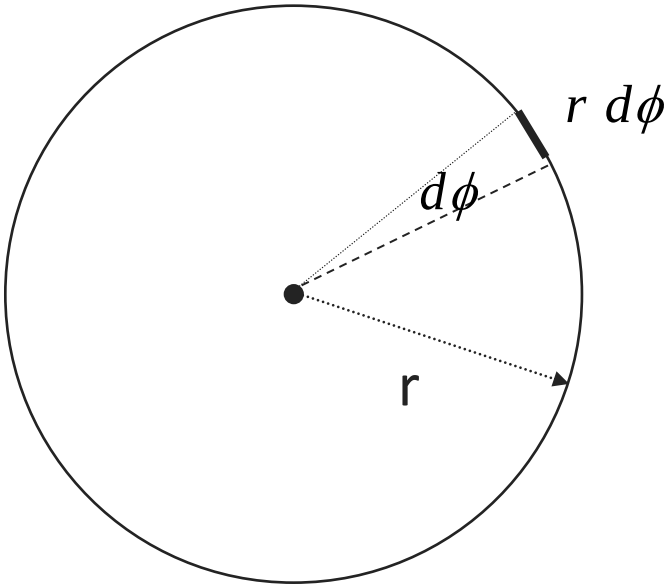
Foreshortening

- A big source, viewed at a glancing angle, must produce the same effect as a small source viewed frontally.
- This phenomenon is known as *foreshortening*.



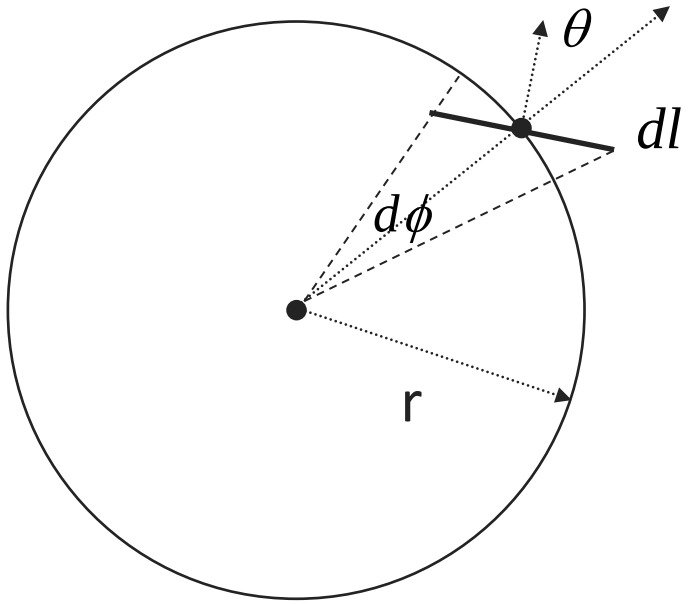
Solid Angle

- Arc length



Solid Angle

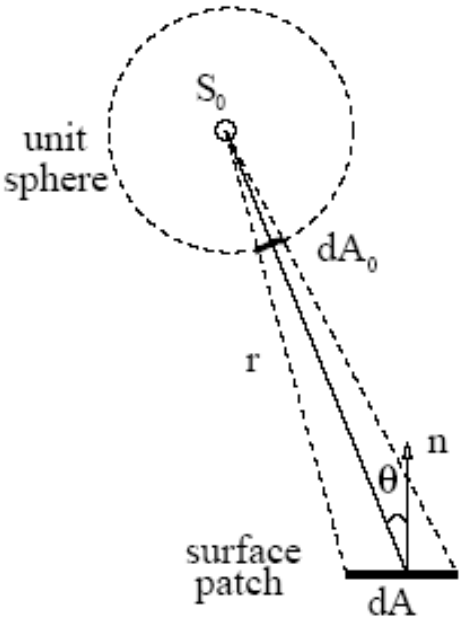
- Similarly, solid angle due to a line segment is



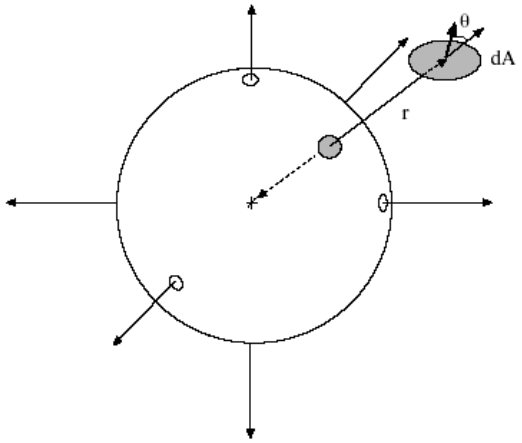
$$d\phi = \frac{dl\cos\theta}{r}$$

Solid Angle

- *Solid angle* is defined by the projected area of a surface patch onto a unit sphere of a point.



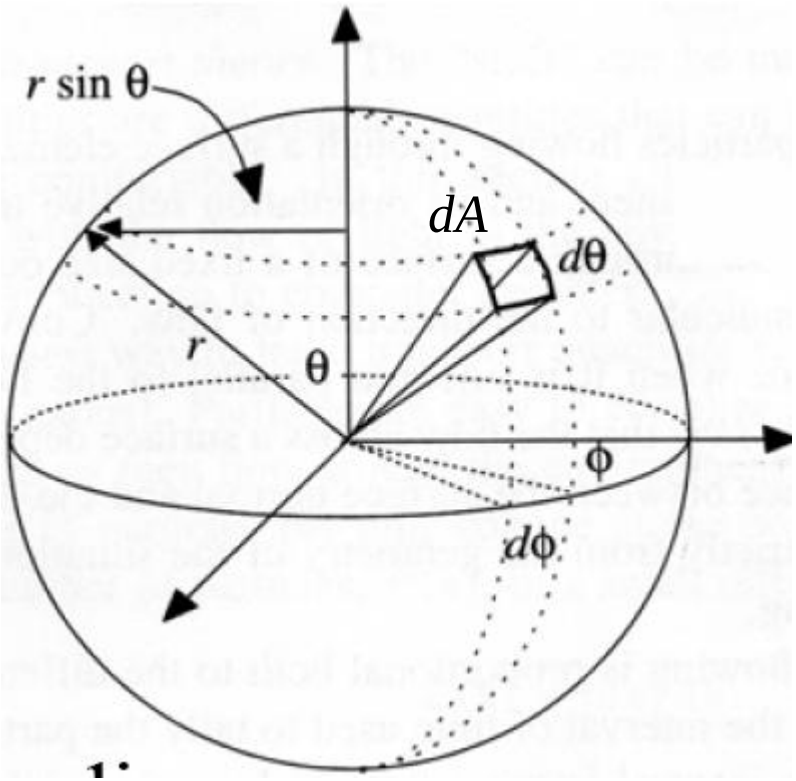
$$d\omega = dA_0 = \frac{dA\cos\theta}{r^2}$$



(Solid angle is subtended by a point and a surface patch.)

Solid Angle

- *Solid angle* is defined by the projected area of a surface patch onto a unit sphere of a point.



$$dA = rd\theta r\sin\theta d\phi = r^2\sin\theta d\theta d\phi$$

$$TotalArea = \int_{\phi=0}^{2\pi}\int_{\theta=0}^{\pi} r^2\sin\theta d\theta d\phi = 4\pi r^2$$

$$dw = \frac{dA}{r^2} = \sin\theta d\theta d\phi$$

Radiance

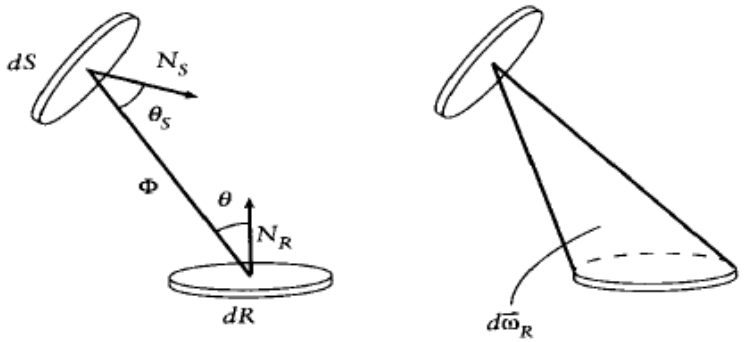
- The distribution of light in space is a function of position and direction.
- The appropriate unit for measuring the distribution of light in space is *radiance*, which is defined as the *power* (the amount of energy per unit time) traveling at some point in a specified direction, per unit area *perpendicular to the direction of travel*, per unit solid angle.
- In short, radiance is the amount of light radiated from a point… (into a unit solid angle, from a unit area).

Radiance = Power / (solid angle x foreshortened area)

W/sr/m2
W is Watt, sr is steradian, m2 is meter-squared

Radiance

- Radiance from dS to dR

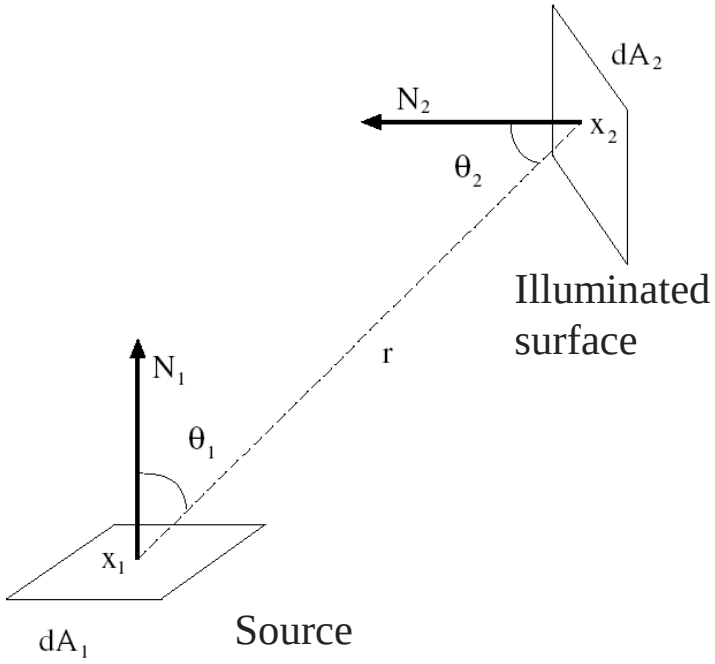


*Radiance = Power / (solid angle * foreshortened area)*

Radiance

- Example: Infinitesimal source and surface patches

Radiance = Power / (solid angle * foreshortened area)



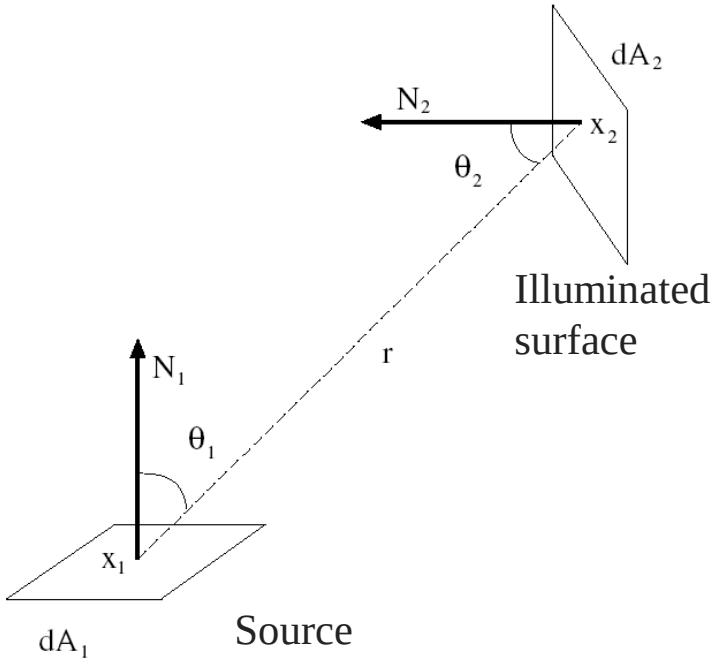
Radiance at x1 leaving to x2

$$L(\mathbf{x}_1, \mathbf{x}_1 \rightarrow \mathbf{x}_2) = \frac{d\psi}{dw \cos\theta_1 dA_1} = \frac{r^2 d\psi}{dA_2 \cos\theta_2 \cos\theta_1 dA_1}$$

$$dw = \frac{dA_2 \cos\theta_2}{r^2}$$

Radiance

Radiance = Power / (solid angle * foreshortened area)



Power at x1 leaving to x2

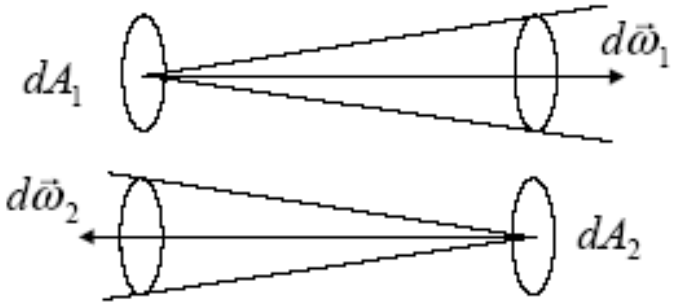
$$d\psi = L(\mathbf{x}_1, \mathbf{x}_1 \rightarrow \mathbf{x}_2)\, dw \cos\theta_1 dA_1$$

$$= \frac{L(\mathbf{x}_1, \mathbf{x}_1 \rightarrow \mathbf{x}_2)\, dA_2 \cos\theta_2 \cos\theta_1 dA_1}{r^2}$$

$$dw = \frac{dA_2 \cos\theta_2}{r^2}$$

Radiance

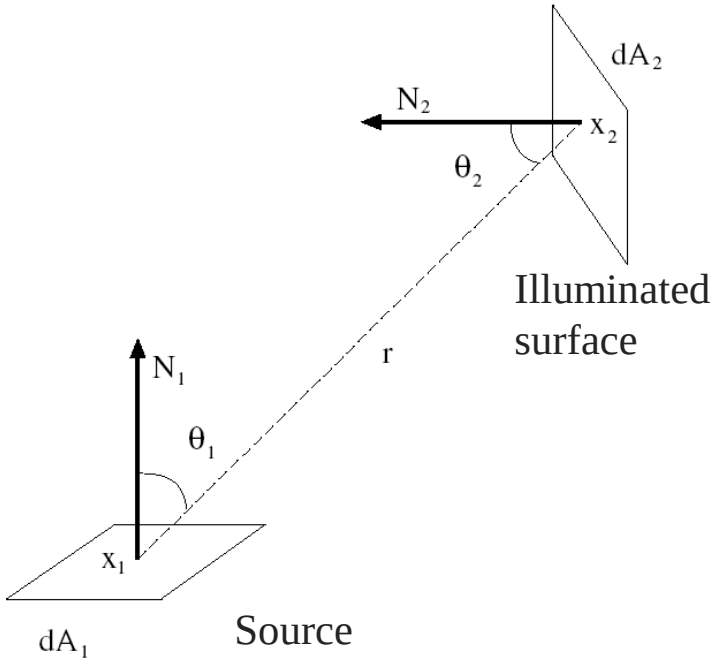
- If the medium is vacuum, power is preserved.



$$L_1 d\vec{\omega}_1 dA_1 = L_2 d\vec{\omega}_2 dA_2$$

Radiance

- The medium is vacuum, that is, it does not absorb energy. Therefore, the power reaching point x2 is equal to the power leaving for x2 from x1.



Power at x2 from direction x1 is

$$d\psi = \frac{L(\mathbf{x}_1, \mathbf{x}_1 \to \mathbf{x}_2) dA_2 \cos\theta_2 \cos\theta_1 dA_1}{r^2}$$

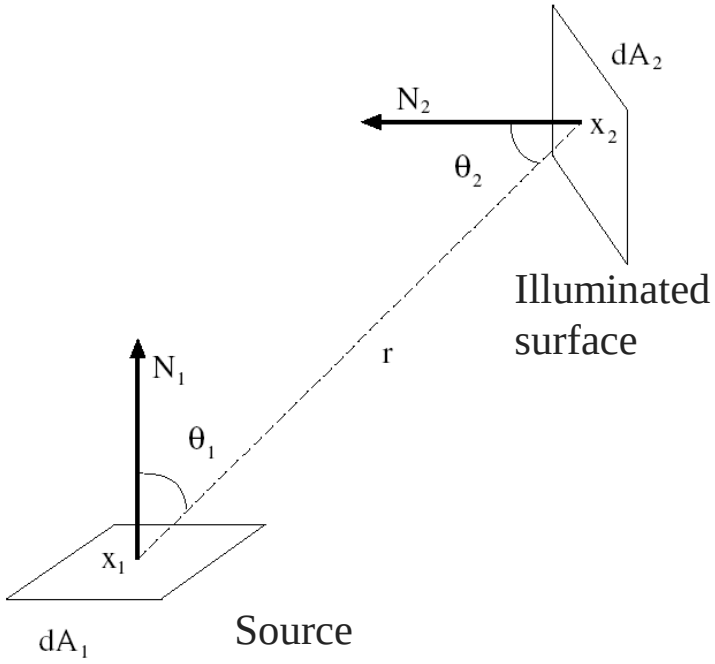
Let the radiance arriving at x2 from the direction of x1 is

$$L(\mathbf{x}_2, \mathbf{x}_1 \to \mathbf{x}_2) = \frac{d\psi}{dw \cos\theta_2 dA_2} = \frac{r^2 d\psi}{dA_1 \cos\theta_1 \cos\theta_2 dA_2}$$

$$dw = \frac{dA_1 \cos\theta_1}{r^2}$$

Radiance

- **Radiance is constant along a straight line.**



$$L(\mathbf{x}_1, \mathbf{x}_1 \rightarrow \mathbf{x}_2) = L(\mathbf{x}_2, \mathbf{x}_1 \rightarrow \mathbf{x}_2)$$

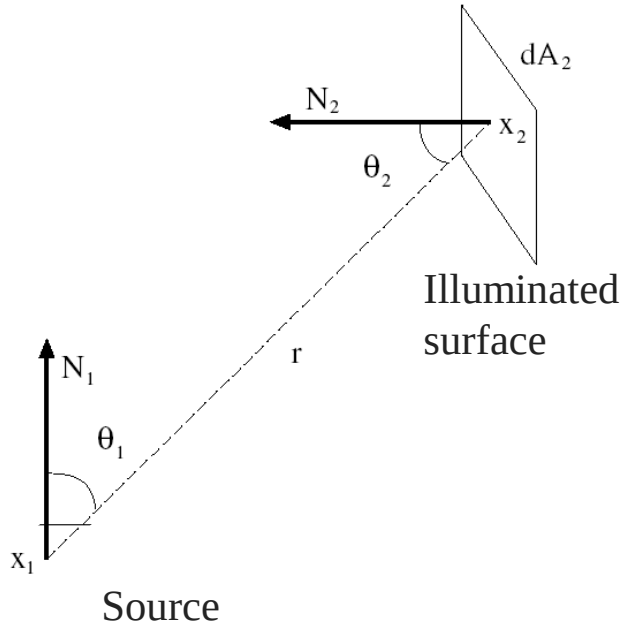
Point Source

- Many light sources are physically small compared with the environment in which they stand.
- Such a light source is approximated as an extremely small sphere, in fact, a point.
- Such a light source is known as a *point source*.

Radiance Intensity

- If the source is a *point source*, we use radiance intensity.

Radiance intensity = Power / (solid angle)



$$I = \frac{d\psi}{dw} = \frac{r^2 d\psi}{dA_2 \cos\theta_2}$$

$$dw = \frac{dA_2 \cos\theta_2}{r^2}$$

Light at Surfaces

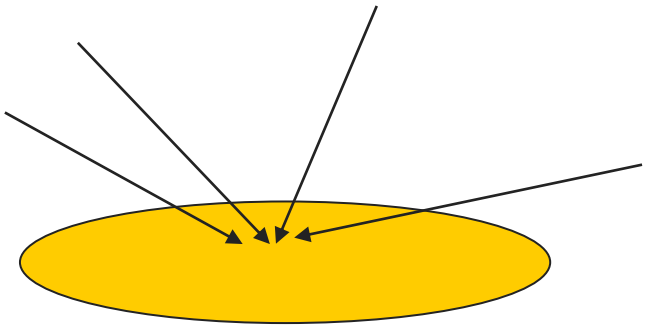
- When light strikes a surface, it may be absorbed, transmitted, or scattered; usually, combination of these effects occur.
- It is common to assume that all effects are local and can be explained with a **local interaction model**. In this model:
  - The radiance leaving a point on a surface is due only to radiance arriving at this point.
  - Surfaces do not generate light internally and treat sources separately.
  - Light leaving a surface at a given wavelength is due to light arriving at that wavelength.

Light at Surfaces

- In the local interaction model, *fluorescence*, [absorb light at one wavelength and then radiate light at a different wavelength], and *emission* [e.g., warm surfaces emits light in the visible range] are neglected.

Irradiance

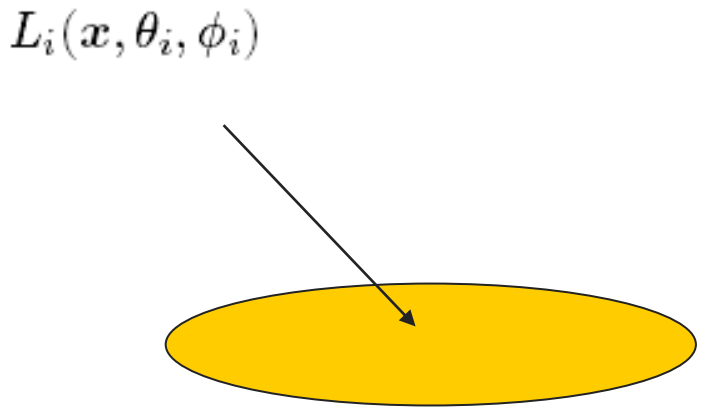
- *Irradiance* is the total incident power per unit area.



Irradiance = Power / Area

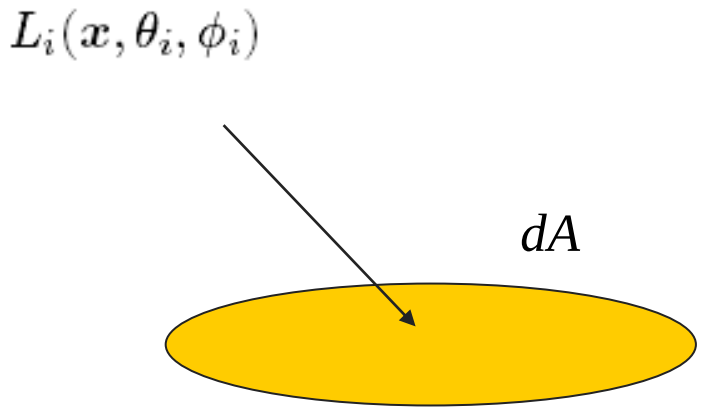
Irradiance

- What is the irradiance due to source from angle (θ_i, ϕ_i)?

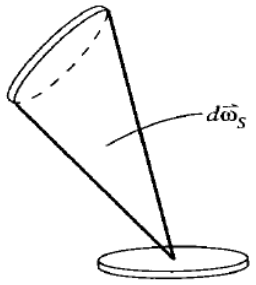


Irradiance

- What is the irradiance due to source from angle (θ_i, ϕ_i)?



$$Irradiance = \frac{d\psi}{dA} = \frac{L_i(x,\theta_i,\phi_i)dw\cos\theta_i dA}{dA} = L_i(x,\theta_i,\phi_i)dw\cos\theta_i$$



Irradiance due to a Point Source

- For a point source,

$$\text{Radiance_Intensity} = \frac{\text{power}}{\text{solid_angle_source}}$$

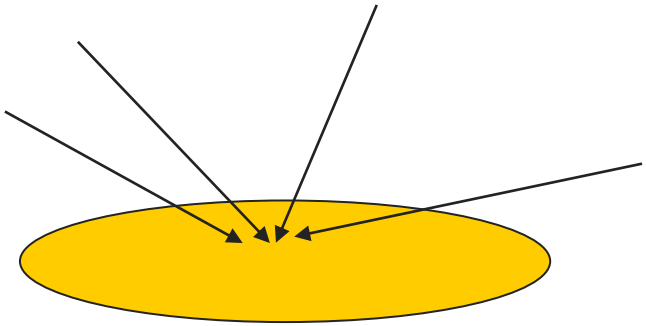
$$I = \frac{d\psi}{dw} = \frac{r^2 d\psi}{dA_i \cos\theta_i}$$

$$dw = \frac{dA_i \cos\theta_i}{r^2}$$

$$\Rightarrow \quad d\psi = I \frac{dA_i \cos\theta_i}{r^2} \qquad \Rightarrow \quad Irradiance = \frac{d\psi}{dA_i} = I \frac{\cos\theta_i}{r^2}$$

Irradiance

- What is the total irradiance?

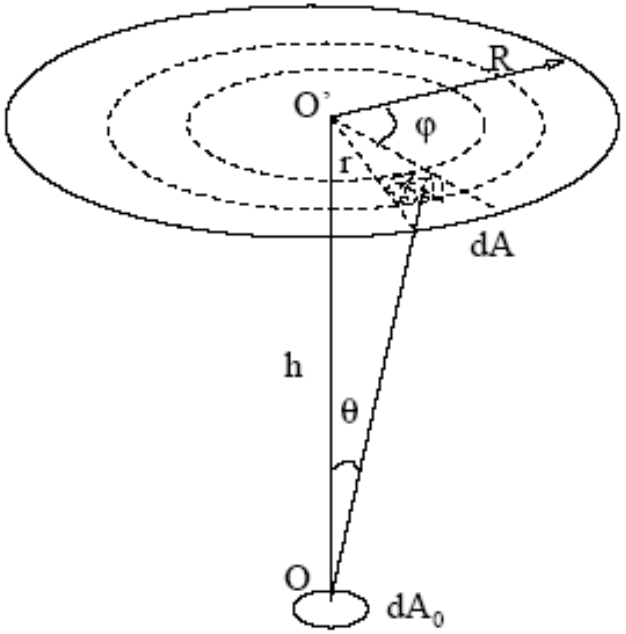


Integrate over the whole hemisphere.

Exercise: Suppose the radiance is constant from all directions. Calculate the irradiance.

Irradiance

- Exercise: Calculate the irradiance at O due to a plate source at O’.



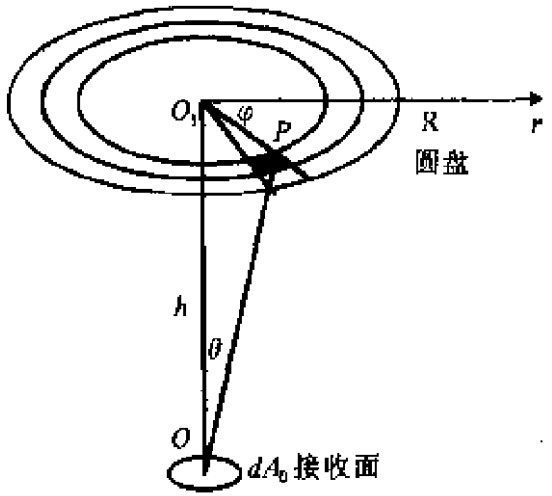
Irradiance due to a Surface Source

$$d\omega = \frac{dA_0 \cos\theta}{(PO)^2}$$

$$d^2P = LdA.d\omega = L.r\cos\theta dr d\varphi \bullet \frac{dA_0 \cos\theta}{(PO)^2}$$

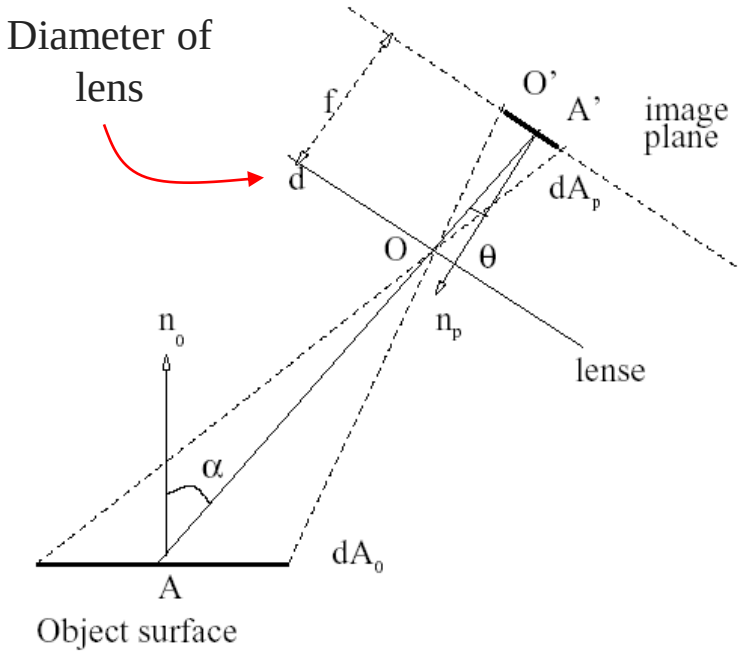
$$dP = \int_0^R \int_0^{2\pi} d^2P = L\frac{\pi R^2}{h^2 + R^2} dA_0$$

$$E = \frac{dP}{dA_0} = L\frac{\pi R^2}{h^2 + R^2}$$



$$\cos\theta = \frac{h}{PO}$$

The Relationship Between Image Intensity and Object Radiance



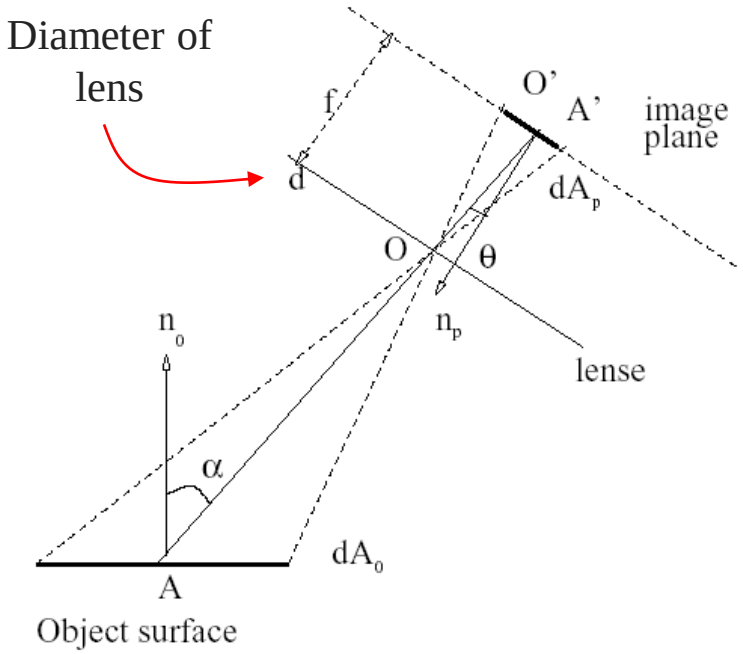
We assume that there is no power loss in the lens.

The power emitted to the lens is

$$d\psi = L_{object} dA_0 \cos\alpha dw_0$$

Radiance of object

The Relationship Between Image Intensity and Object Radiance



The solid angle for the entire lens is

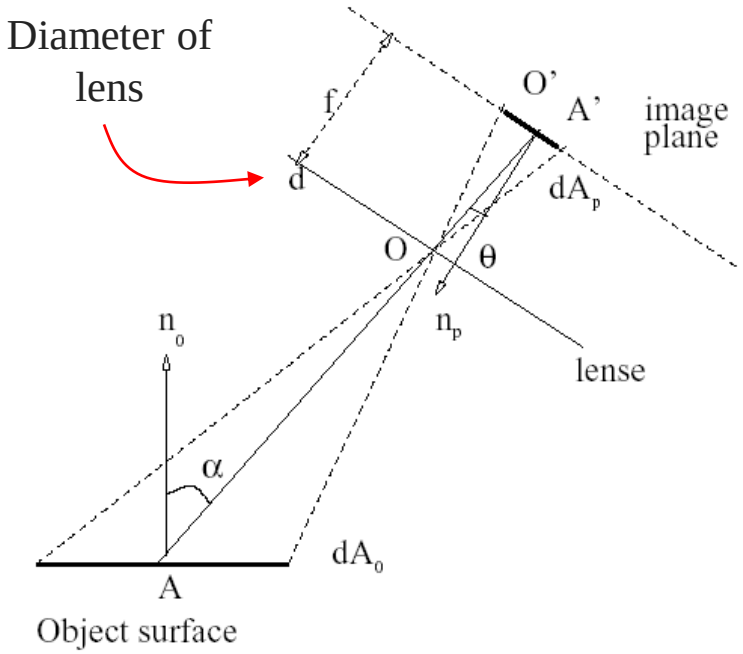
$$dw_0 = \frac{\left(\pi d^2 / 4\right)\cos\theta}{r^2}$$

The power emitted to the lens is

$$d\psi = L_{object} dA_0 \cos\alpha dw_0$$

$$= L_{object} dA_0 \cos\alpha \frac{\pi d^2 \cos\theta}{4r^2}$$

The Relationship Between Image Intensity and Object Radiance



The solid angle at O can be written in two ways.

$$\frac{dA_0 \cos\alpha}{r^2} = \frac{dA_p \cos\theta}{|OA'|^2}$$

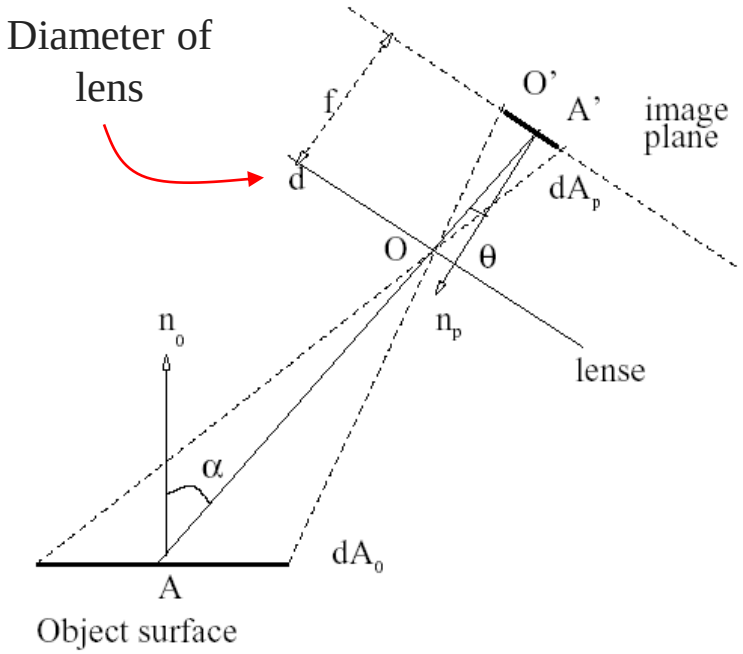
Note that

$$|OA'| = f / \cos\theta$$

Therefore

$$\frac{dA_0 \cos\alpha}{r^2} = \frac{dA_p \cos^3\theta}{f^2}$$

The Relationship Between Image Intensity and Object Radiance



Combine

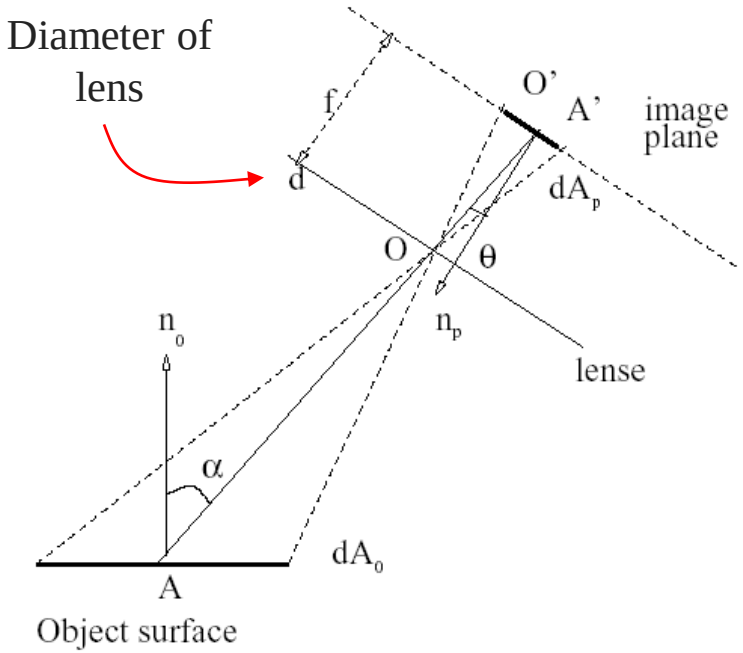
$$\frac{dA_0 \cos\alpha}{r^2} = \frac{dA_p \cos^3\theta}{f^2}$$

$$d\psi = L_{object} dA_0 \cos\alpha \frac{\pi d^2 \cos\theta}{4r^2}$$

to get

$$d\psi = \left(\frac{\pi}{4}\right) L_{object} \left(\frac{d}{f}\right)^2 \cos^4\theta dA_p$$

The Relationship Between Image Intensity and Object Radiance



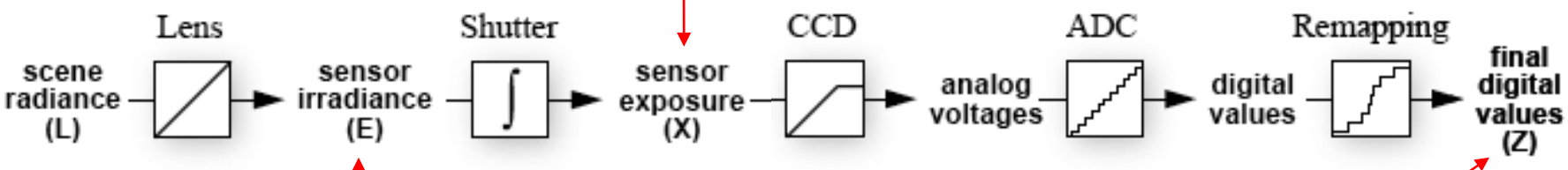
Therefore the irradiance on the image plane is

$$Irradiance = \frac{d\psi}{dA_p} = \left(\frac{\pi}{4}\right) L_{object} \left(\frac{d}{f}\right)^2 \cos^4\theta$$

The irradiance is converted to pixel intensities, which is directly proportional to the radiance of the object.

From light rays to pixel values

$$X = E \cdot \Delta t$$



$$E = \left[\frac{\pi}{4} \left(\frac{d}{f} \right)^2 \cos^4 \alpha \right] L$$

$$Z = f\left(E \cdot \Delta t\right)$$

- Camera response function: the mapping f from irradiance to pixel values
  - Useful if we want to estimate material properties
  - Enables us to create *high dynamic range (HDR) images*
  - Classic reference: P. E. Debevec and J. Malik, *Recovering High Dynamic Range Radiance Maps from Photographs*, *SIGGRAPH 97*

Photometry

Radiometry：energy，objective

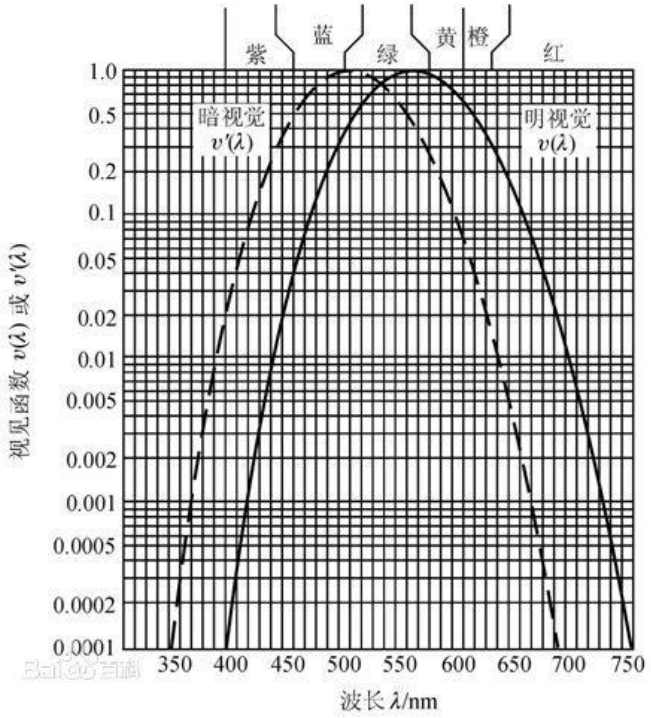
Photometry：visible intensity，subjective

Visibility function V_λ

$\phi_\lambda = V_\lambda . P$

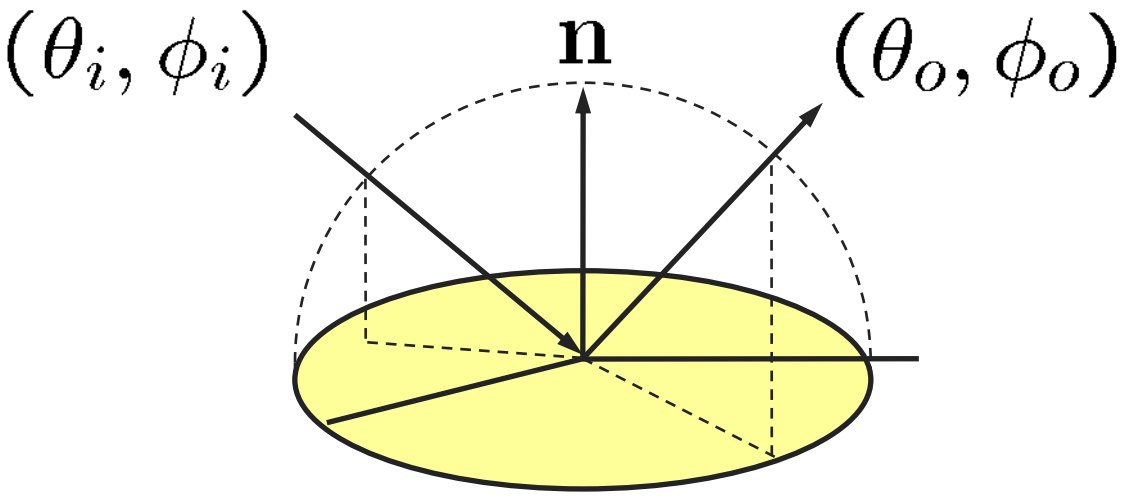
Definition：visible energy=683lm @λ=555nm，P=1W

$$\phi_\lambda = 683 \int_{\lambda_a}^{\lambda_b} V_\lambda \varphi_\lambda d\lambda$$



Surface Characteristics

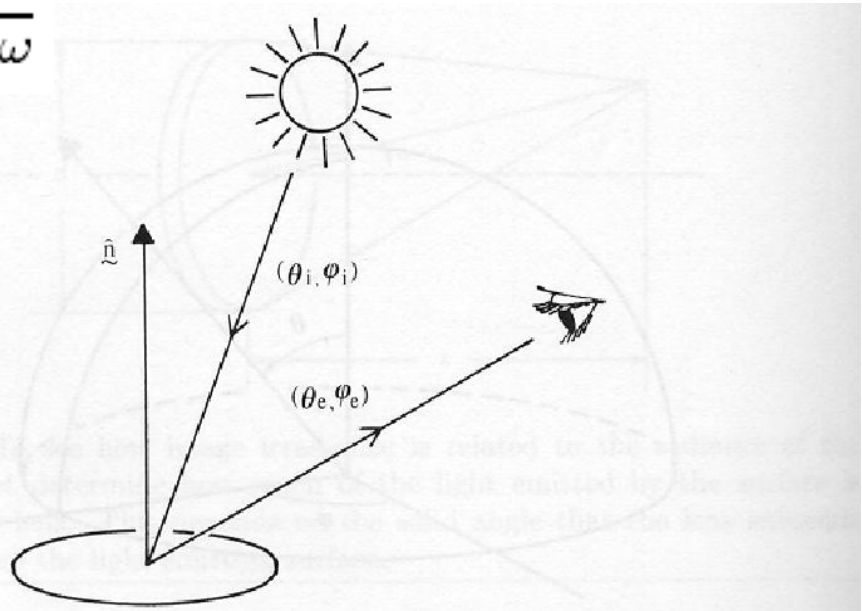
- We want to describe the relationship between incoming light and reflected light.
- This is a function of both the direction in which light arrives at a surface and the direction in which it leaves.



Bidirectional Reflectance Distribution Function (BRDF)

- BRDF is defined as the ratio of the radiance in the outgoing direction to the incident irradiance.

$$\rho_{bd}(\theta_o, \phi_o, \theta_i, \phi_i) = \frac{L_o(\boldsymbol{x}, \theta_o, \phi_o)}{L_i(\boldsymbol{x}, \theta_i, \phi_i) \cos\theta_i d\omega}$$



Horn, 1986

Figure 10-7. The bidirectional reflectance distribution function is the ratio of the radiance of the surface patch as viewed from the direction (θ_e, ϕ_e) to the irradiance resulting from illumination from the direction (θ_i, ϕ_i).

Bidirectional Reflectance Distribution Function (BRDF)

- The radiance leaving a surface due to irradiance *in a particular direction* is easily obtained from the definition of BRDF:

$$L_o(\boldsymbol{x}, \theta_o, \phi_o) = \rho_{bd}(\theta_o, \phi_o, \theta_i, \phi_i) L_i(\boldsymbol{x}, \theta_i, \phi_i) \cos\theta_i d\omega$$

Bidirectional Reflectance Distribution Function (BRDF)

- The radiance leaving a surface due to irradiance *in all incoming directions* is

$$L_o(\boldsymbol{x}, \theta_o, \phi_o) = \int_\Omega \rho_{bd}(\theta_o, \phi_o, \theta_i, \phi_i) L_i(\boldsymbol{x}, \theta_i, \phi_i) \cos\theta_i d\boldsymbol{\omega}$$

where \Omega is the incoming hemisphere.

Lambertian Surface

- A *Lambertian surface* has constant BRDF.

constant

$$\rho_{bd}(\theta_o, \phi_o, \theta_i, \phi_i) = \frac{L_o(x, \theta_o, \phi_o)}{L_i(x, \theta_i, \phi_i) \cos\theta_i d\omega}$$

Lambertian Surface

- A Lambertian surface looks equally bright from any view direction.
- The image intensities of the surface only changes with the illumination directions.

constant

$$\rho_{bd}(\theta_o, \phi_o, \theta_i, \phi_i) = \frac{L_o(\boldsymbol{x}, \theta_o, \phi_o)}{L_i(\boldsymbol{x}, \theta_i, \phi_i) \cos\theta_i d\omega}$$

Lambertian Surface

- For a Lambertian surface, the outgoing radiance is proportional to the incident radiance.

constant

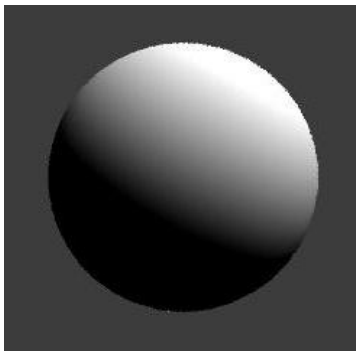
$$\rho_{bd}(\theta_o, \phi_o, \theta_i, \phi_i) = \frac{L_o(x, \theta_o, \phi_o)}{L_i(x, \theta_i, \phi_i) \cos\theta_i d\omega}$$

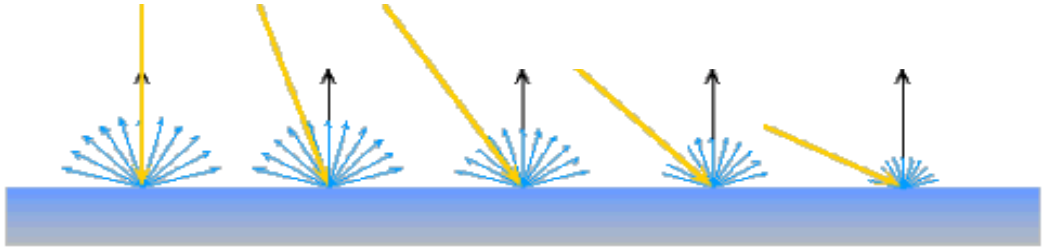
- If the light source is a point source, a pixel intensity will only be a function of $cos\theta_i$

Remember, for a point source $Irradiance = \frac{d\psi}{dA_i} = I\frac{\cos\theta_i}{r^2}$

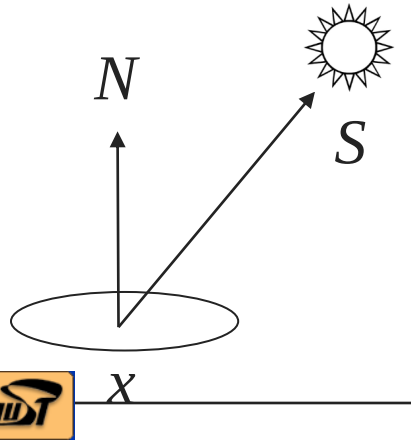
Diffuse reflection: Lambert's law

- Viewed brightness does not depend on viewing direction, but it *does* depend on direction of illumination





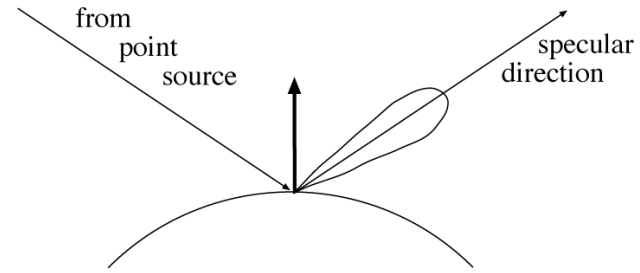
$$B(x) = \rho_d(x)\left(N(x)\cdot S_d(x)\right)$$

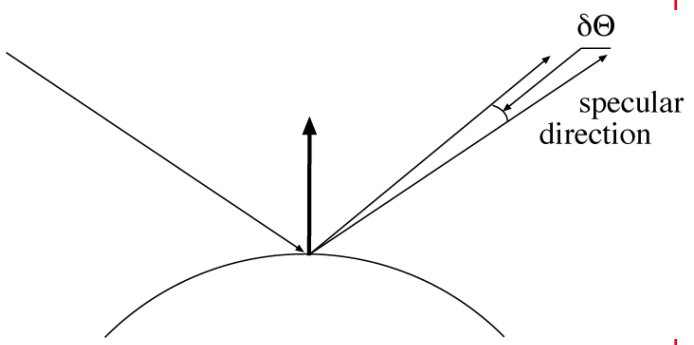


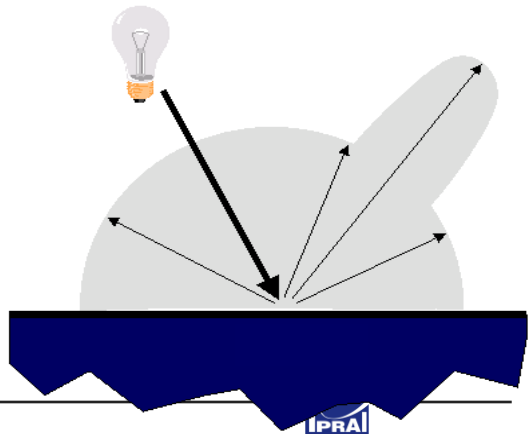
B: radiosity
ρ: albedo
N: unit normal
S: source vector (magnitude proportional to intensity of the source)

Specular reflection

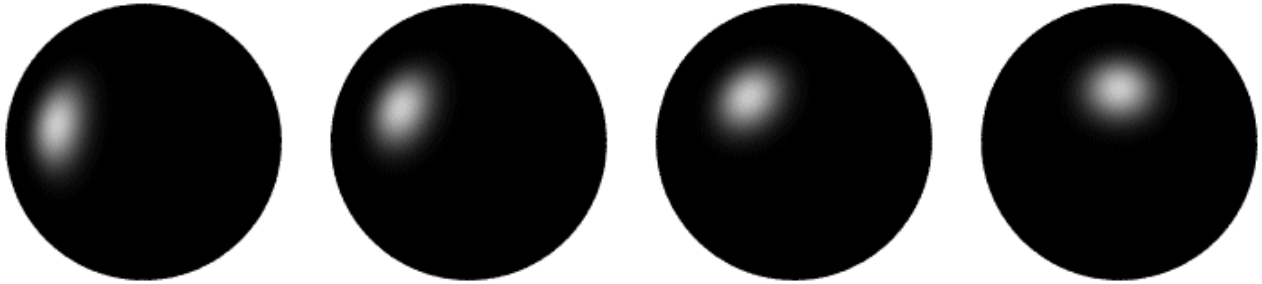
- Radiation arriving along a source direction leaves along the specular direction (source direction reflected about normal)
- Some fraction is absorbed, some reflected
- On real surfaces, energy usually goes into a lobe of directions
- Phong model: reflected energy falls of with $\cos^n(\delta\theta)$
- Lambertian + specular model: sum of diffuse and specular term



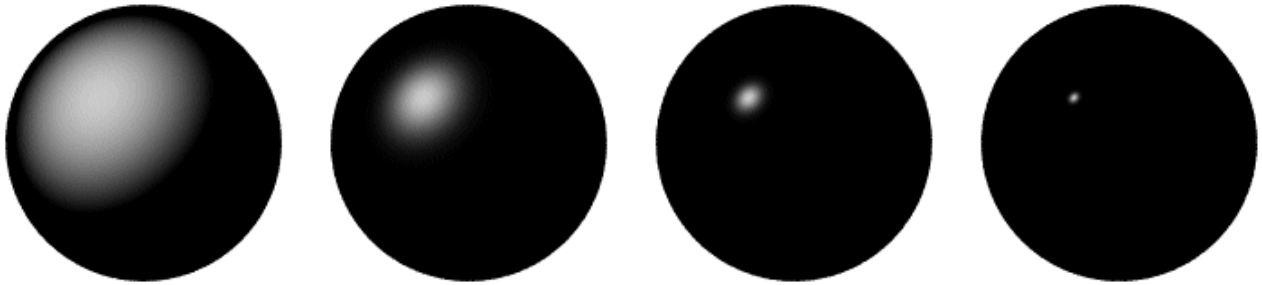




Specular reflection

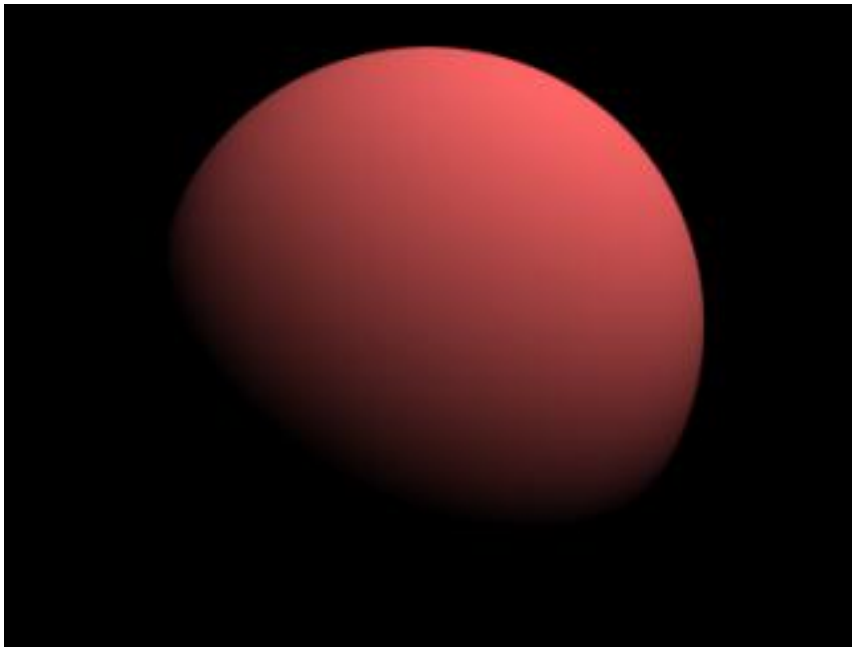


Moving the light source

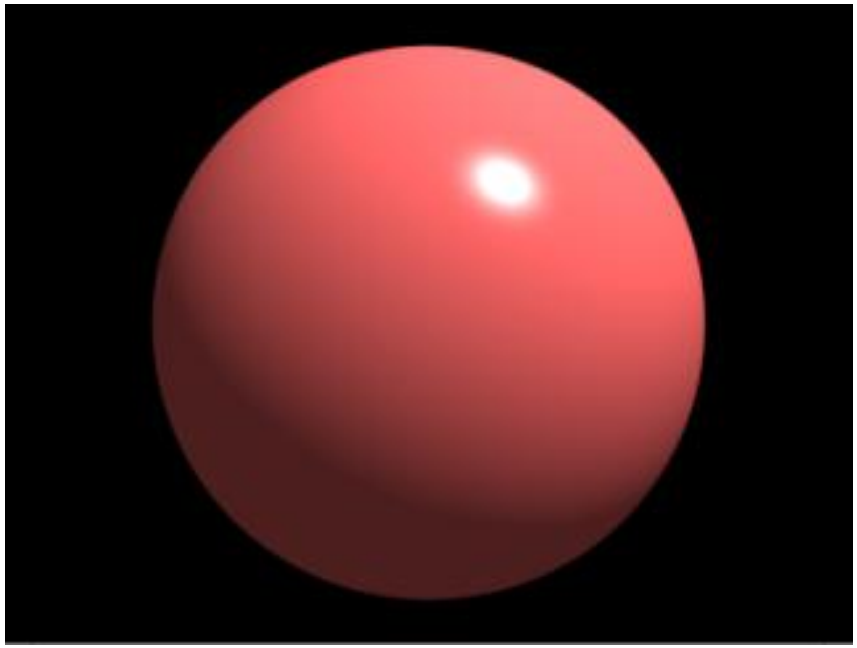


Changing the exponent

Lambertian + Specular Model



Lambertian



Lambertian + Specular

What differences in intensity tell us about shape

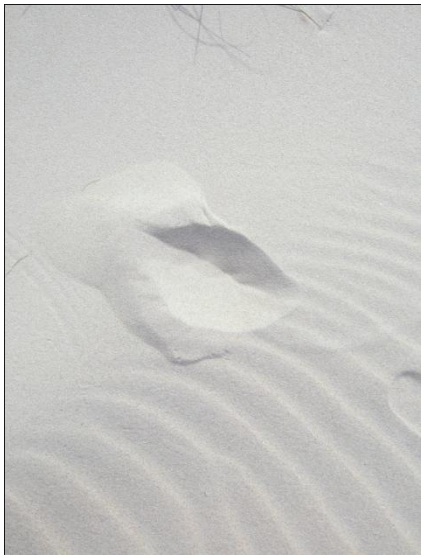
Changes in surface normal

Texture

Proximity

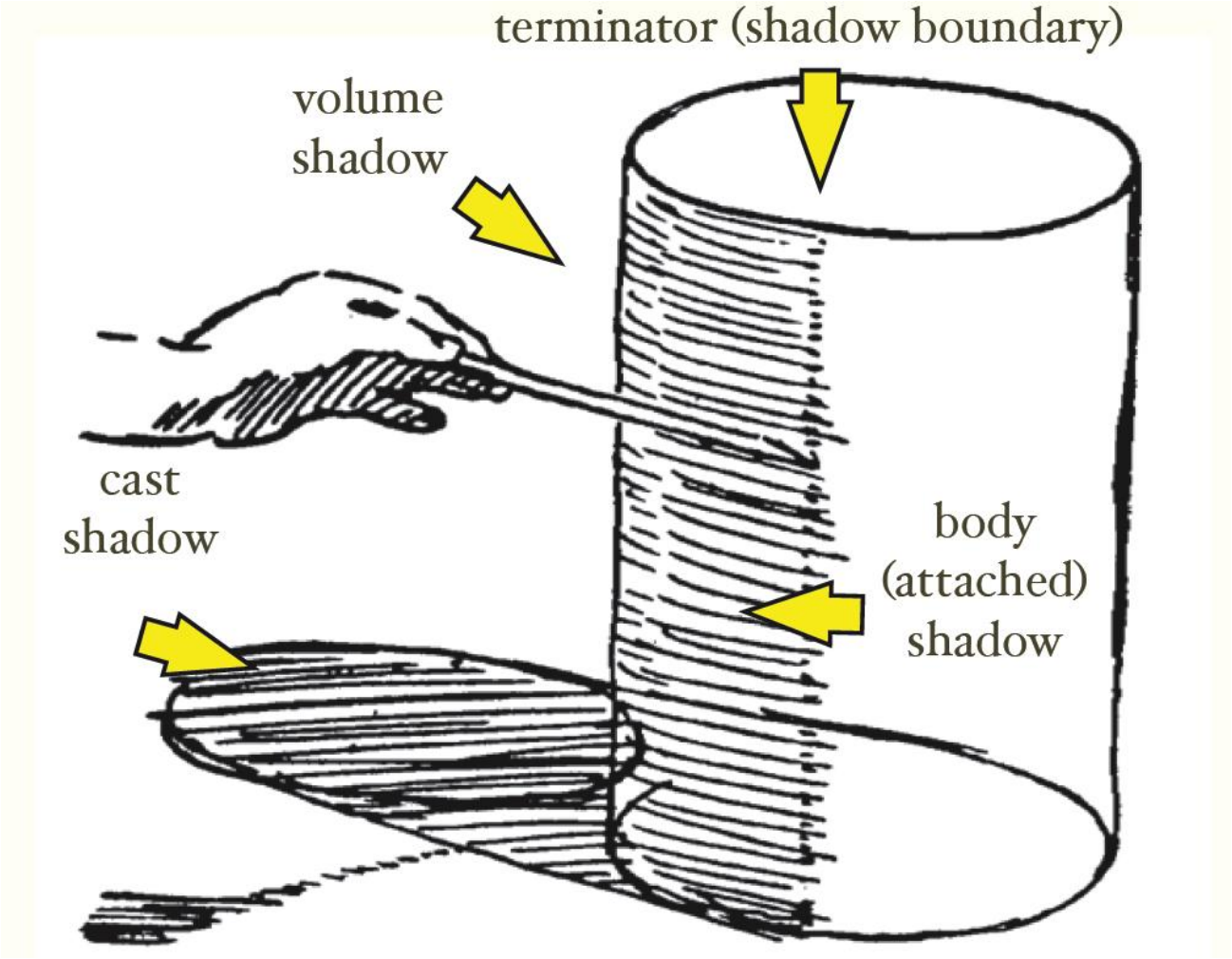
Indents and bumps

Grooves and creases





Shadows as cues



From Koenderink slides on image texture and the flow of light

Photometric stereo (shape from shading)

- Can we reconstruct the shape of an object based on shading cues?



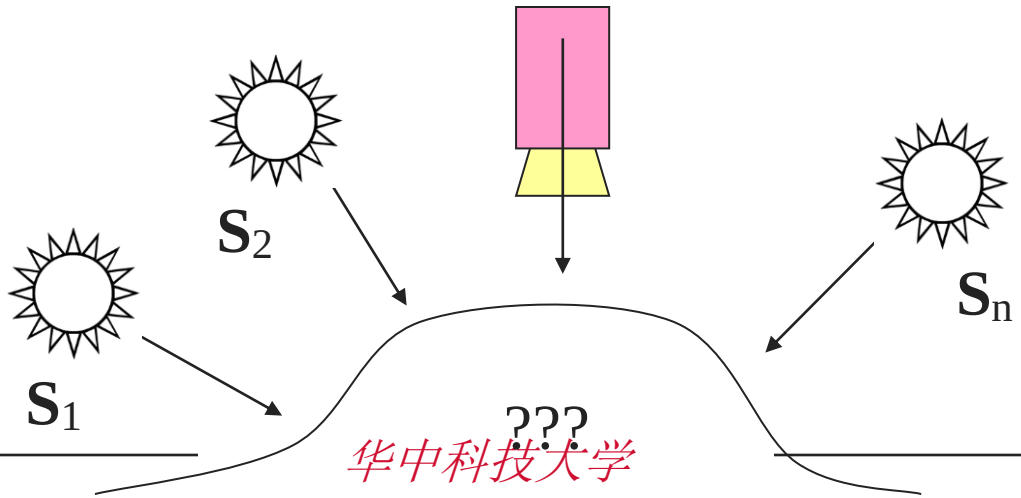
Luca della Robbia, *Cantoria*, 1438

Photometric stereo

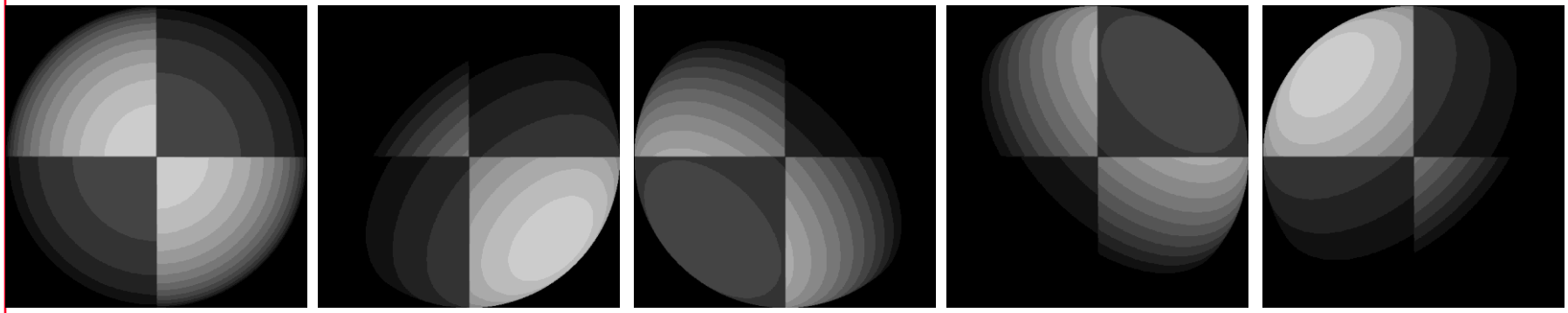
Assume:

- A Lambertian object
- A *local shading model* (each point on a surface receives light only from sources visible at that point)
- A set of *known* light source directions
- A set of pictures of an object, obtained in exactly the same camera/object configuration but using different sources
- Orthographic projection

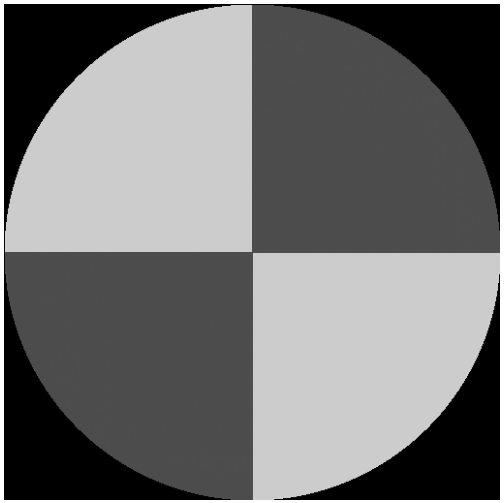
Goal: reconstruct object shape and albedo



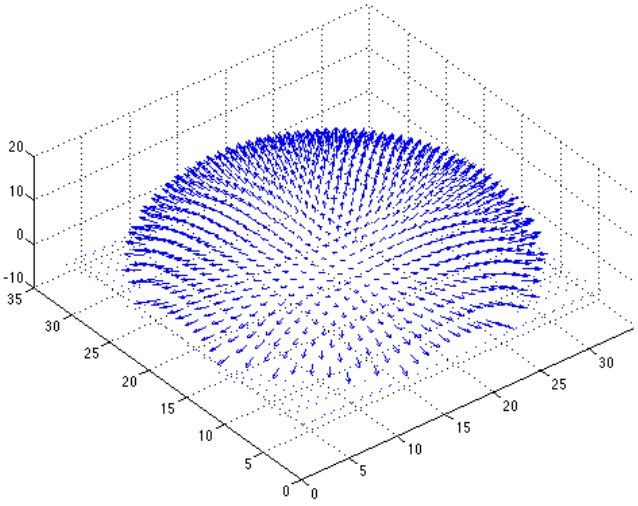
Example 1



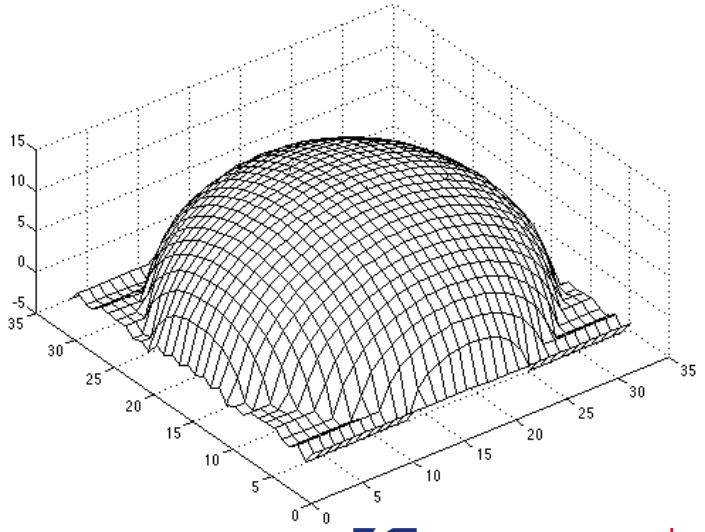
Recovered albedo



Recovered normal field



Recovered surface model

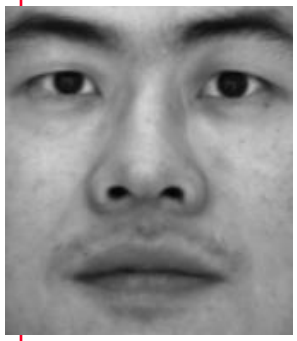


Example 2

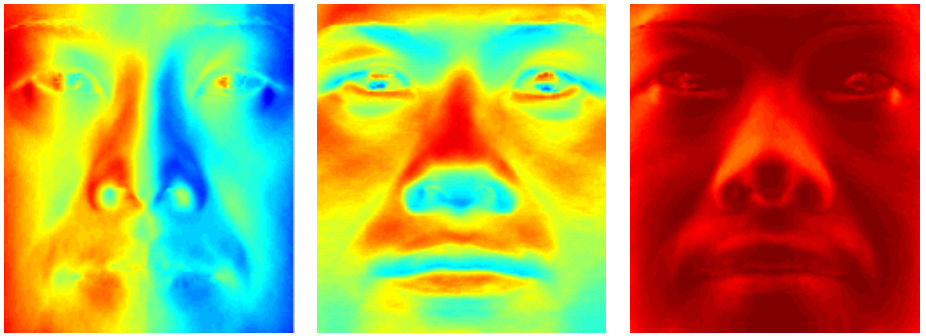
Input



Recovered albedo



Recovered normal field



x y z

Recovered surface model

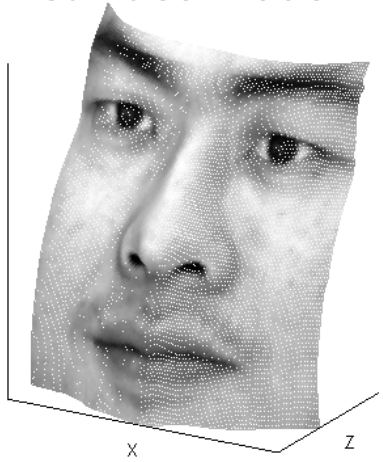


Image model

- **Known:** source vectors $\mathbf{S}_j$ and pixel values $I_j(x,y)$
- **Unknown:** surface normal $\mathbf{N}(x,y)$ and albedo $\rho(x,y)$

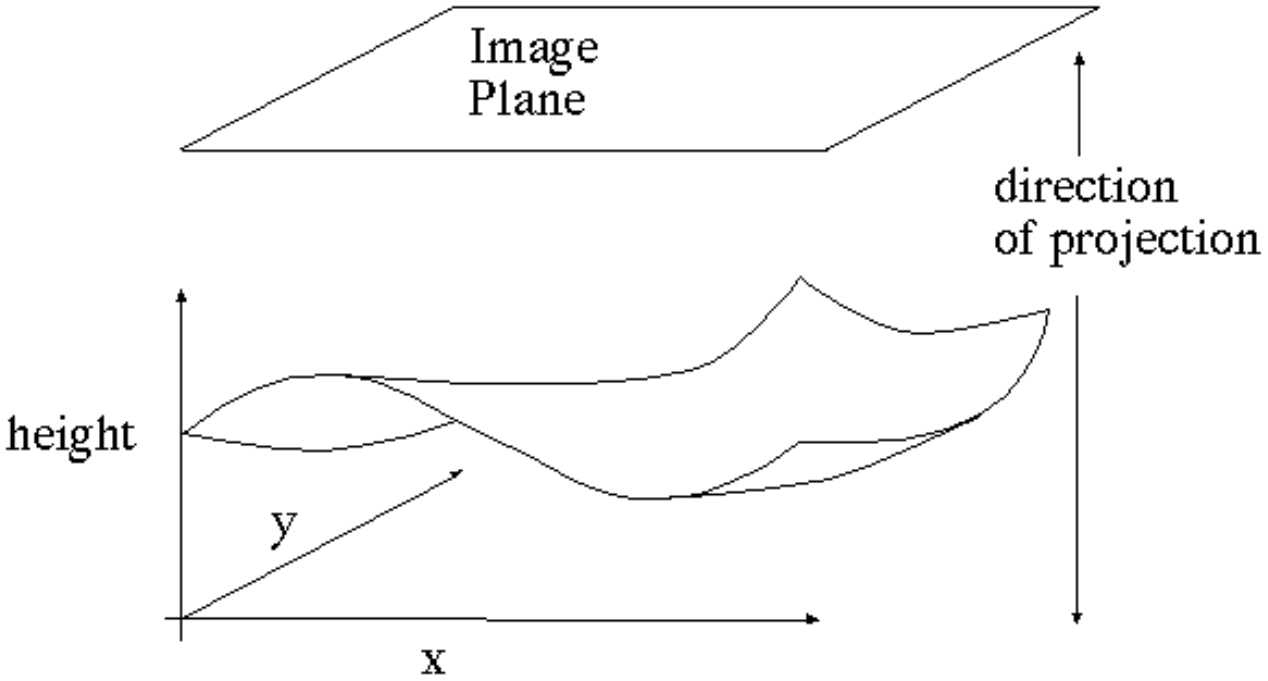


Image model

- **Known:** source vectors $\mathbf{S}_j$ and pixel values $I_j(x,y)$
- **Unknown:** surface normal $\mathbf{N}(x,y)$ and albedo $\rho(x,y)$
- Assume that the response function of the camera is a linear scaling by a factor of k
- Lambert's law:

$$
\begin{aligned}
I_j(x,y) &= k\,\rho(x,y)\left(\mathbf{N}(x,y)\cdot\mathbf{S}_j\right) \\
&= \left(\rho(x,y)\mathbf{N}(x,y)\right)\cdot\left(k\mathbf{S}_j\right) \\
&= \mathbf{g}(x,y)\cdot\mathbf{V}_j
\end{aligned}
$$

Least squares problem

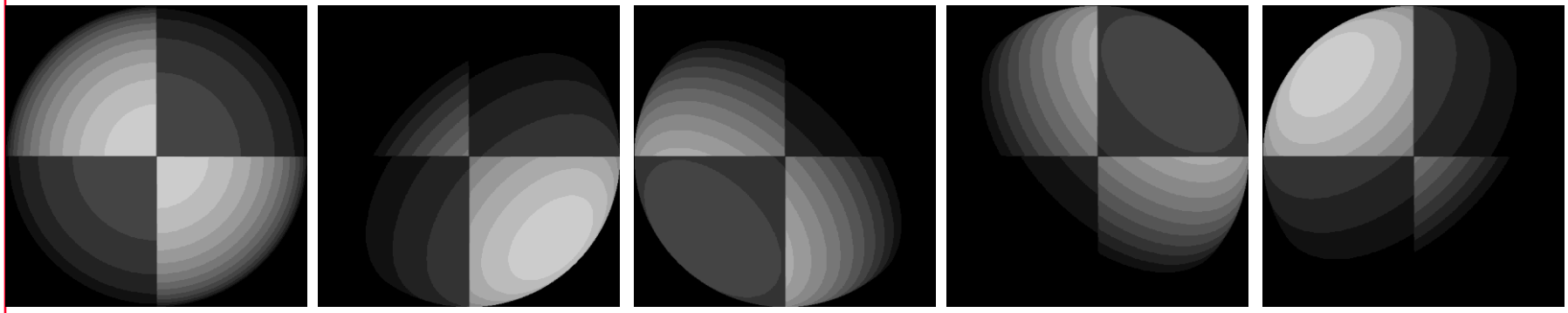
- For each pixel, set up a linear system:

$$\begin{bmatrix} I_1(x,y) \\ I_2(x,y) \\ \vdots \\ I_n(x,y) \end{bmatrix} = \begin{bmatrix} \mathbf{V}_1^T \\ \mathbf{V}_2^T \\ \vdots \\ \mathbf{V}_n^T \end{bmatrix} \mathbf{g}(x,y)$$

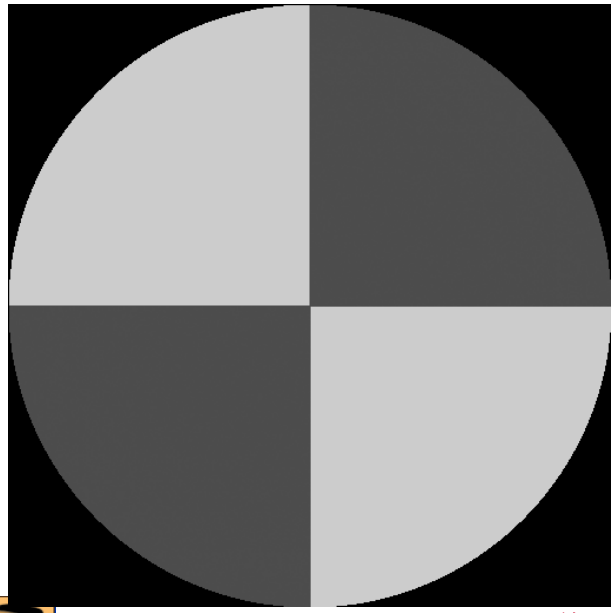
$(n \times 1)$ known — $(n \times 3)$ known — (3×1) unknown

- Obtain least-squares solution for $\mathbf{g}(x,y)$ (which we defined as $\mathbf{N}(x,y)\,\rho(x,y)$)
- Since $\mathbf{N}(x,y)$ is the unit normal, $\rho(x,y)$ is given by the magnitude of $\mathbf{g}(x,y)$
- Finally, $\mathbf{N}(x,y) = \mathbf{g}(x,y) / \rho(x,y)$

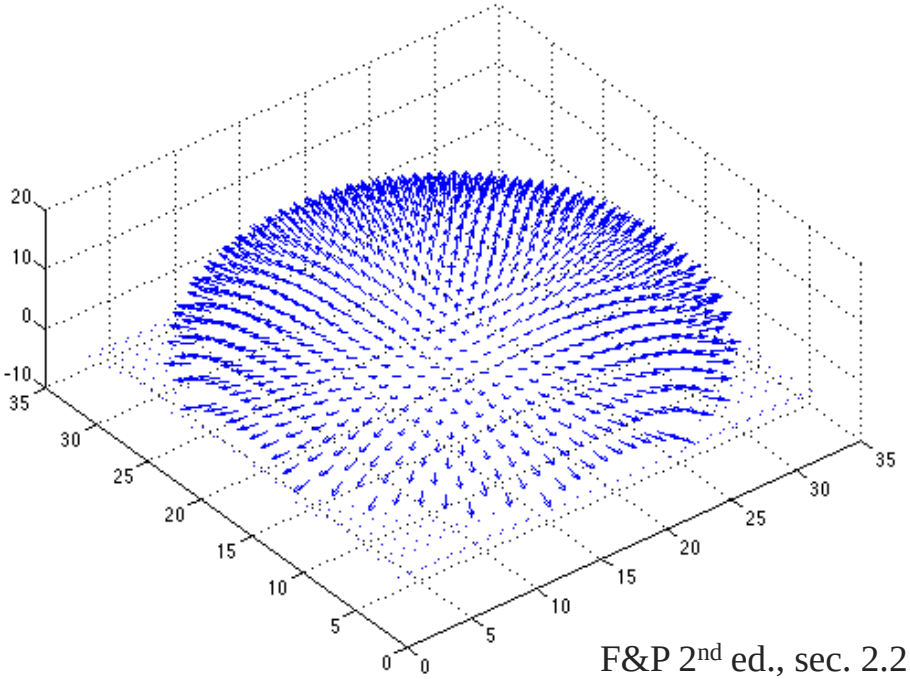
Synthetic example



Recovered albedo



Recovered normal field



华中

F&P 2nd ed., sec. 2.2.4

Recovering a surface from normals

Recall the surface is written as

$$(x, y, f(x, y))$$

This means the normal has the form:

$$\mathbf{N}(x, y) = \frac{1}{\sqrt{f_x^2 + f_y^2 + 1}} \begin{pmatrix} f_x \\ f_y \\ 1 \end{pmatrix}$$

If we write the estimated vector g as

$$\mathbf{g}(x, y) = \begin{pmatrix} g_1(x, y) \\ g_2(x, y) \\ g_3(x, y) \end{pmatrix}$$

Then we obtain values for the partial derivatives of the surface:

$$f_x(x, y) = g_1(x, y) / g_3(x, y)$$
$$f_y(x, y) = g_2(x, y) / g_3(x, y)$$

Recovering a surface from normals

We can now recover the surface height at any point by integration along some path, e.g.

$$f(x,y) = \int_0^x f_x(s,0)\,ds + \int_0^y f_y(x,t)\,dt + C$$

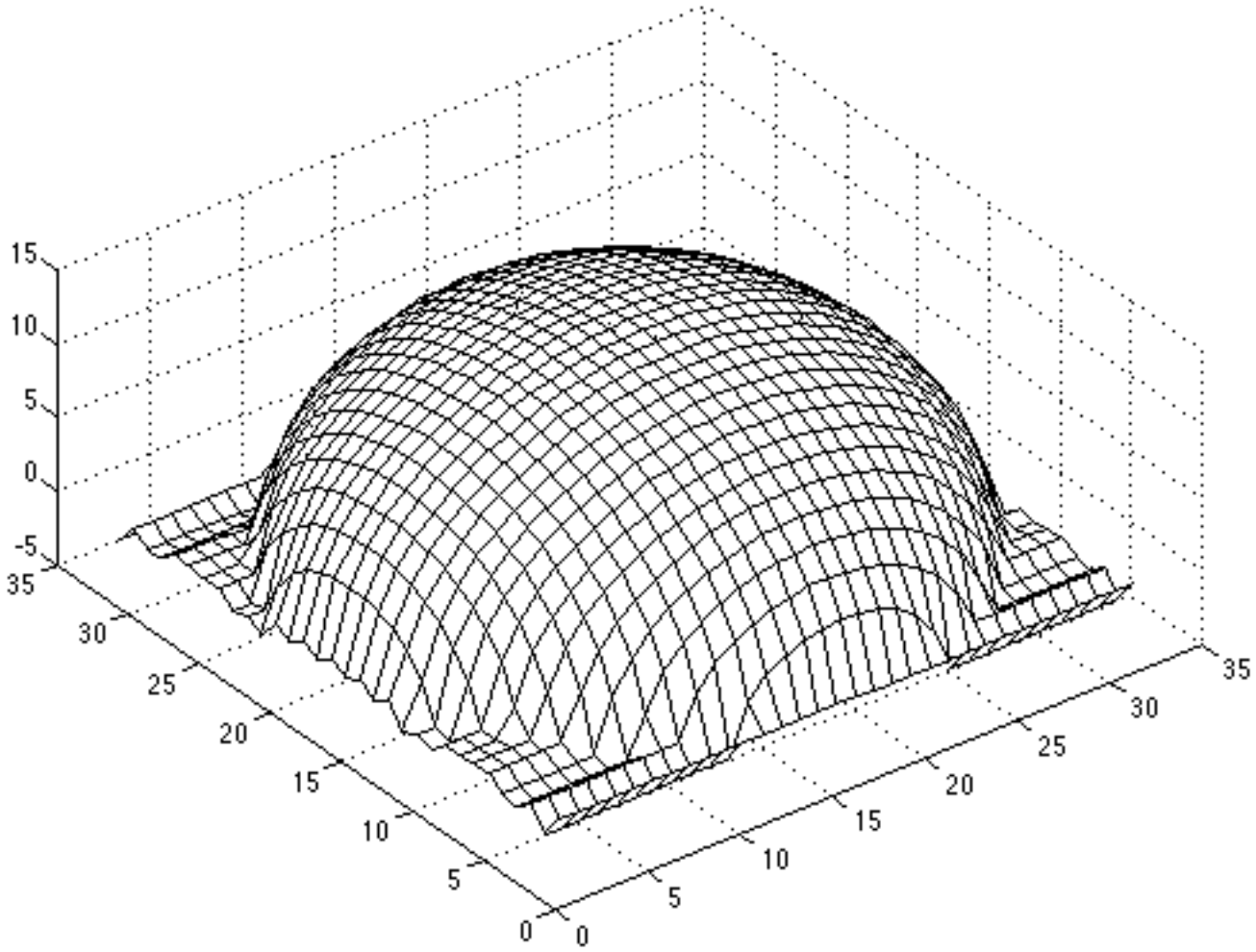
(for robustness, should take integrals over many different paths and average the results)

Integrability: for the surface f to exist, the mixed second partial derivatives must be equal:

$$\frac{\partial}{\partial y}(g_1(x,y)/g_3(x,y)) = \frac{\partial}{\partial x}(g_2(x,y)/g_3(x,y))$$

(in practice, they should at least be similar)

Surface recovered by integration



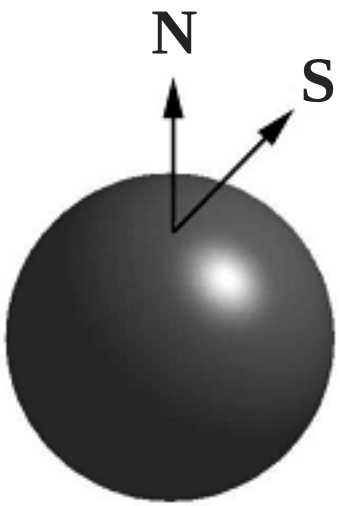
Limitations

- Orthographic camera model
- Simplistic reflectance and lighting model
- No shadows
- No interreflections
- No missing data
- Integration is tricky

Finding the direction of the light source

$$I(x,y) = \mathbf{N}(x,y) \cdot \mathbf{S}(x,y)$$

Full 3D case:

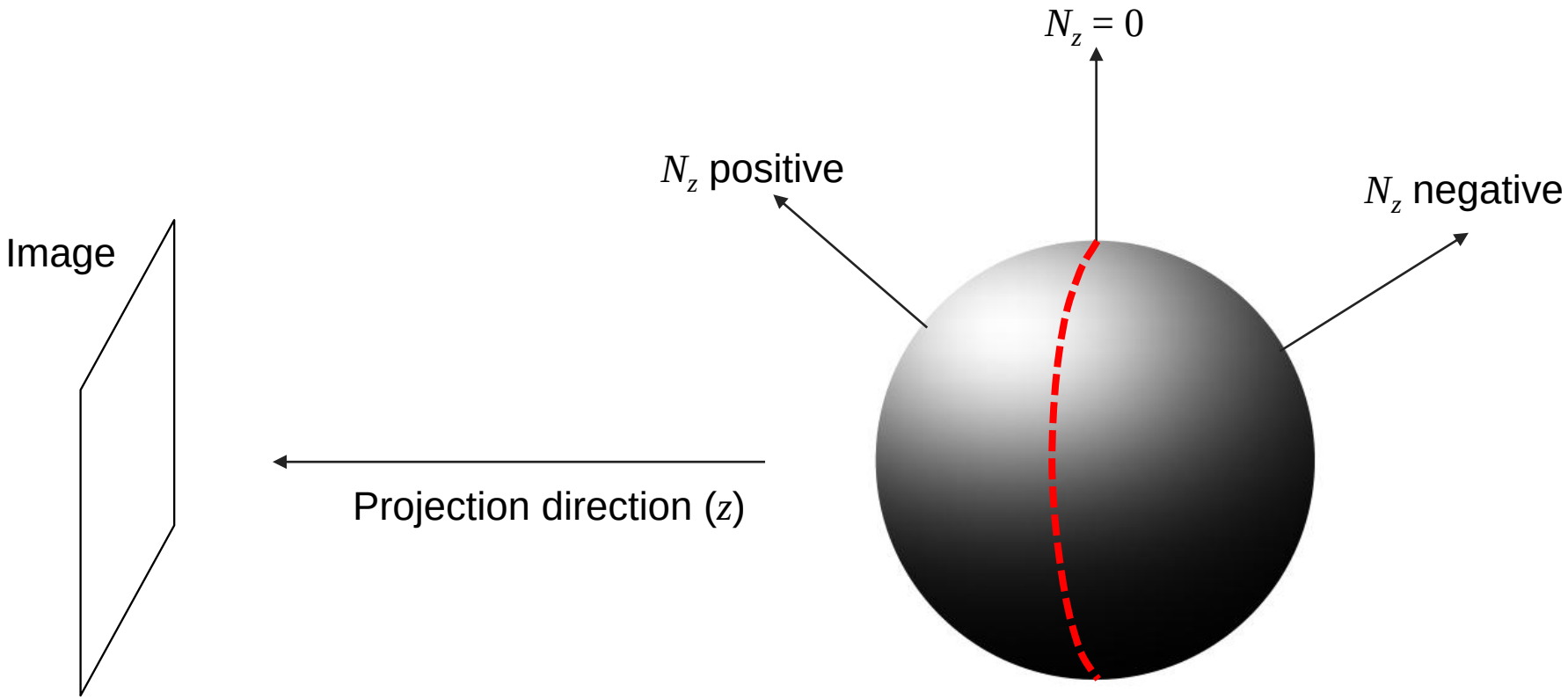


$$\begin{pmatrix} N_x(x_1,y_1) & N_y(x_1,y_1) & N_z(x_1,y_1) \\ N_x(x_2,y_2) & N_y(x_2,y_2) & N_z(x_2,y_2) \\ \vdots & \vdots & \vdots \\ N_x(x_n,y_n) & N_y(x_n,y_n) & N_z(x_n,y_n) \end{pmatrix} \begin{pmatrix} S_x \\ S_y \\ S_z \end{pmatrix} = \begin{pmatrix} I(x_1,y_1) \\ I(x_2,y_2) \\ \vdots \\ I(x_n,y_n) \end{pmatrix}$$

P. Nillius and J.-O. Eklundh, “Automatic estimation of the projected light source direction,” CVPR2001

Finding the direction of the light source

Consider points on the *occluding contour*:

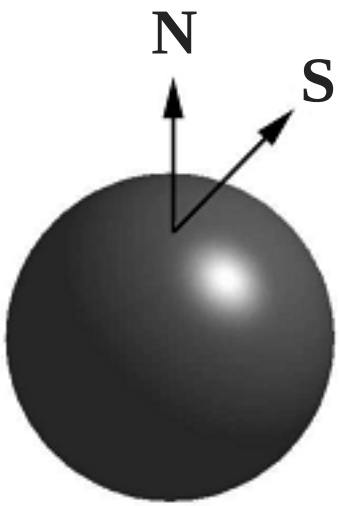


P. Nillius and J.-O. Eklundh, "Automatic estimation of the projected light source direction,"

Finding the direction of the light source

$$I(x,y) = \mathbf{N}(x,y) \cdot \mathbf{S}(x,y)$$

Full 3D case:



$$\begin{pmatrix} N_x(x_1,y_1) & N_y(x_1,y_1) & N_z(x_1,y_1) \\ N_x(x_2,y_2) & N_y(x_2,y_2) & N_z(x_2,y_2) \\ \vdots & \vdots & \vdots \\ N_x(x_n,y_n) & N_y(x_n,y_n) & N_z(x_n,y_n) \end{pmatrix} \begin{pmatrix} S_x \\ S_y \\ S_z \end{pmatrix} = \begin{pmatrix} I(x_1,y_1) \\ I(x_2,y_2) \\ \vdots \\ I(x_n,y_n) \end{pmatrix}$$

For points on the *occluding contour*, $N_z = 0$:

$$\begin{pmatrix} N_x(x_1,y_1) & N_y(x_1,y_1) \\ N_x(x_2,y_2) & N_y(x_2,y_2) \\ \vdots & \vdots \\ N_x(x_n,y_n) & N_y(x_n,y_n) \end{pmatrix} \begin{pmatrix} S_x \\ S_y \end{pmatrix} = \begin{pmatrix} I(x_1,y_1) \\ I(x_2,y_2) \\ \vdots \\ I(x_n,y_n) \end{pmatrix}$$

P. Nillius and J.-O. Eklundh, “Automatic estimation of the projected light source direction,”

Finding the direction of the light source



P. Nillius and J.-O. Eklundh, “Automatic estimation of the projected light source direction,”

Application: Detecting composite photos

Real photo

Fake photo

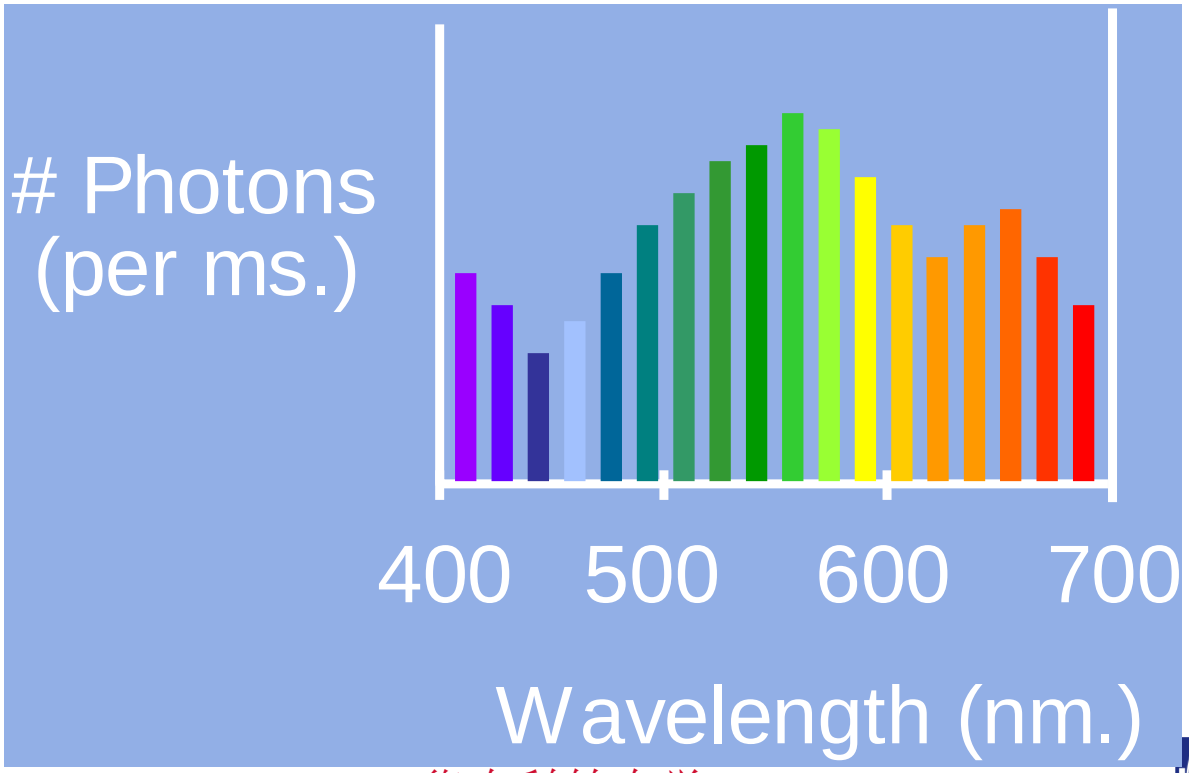




M. K. Johnson and H. Farid, Exposing Digital Forgeries by Detecting Inconsistencies in Lighting, ACM Multimedia and Security Workshop, 2005.

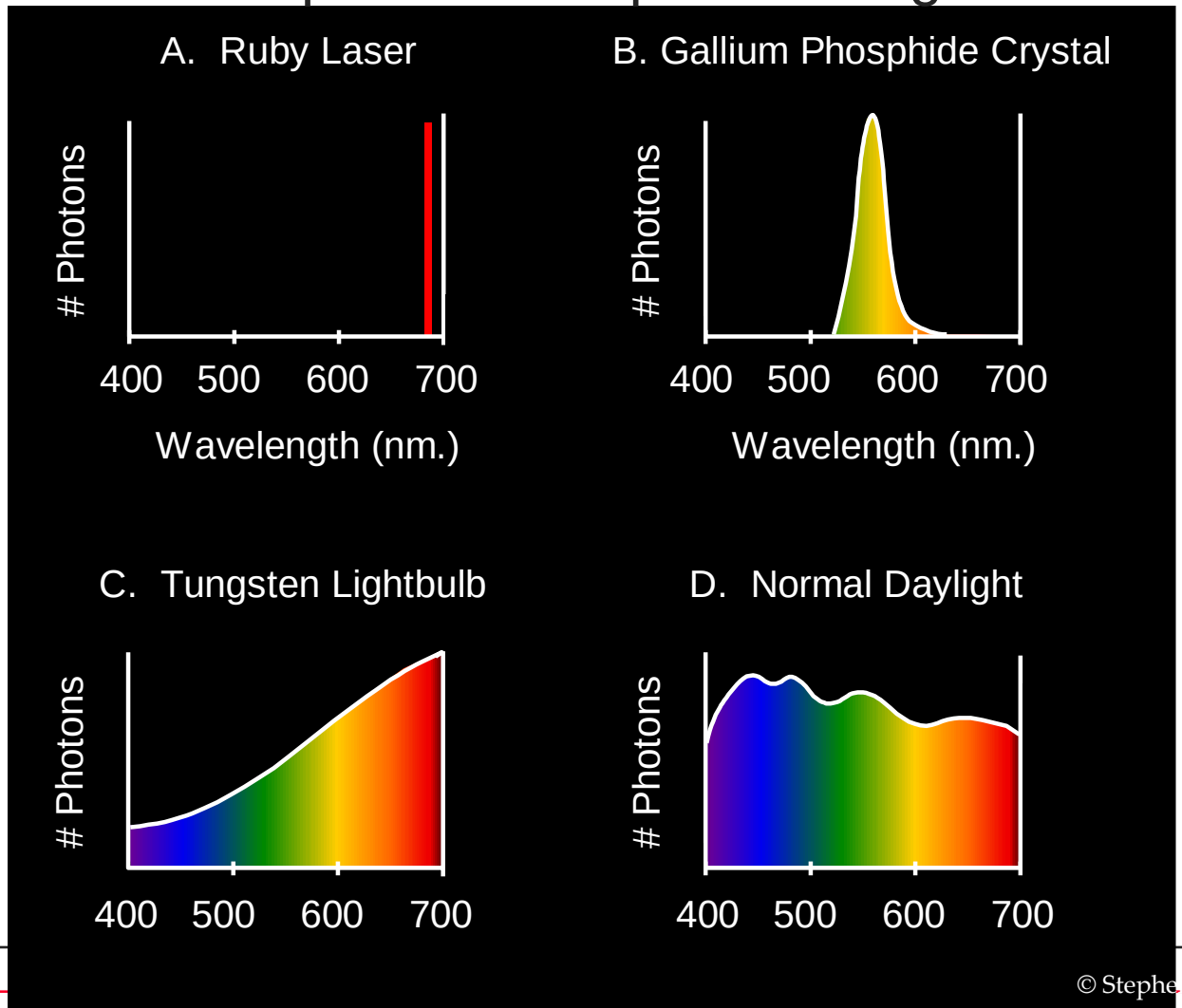
The Physics of Light

Any patch of light can be completely described physically by its spectrum: the number of photons (per time unit) at each wavelength 400 - 700 nm.



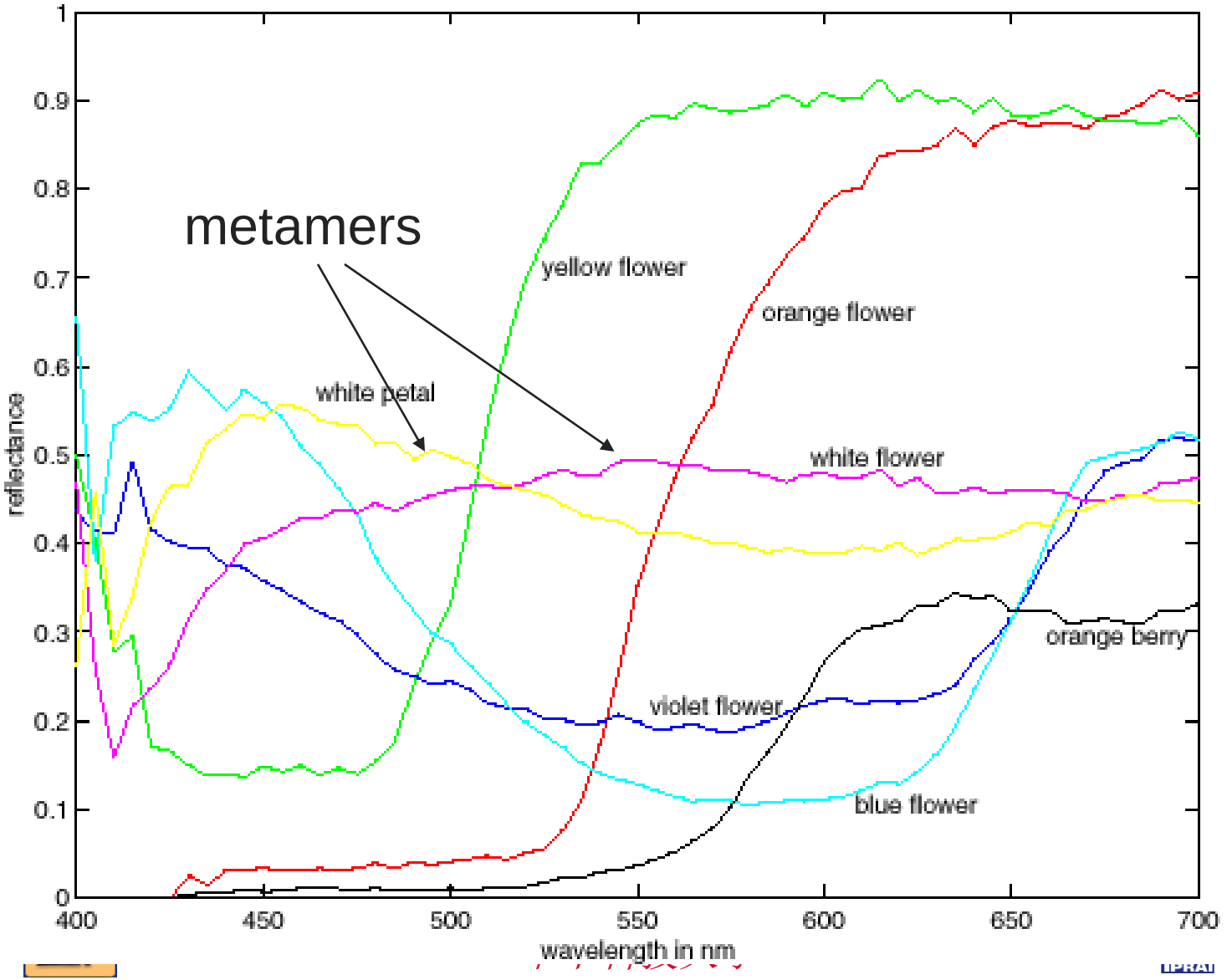
The Physics of Light

Some examples of the spectra of light sources



© Stephen

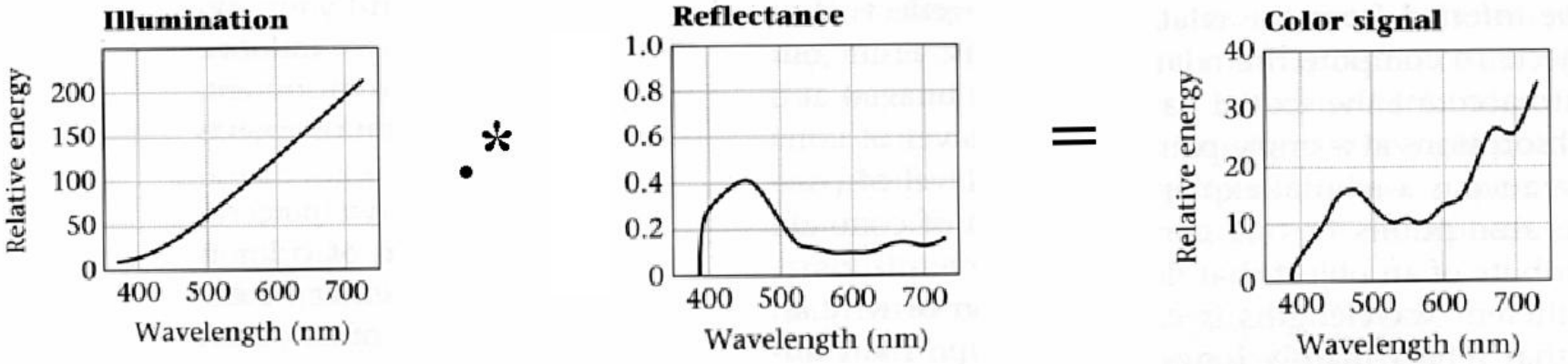
More Spectra



Interaction of light and surfaces



Reflected color is the result of interaction of light source spectrum with surface reflectance



Interaction of light and surfaces

What is the observed color of any surface under monochromatic light?



Olafur Eliasson, *Room for one color*

华中科技大学

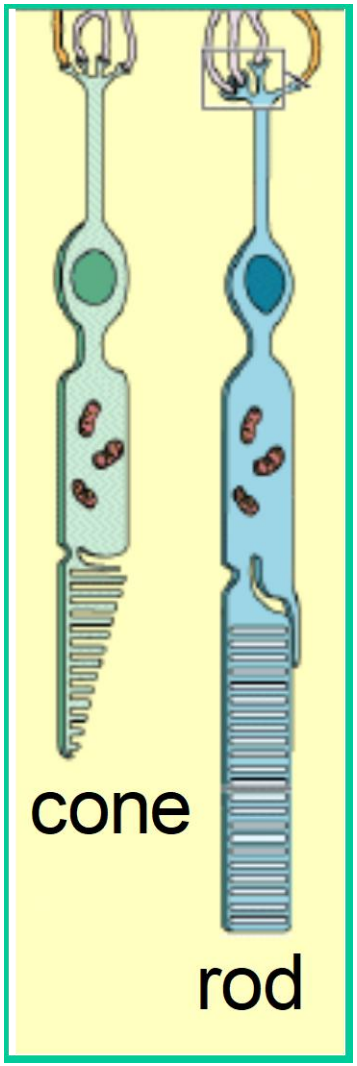
Two types of light-sensitive receptors

Cones

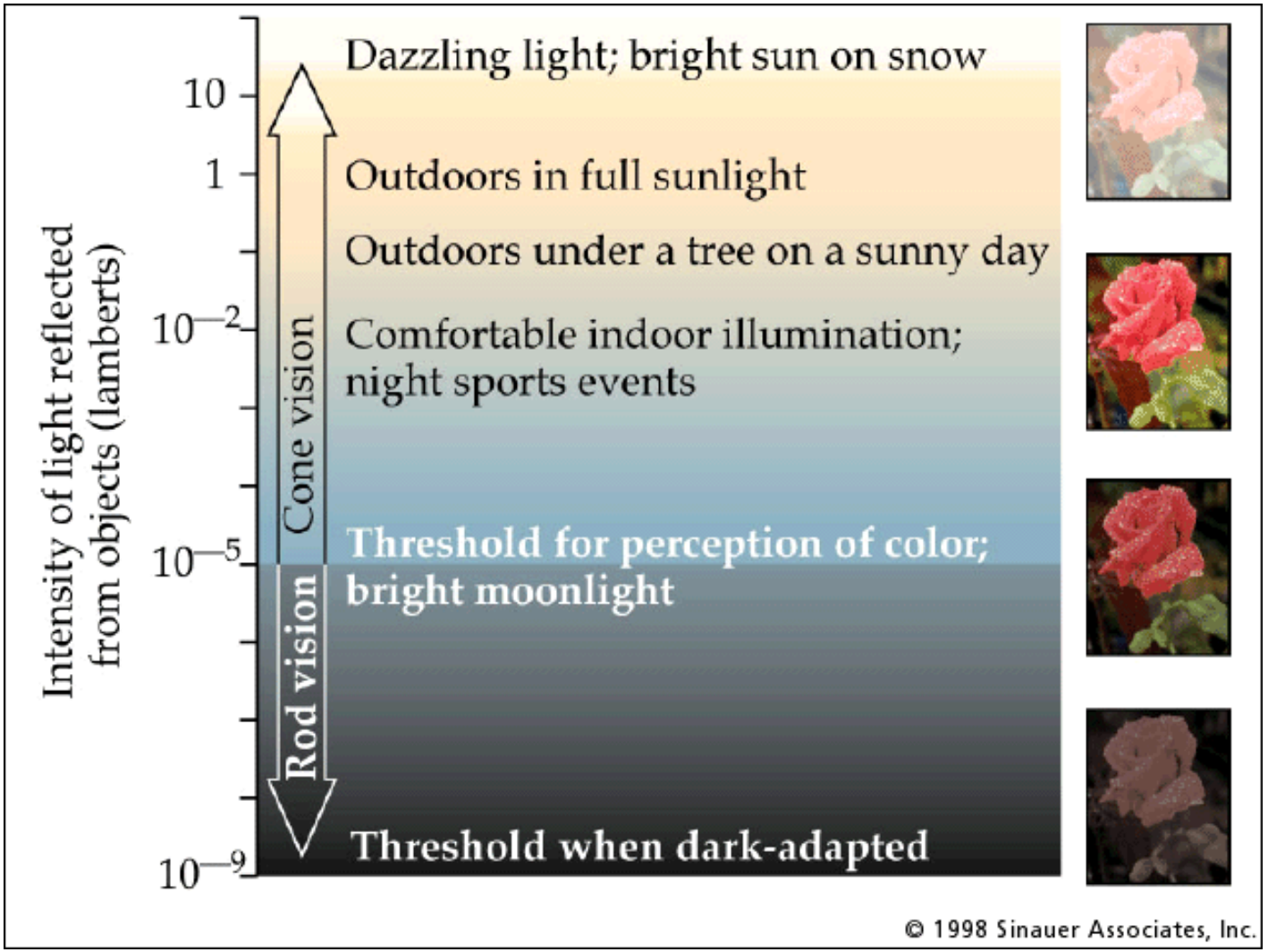
- cone-shaped
- less sensitive
- operate in high light
- color vision

Rods

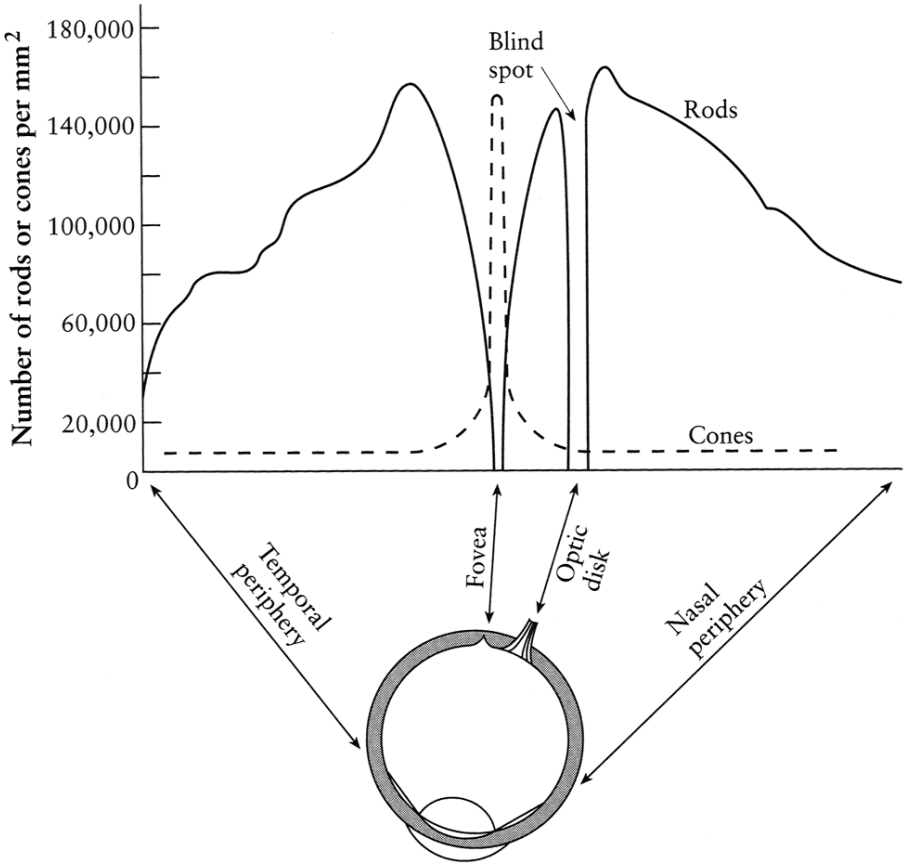
- rod-shaped
- highly sensitive
- operate at night
- gray-scale vision



Rod / Cone sensitivity



Distribution of Rods and Cones

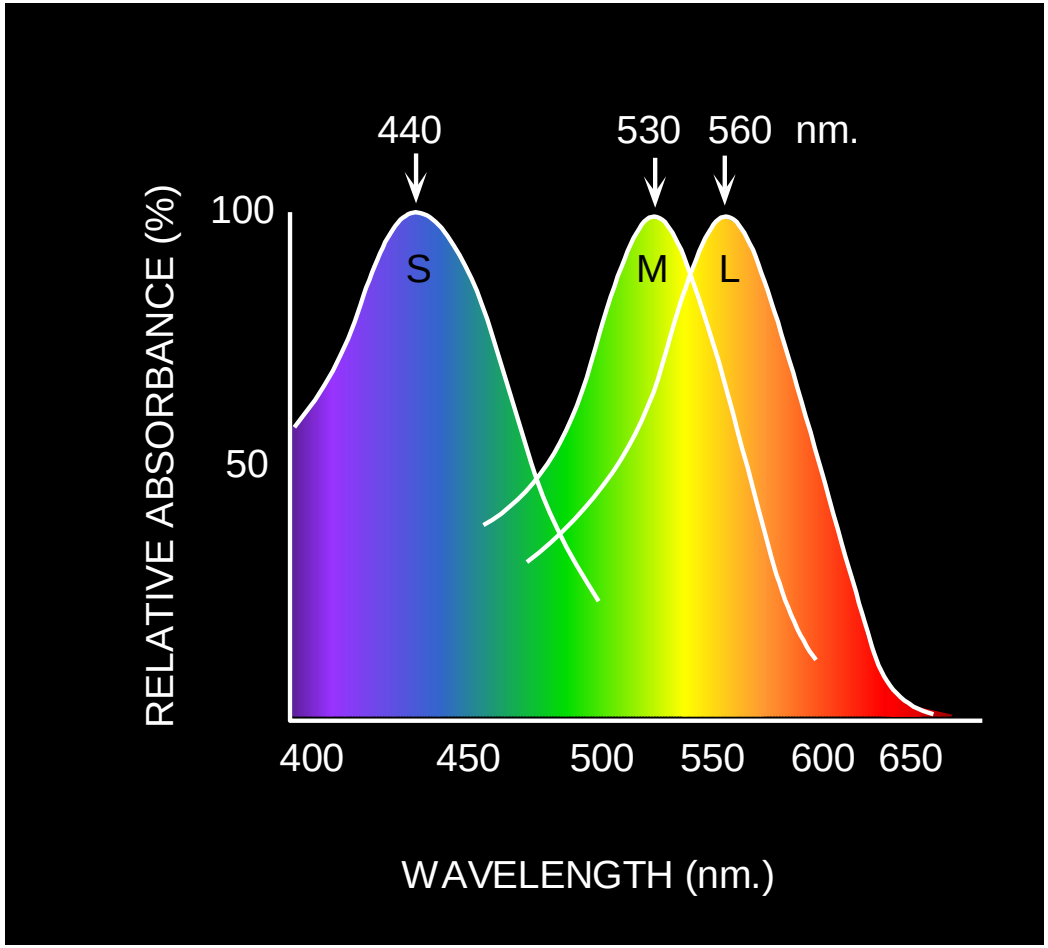


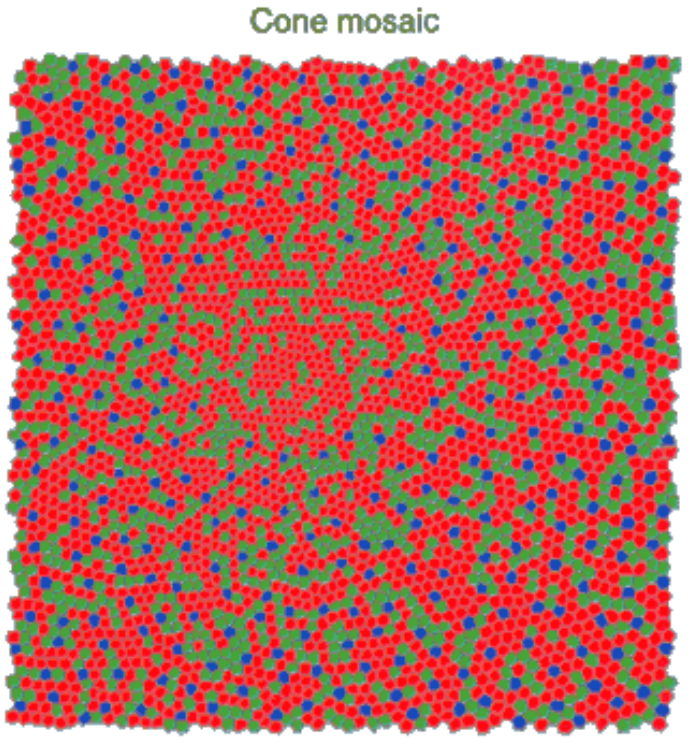
Night Sky: why are there more stars off-center?

Averted vision: http://en.wikipedia.org/wiki/Averted_vision

Physiology of Color Vision

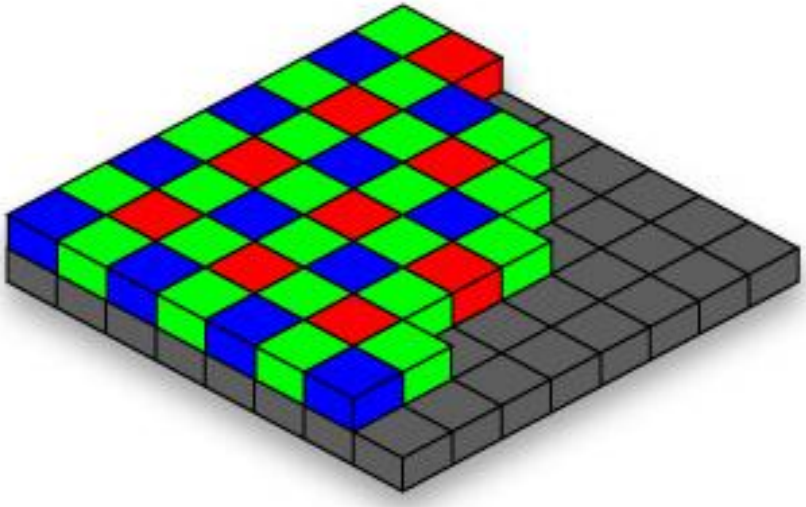
Three kinds of cones:



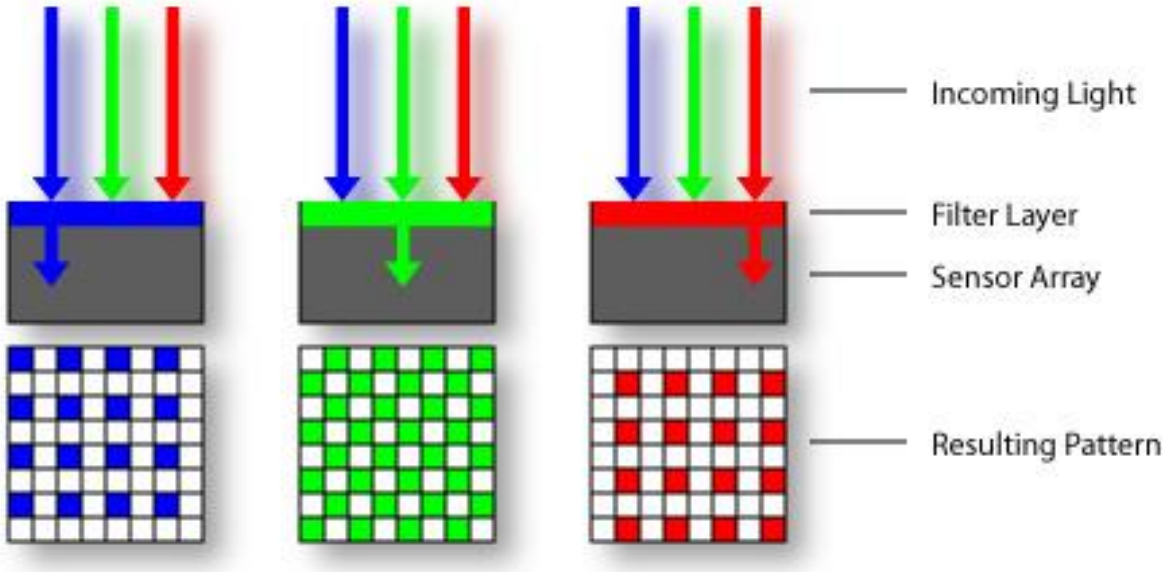


- Why are M and L cones so close?
- Why are there 3?

Practical Color Sensing: Bayer Grid



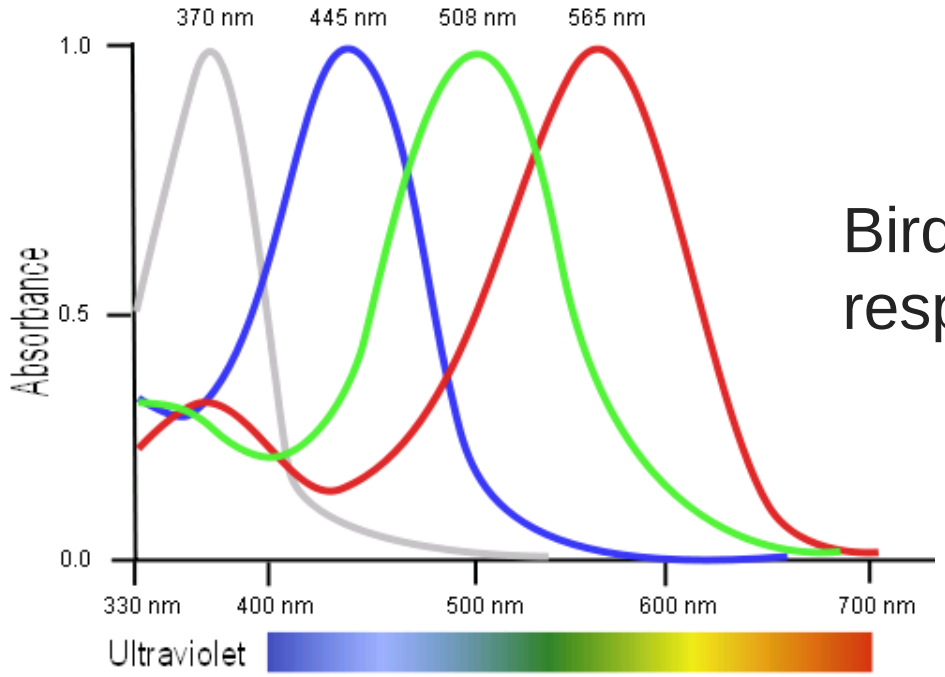




Estimate RGB
at ‘G’ cells from
neighboring values

Slide by Steve Seitz

Tetrachromatism



Bird cone responses

Most birds, and many other animals, have cones for ultraviolet light.

Some humans, mostly female, seem to have slight tetrachromatism.

Physiology of Color Vision: Fun facts

"M" and "L" pigments are encoded on the X-chromosome

- That's why men are more likely to be color blind
- "L" gene has high variation, so some women may be *tetrachromatic*

Some animals have one (night animals), two (e.g., dogs), four (fish, birds), five (pigeons, some reptiles/amphibians), or even 12 (mantis shrimp) types of cones

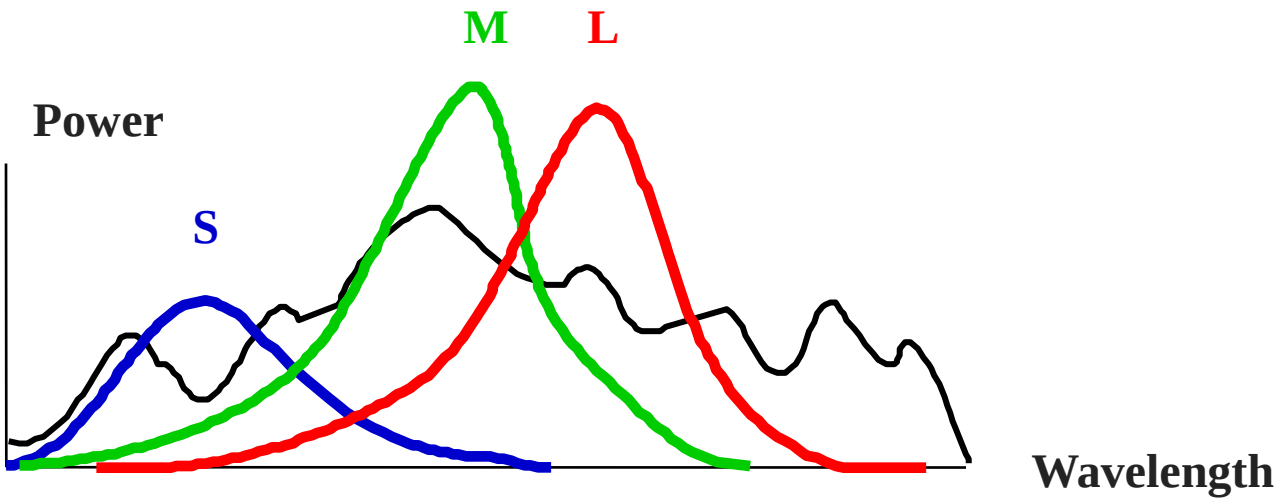
http://ngm.nationalgeographic.com/2016/02/evolution-of-eyes-text

http://en.wikipedia.org/wiki/Color_vision

华中科技大学

Slide by D. Hoiem

Color perception



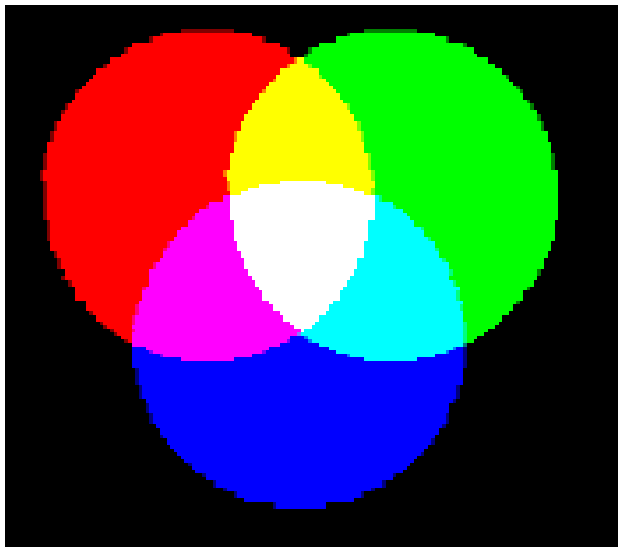
Rods and cones act as *filters* on the spectrum

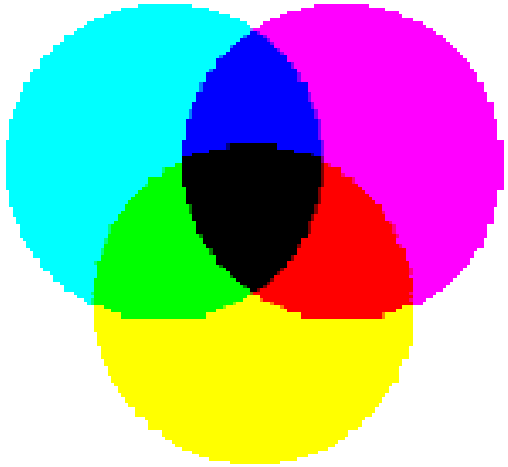
To get the output of a filter, multiply its response curve by the spectrum, integrate over all wavelengths

Each cone yields one number

- How can we represent an entire spectrum with three numbers?
- We can't! Most of the information is lost
  - As a result, two different spectra may appear indistinguishable
    - » such spectra are known as **metamers**

颜色模型

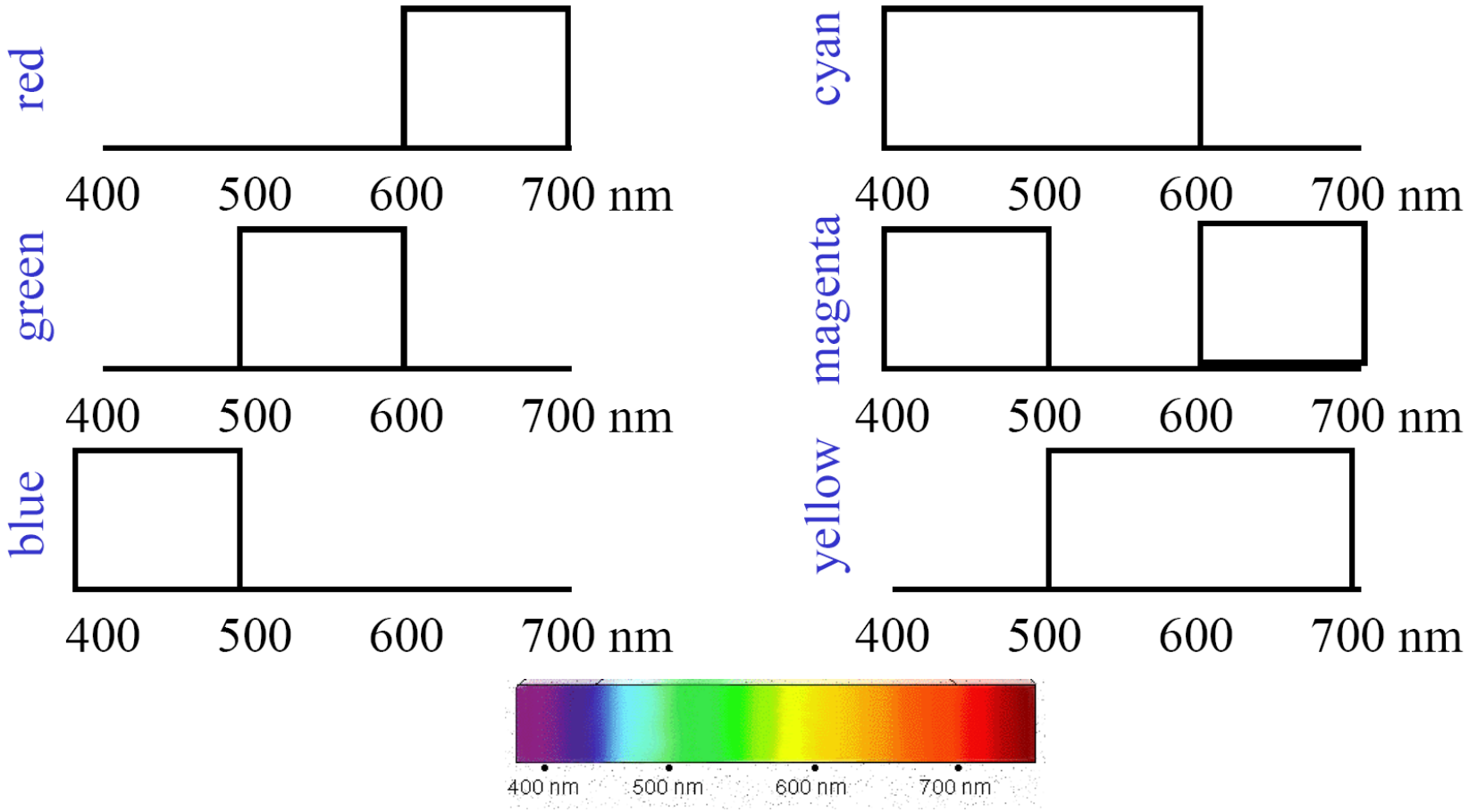




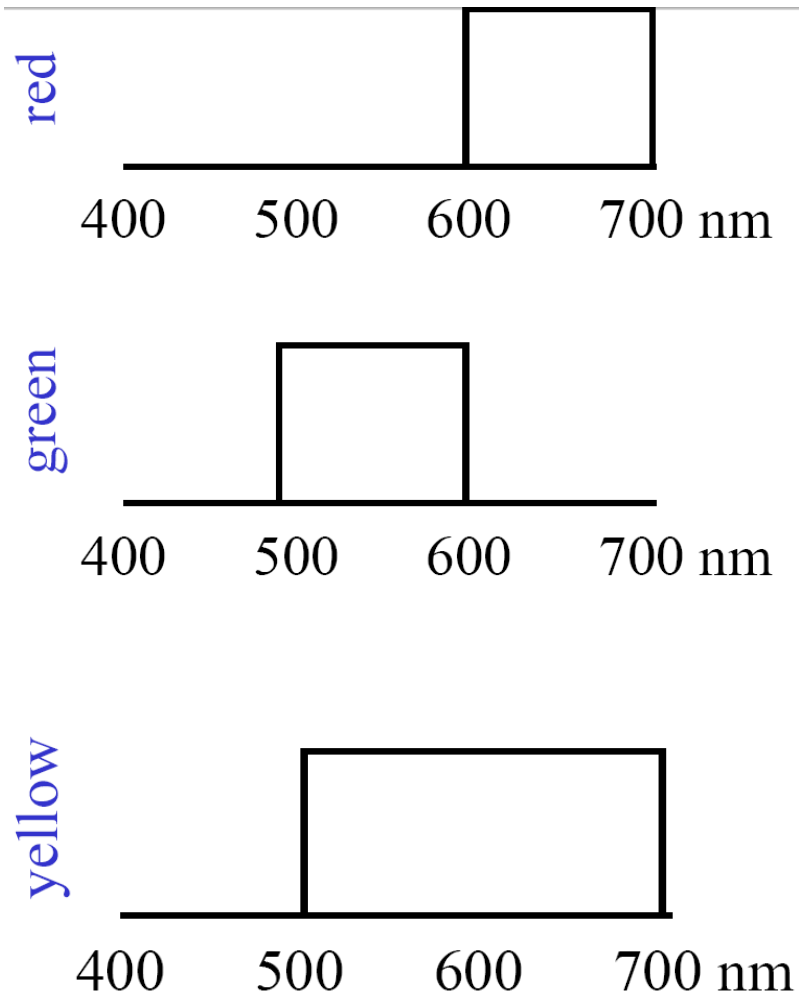
RGB,HIS,YIQ

Color mixing

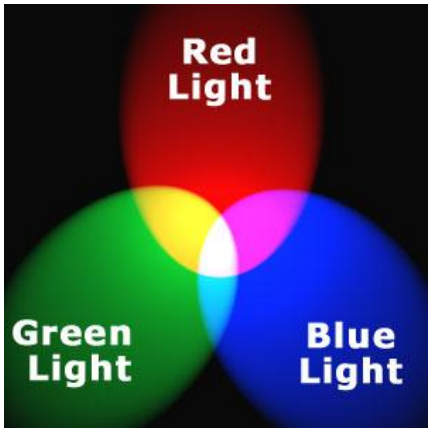
Cartoon spectra for color names:



Additive color mixing

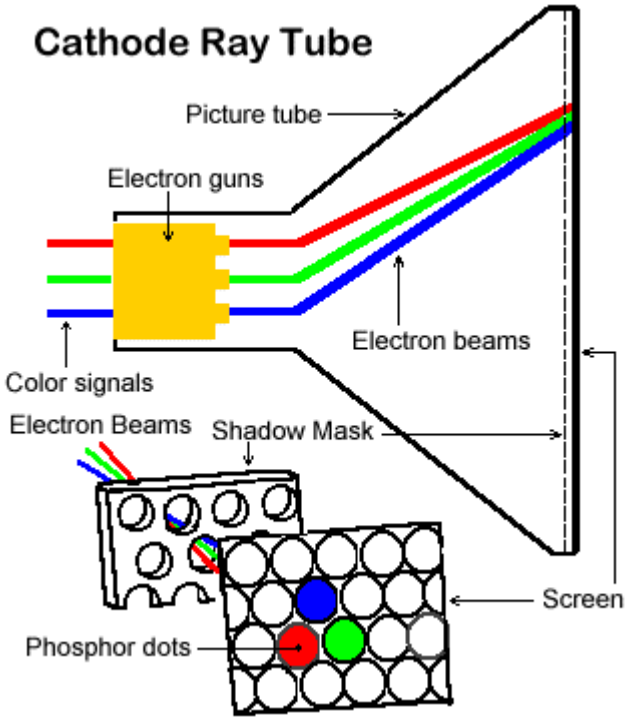


Colors combine by *adding* color spectra



Light *adds* to black.

Examples of additive color systems



CRT phosphors



multiple projectors

Superposition

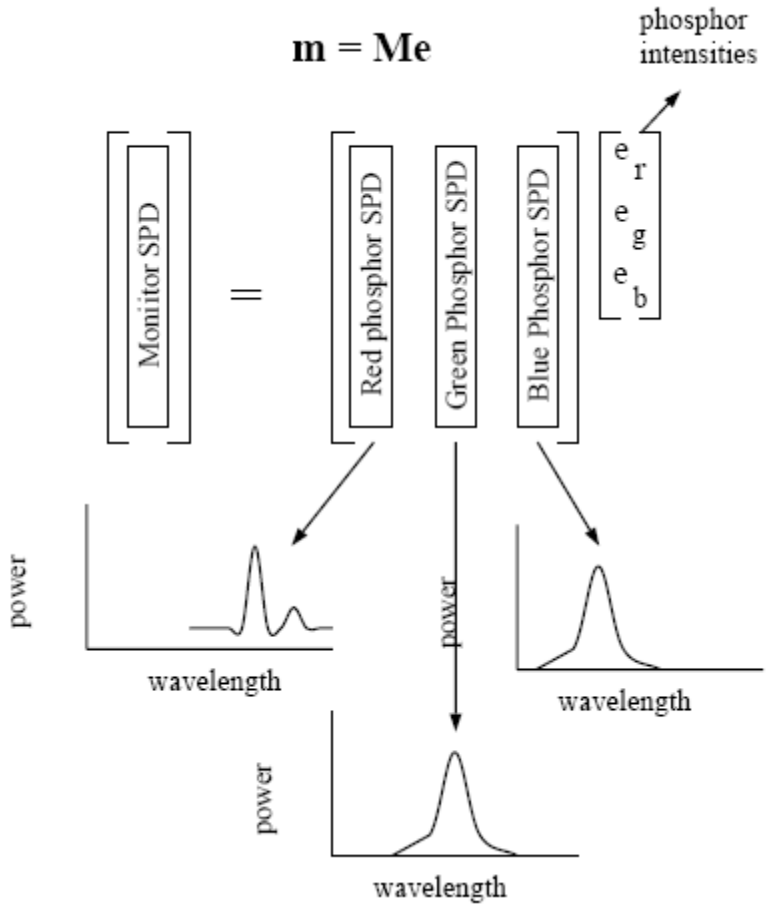
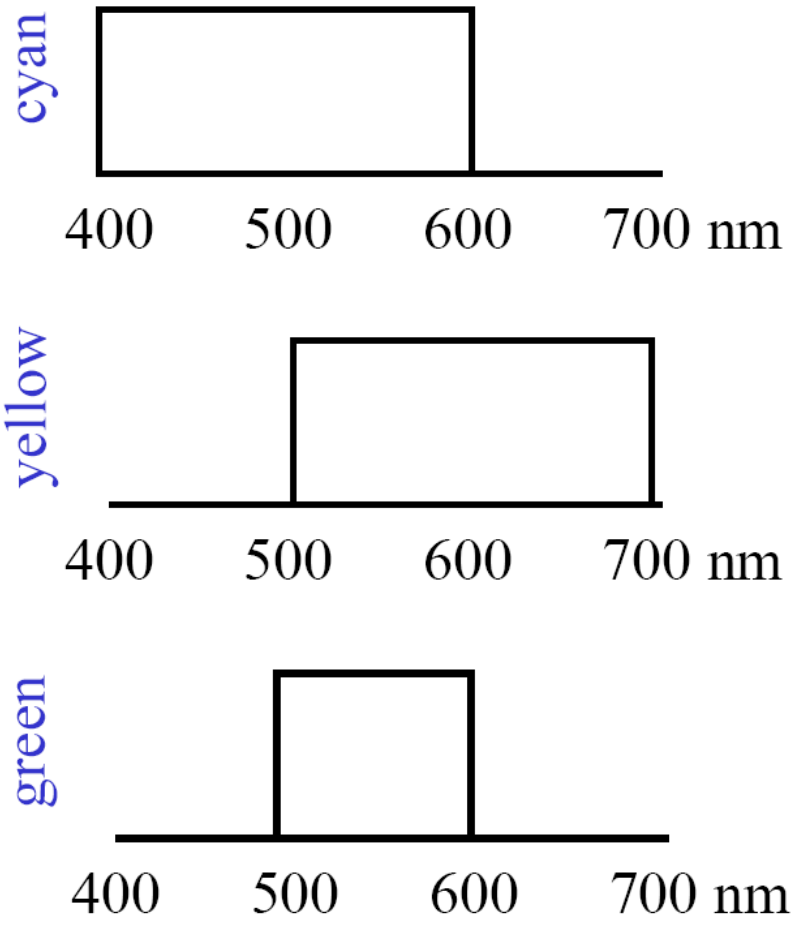


Figure from B. Wandell, 1996

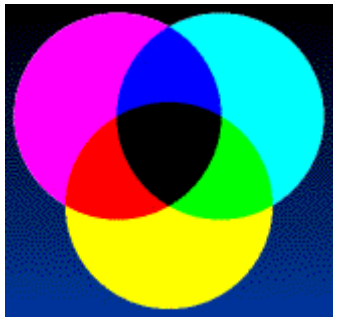
Additive mixing:

The spectral power distribution of the mixture is the sum of the spectral power distributions of the components.

Subtractive color mixing



Colors combine by *multiplying* color spectra.



Pigments *remove* color from incident light (white).

Examples of subtractive color systems

Printing on paper

Crayons

Most photographic film



Standard color spaces

Use a common set of primaries/color matching functions

Linear color space examples

- RGB
- CIE XYZ

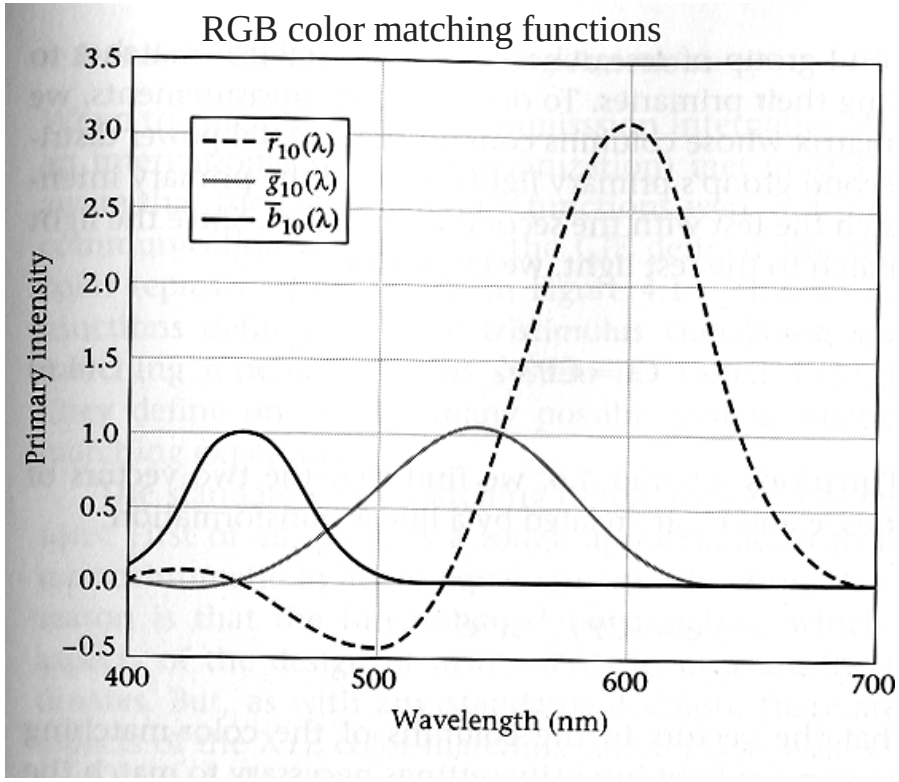
Non-linear color space

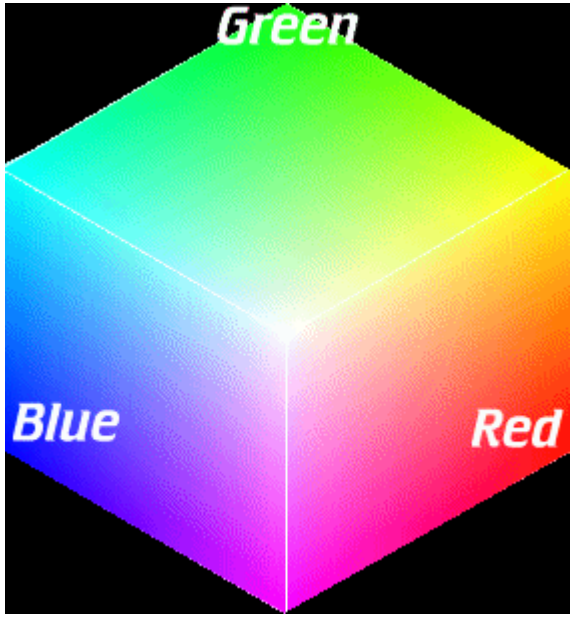
- HSV

RGB color space

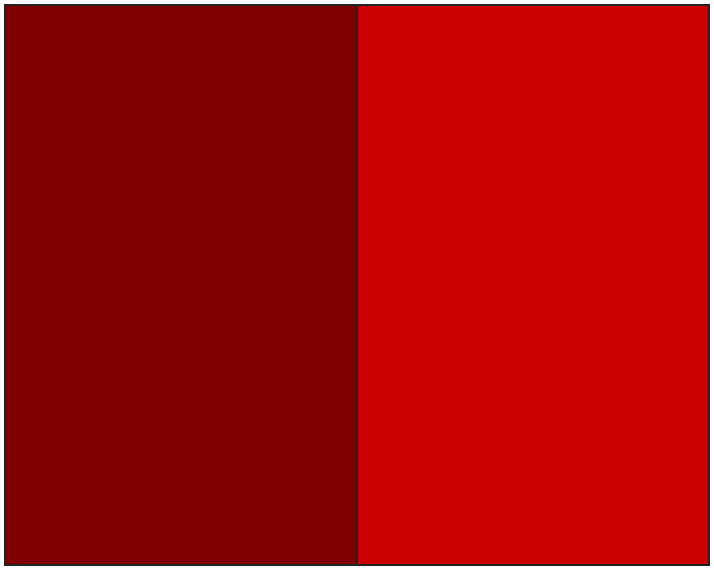
Single wavelength primaries

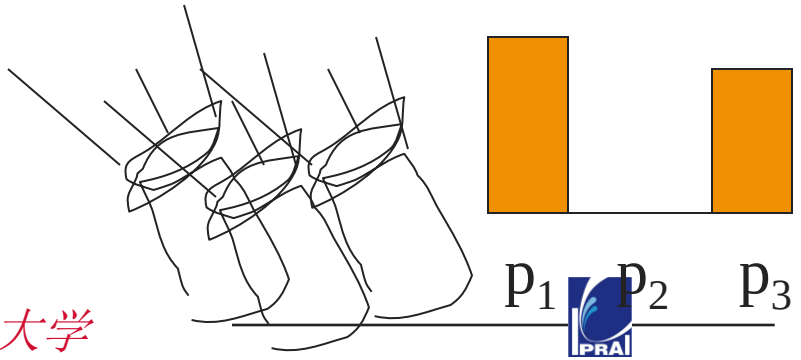
Good for devices (e.g., phosphors for monitor), but not for perception





Color matching experiment 2

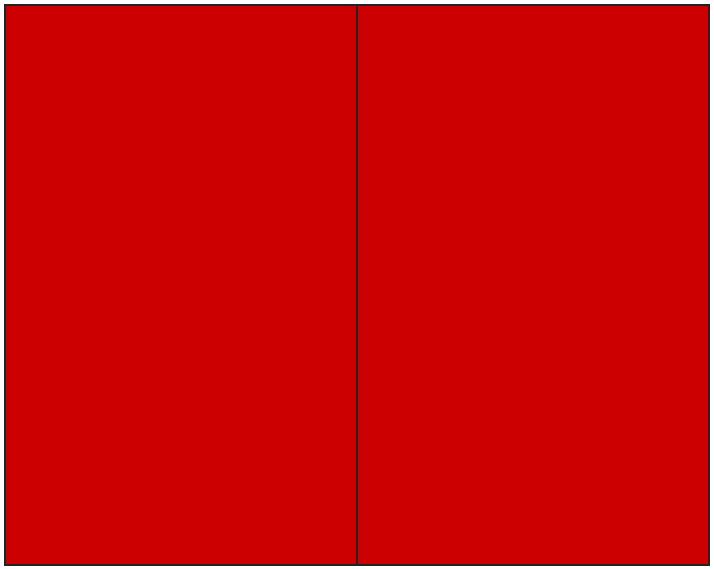




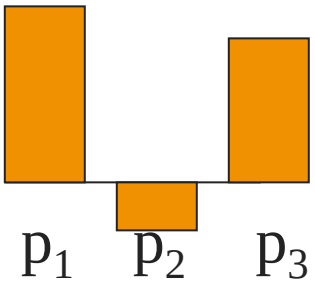
Slide credit: W Freeman

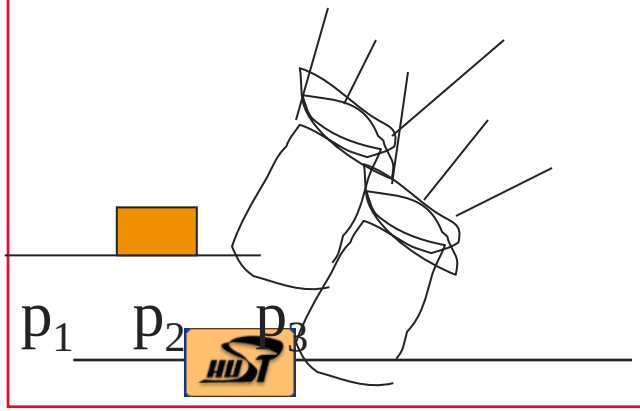
Color matching experiment 2

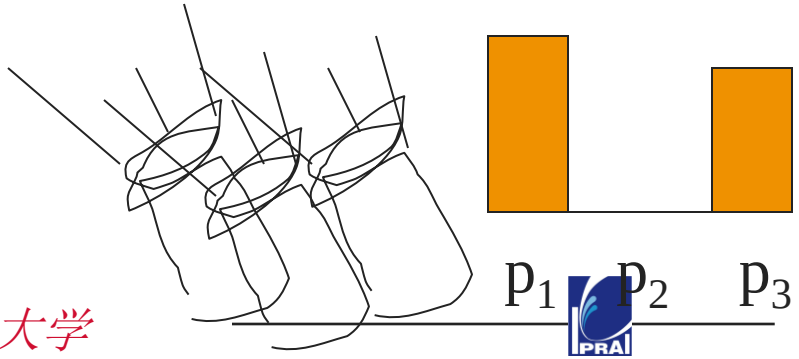
We say a "negative" amount of p_2 was needed to make the match, because we added it to the test color's side.



The primary color amounts needed for a match:







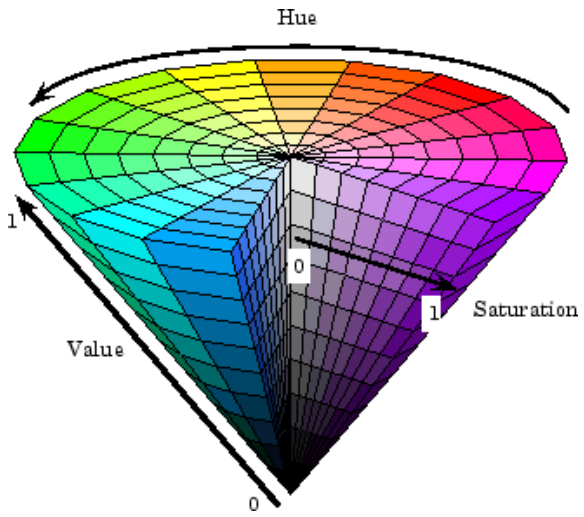
HSV color space

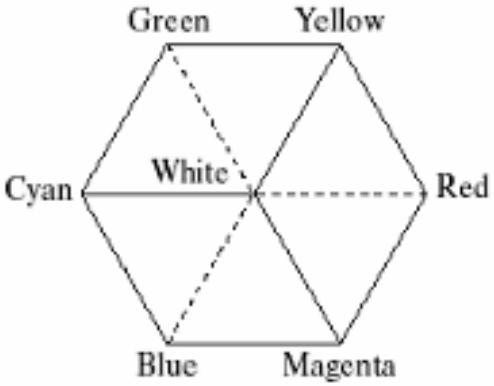
Hue, **S**aturation, **V**alue (Brightness)

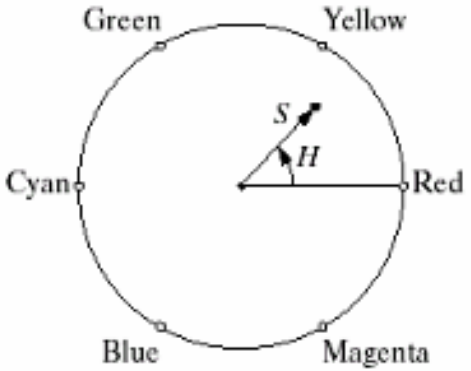
Nonlinear – reflects topology of colors by coding **hue** as an angle

Matlab: `hsv2rgb`, `rgb2hsv`.

OpenCV: cvCvtColor(floatsrc, floathsv, CV_BGR2HSV)







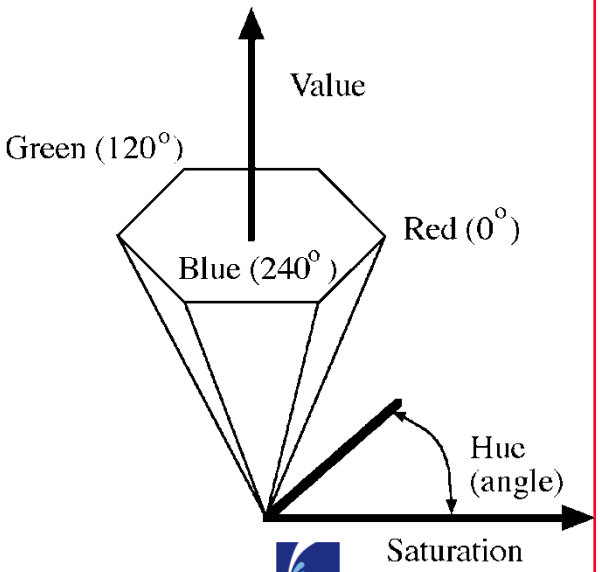


Image from mathworks.com

RGB->HIS

$$\theta = \cos^{-1}\left\{\frac{[(R-G)+(R-B)]/2}{\sqrt{(R-G)^2+(R-B)(G-B)}}\right\}$$

$$H = \begin{cases} \theta & B \le G \\ 360-\theta & B > G \end{cases}$$

$$S = 1 - \frac{3\cdot\min(R,G,B)}{R+G+B}$$

$$I = (R+G+B)/3$$

RGB->HSL,HSV

$$h = \begin{cases} 0^\circ & \text{if } max = min \\ 60^\circ \times \frac{g-b}{max-min} + 0^\circ, & \text{if } max = r \text{ and } g \ge b \\ 60^\circ \times \frac{g-b}{max-min} + 360^\circ, & \text{if } max = r \text{ and } g < b \\ 60^\circ \times \frac{b-r}{max-min} + 120^\circ, & \text{if } max = g \\ 60^\circ \times \frac{r-g}{max-min} + 240^\circ, & \text{if } max = b \end{cases}$$

$$l = \tfrac{1}{2}(max + min)$$

$$s = \begin{cases} 0 & \text{if } l = 0 \text{ or } max = min \\ \frac{max-min}{max+min} = \frac{max-min}{2l}, & \text{if } 0 < l \le \frac{1}{2} \\ \frac{max-min}{2-(max+min)} = \frac{max-min}{2-2l}, & \text{if } l > \frac{1}{2} \end{cases}$$

$$s = \begin{cases} 0, & \text{if } max = 0 \\ \frac{max-min}{max} = 1 - \frac{min}{max}, & \text{otherwise} \end{cases}$$

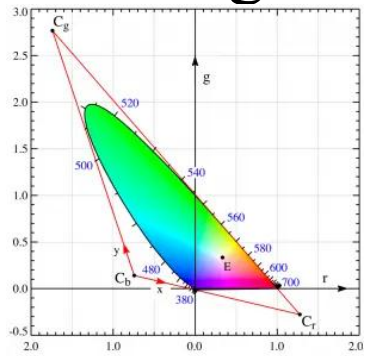
$$v = max$$

CIE XYZ color space

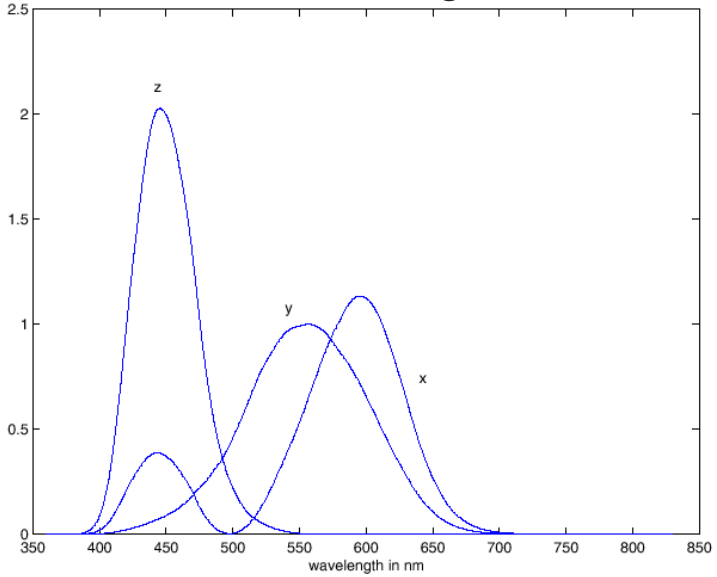
Established by the commission international d'eclairage (CIE), 1931

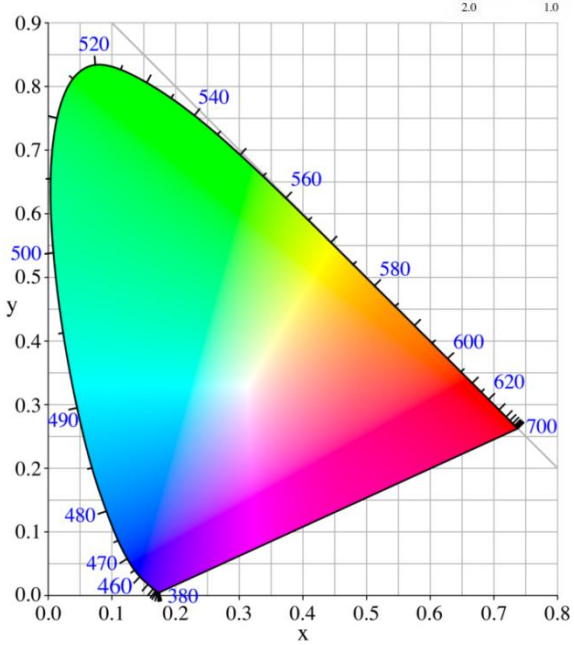
Usually projected to display:

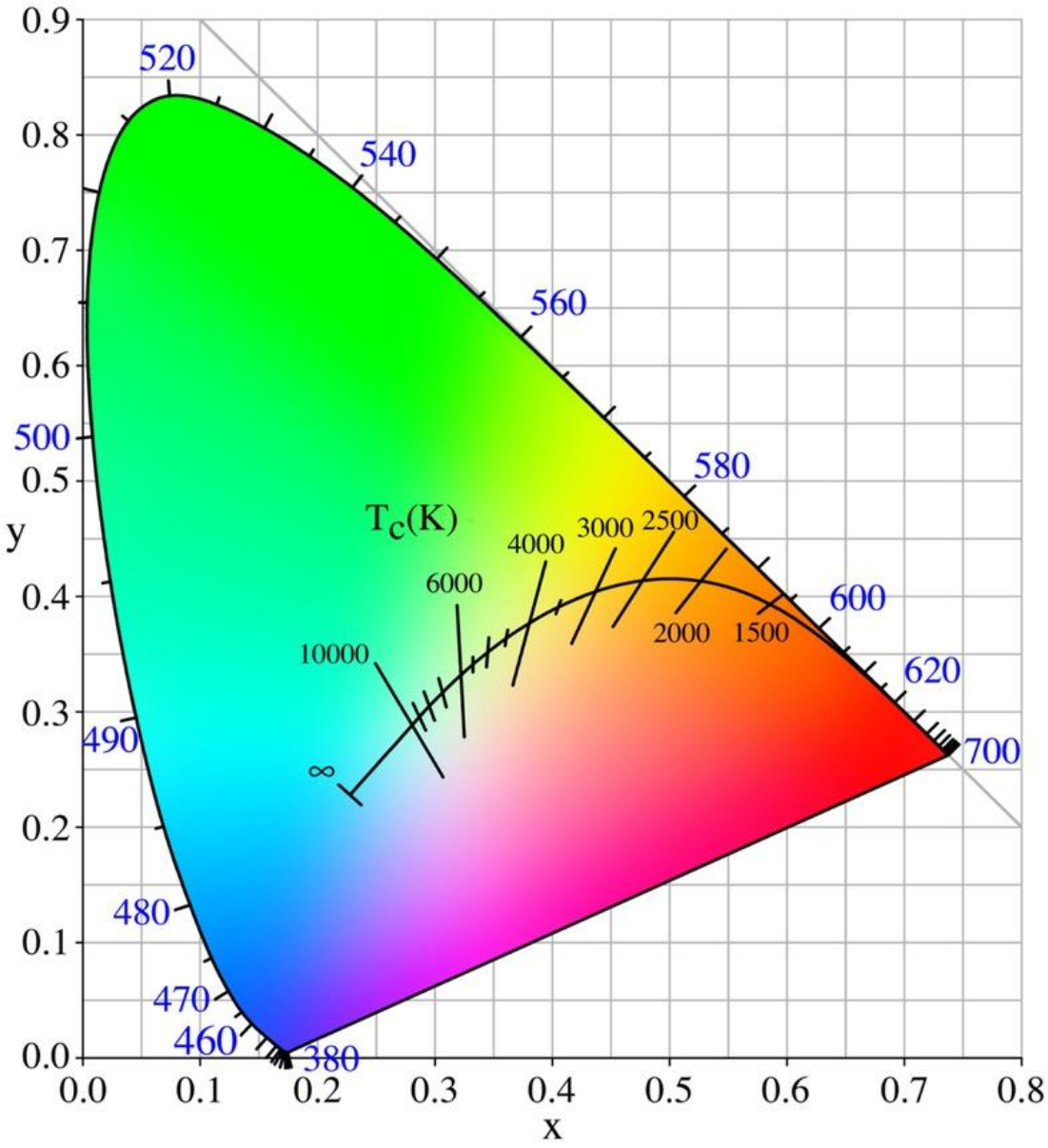
(x,y) = (X/(X+Y+Z), Y/(X+Y+Z))



CIE XYZ Color matching functions







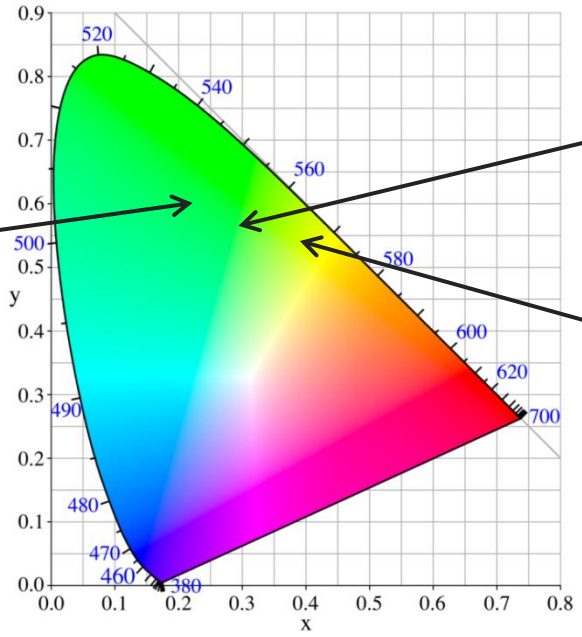
The figure shows:

- The spectral locus (i.e., the colours of monochromatic lights)
- The black-body locus (i.e., the colors of heated black-bodies)
- Some common illuminants

Distances in color space

Are distances between points in a color space perceptually meaningful?





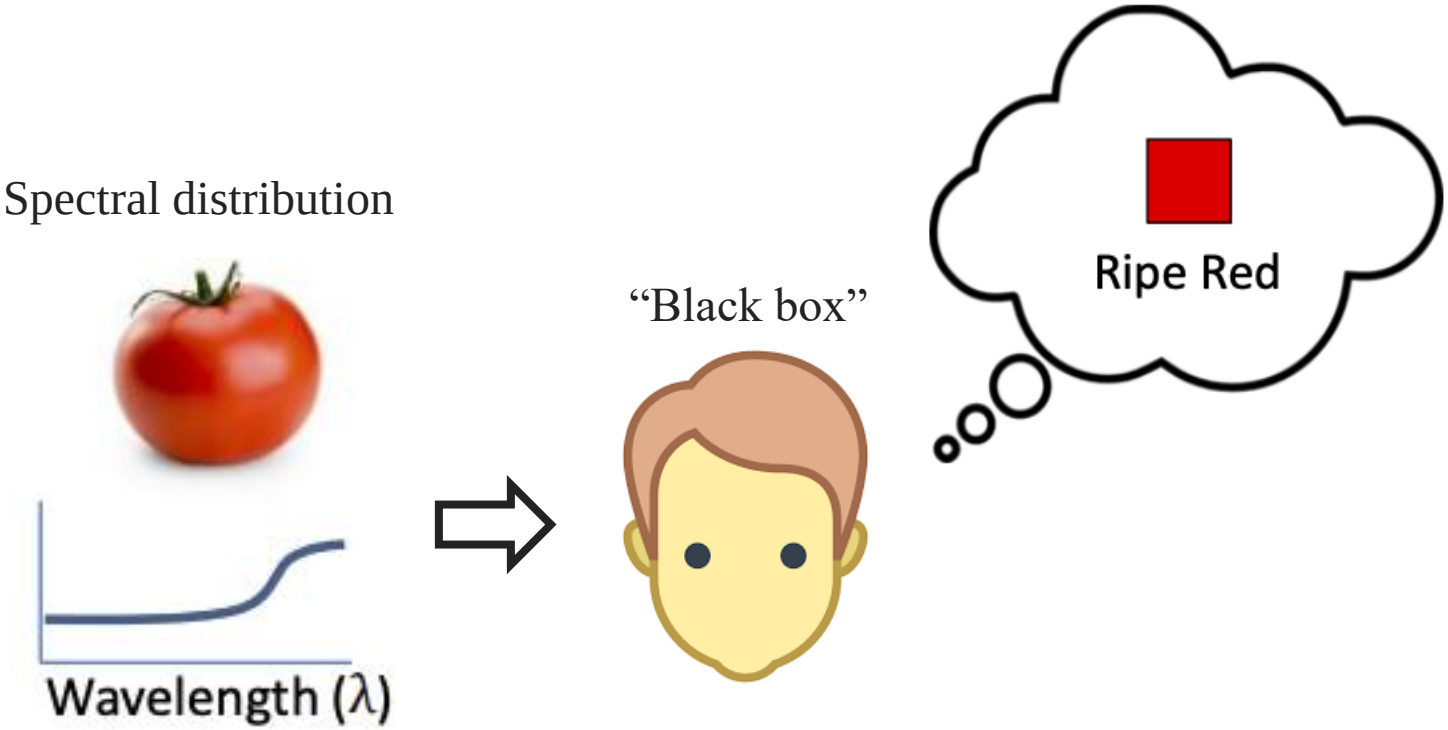




Quantifying color

Spectral distributions go through a “black box” (human visual system) and are perceived as color

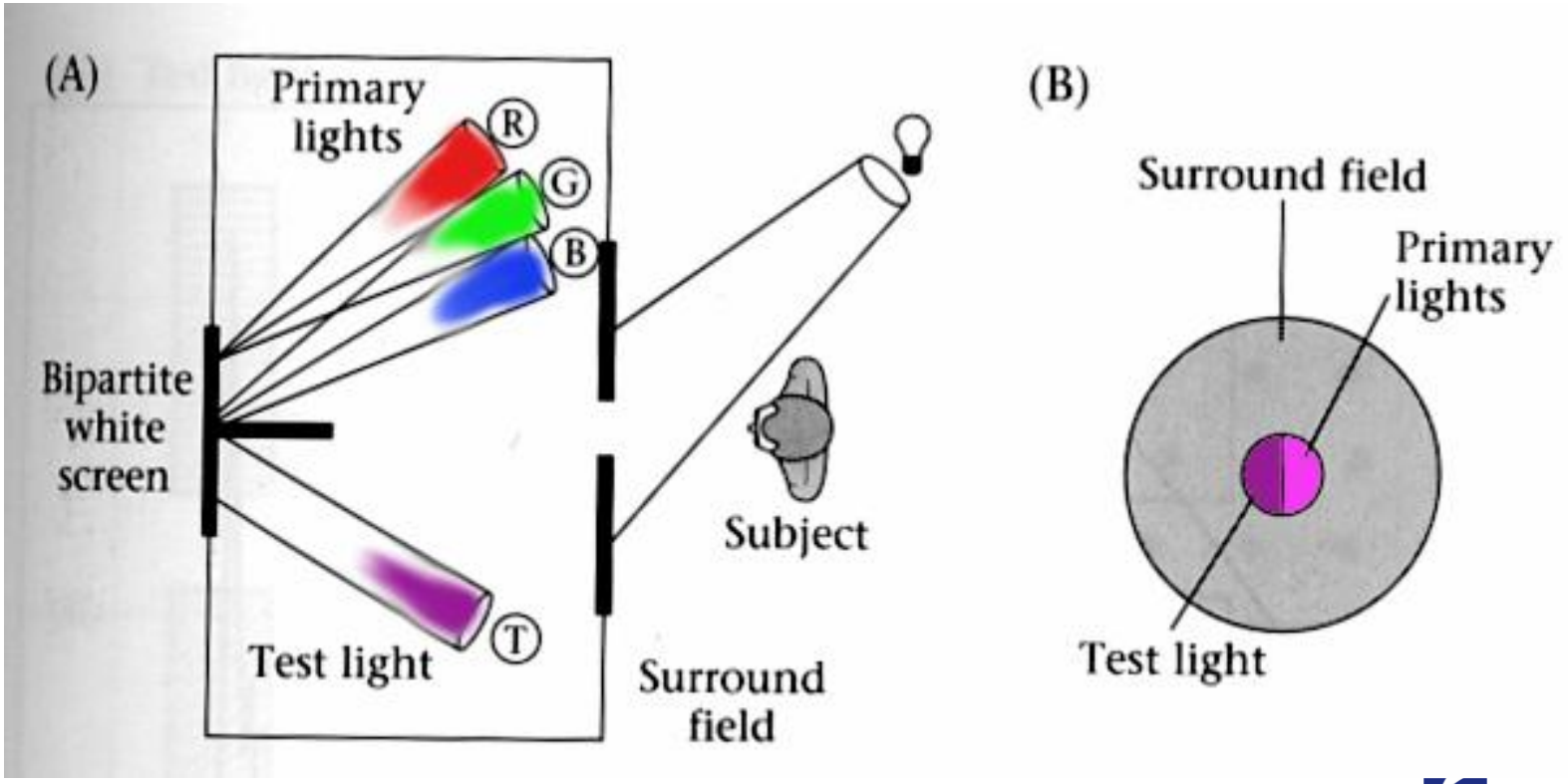
The only way to quantify the “black box” is to perform a human study



Source: M. Brown

Color matching experiments

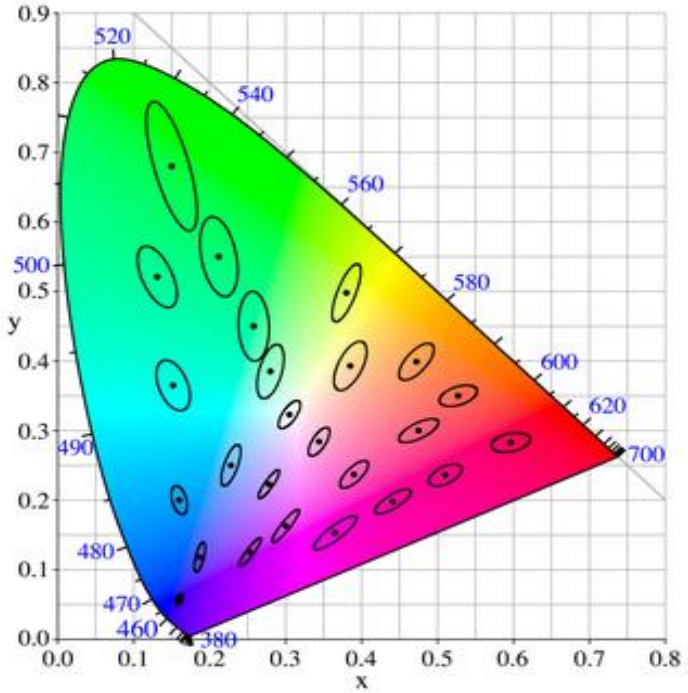
We would like to understand which spectra produce the same color sensation in people under similar viewing conditions



Wandell, Foundations of Vision, 1995

Distances in color space

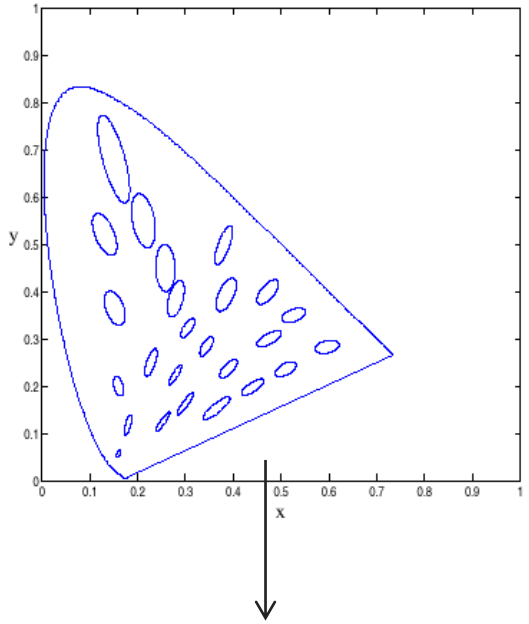
Not necessarily: CIE XYZ is **not** a *uniform* color space, so magnitude of differences in coordinates are poor indicator of color “distance”.



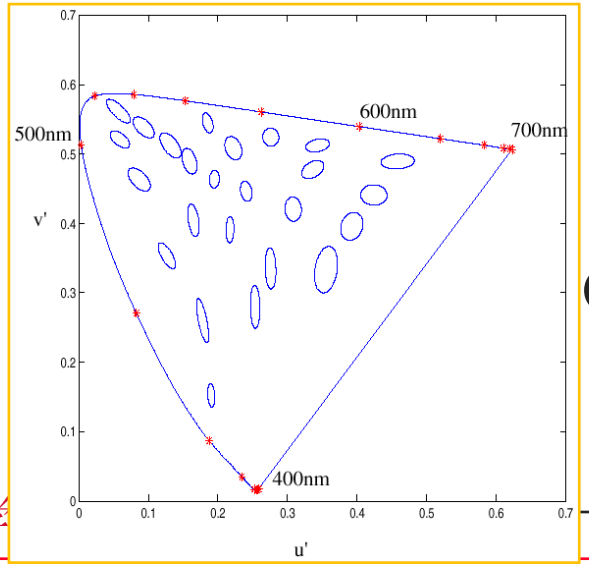
McAdam ellipses:
Just noticeable differences in color

Uniform color spaces

Attempt to correct this limitation by remapping color space so that just-noticeable differences are contained by circles → distances more perceptually meaningful.



CIE XYZ



CIE u'v'

Examples:

CIE u'v'

CIE Lab

Opponent color and chromatism

Lab:

L: Luminance (0,100)

a: green←→ red (-127,127)

b: blue←→yellow (-127,127)

$$\begin{cases} R = \text{gamma}(\frac{r}{255.0}) \\ G = \text{gamma}(\frac{g}{255.0}) \\ B = \text{gamma}(\frac{b}{255.0}) \end{cases} \qquad \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = M * \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

(1)

$$\text{gamma}(x) = \begin{cases} \left(\frac{x + 0.055}{1.055}\right)^{2.4} & (x > 0.04045) \\ \frac{x}{12.92} & (\text{其它}) \end{cases}$$

（2）XYZ转LAB

$$L^* = 116 f(Y/Y_n) - 16$$
$$a^* = 500[f(X/X_n) - f(Y/Y_n)]$$
$$b^* = 200[f(Y/Y_n) - f(Z/Z_n)]$$

$$f(t) = \begin{cases} t^{1/3} & \text{if } t > (\frac{6}{29})^3 \\ \frac{1}{3}\left(\frac{29}{6}\right)^2 t + \frac{4}{29} & \text{otherwise} \end{cases}$$

M=

0.4124,0.3576,0.1805

0.2126,0.7152,0.0722

0.0193,0.1192,0.9505

等同于如下公式：

X = var_R * 0.4124 + var_G * 0.3576 + var_B * 0.1805
Y = var_R * 0.2126 + var_G * 0.7152 + var_B * 0.0722
Z = var_R * 0.0193 + var_G * 0.1192 + var_B * 0.9505

Opponent color and chromatism

Lab:

L: Luminance (0,100)

a: green←→ red (-127,127)

b: blue←→yellow (-127,127)

YUV:

Y=0.3R+0.59G+0.11B

U: B-Y

V: R-Y

Color perception

Color/lightness constancy

The ability of the human visual system to perceive the intrinsic reflectance properties of the surfaces despite changes in illumination conditions



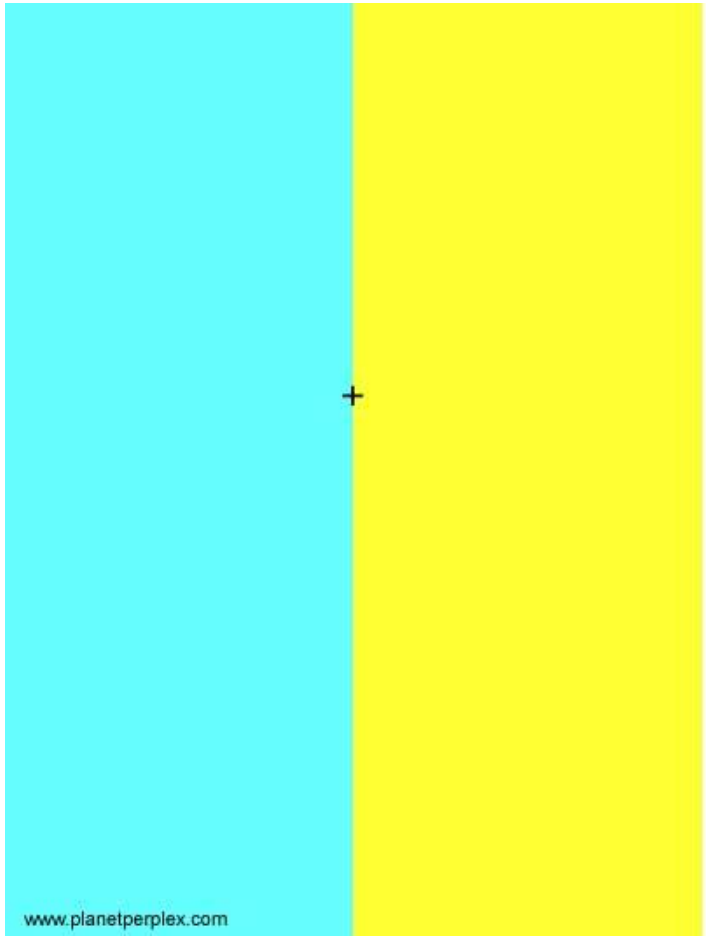
J. S. Sargent, The Daughters of Edward D. Boit, 1882

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Chromatic adaptation

- The visual system changes its sensitivity depending on the luminances prevailing in the visual field
  - The exact mechanism is poorly understood
- Adapting to different brightness levels
  - Changing the size of the iris opening (i.e., the aperture) changes the amount of light that can enter the eye
  - Think of walking into a building from full sunshine
- Adapting to different color temperature
  - The receptive cells on the retina change their sensitivity
  - For example: if there is an increased amount of red light, the cells receptive to red decrease their sensitivity until the scene looks white again
  - We actually adapt better in brighter scenes: This is why candlelit scenes still look yellow

Chromatic adaptation





Color balance, Illuminant estimation and adaptation

What color is the dress?



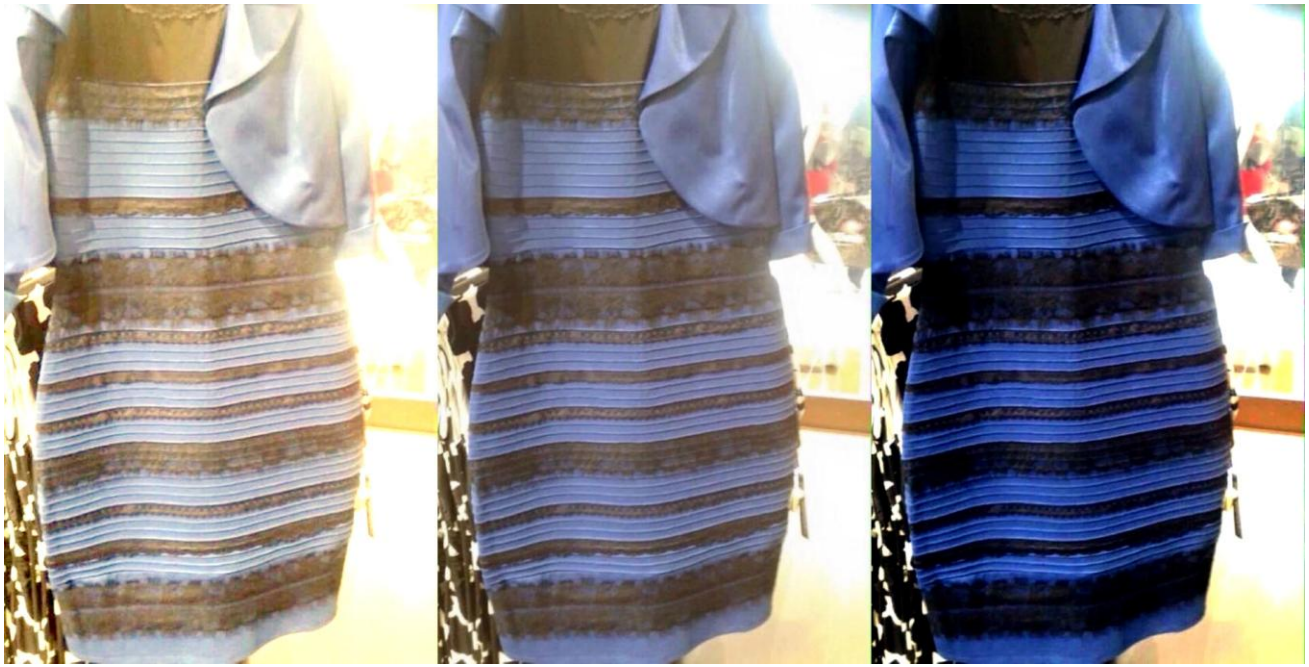
White-Golden

or Blue-Black?

https://www.wired.com/2015/02/science-one-agrees-color-dress/

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What color is the dress?



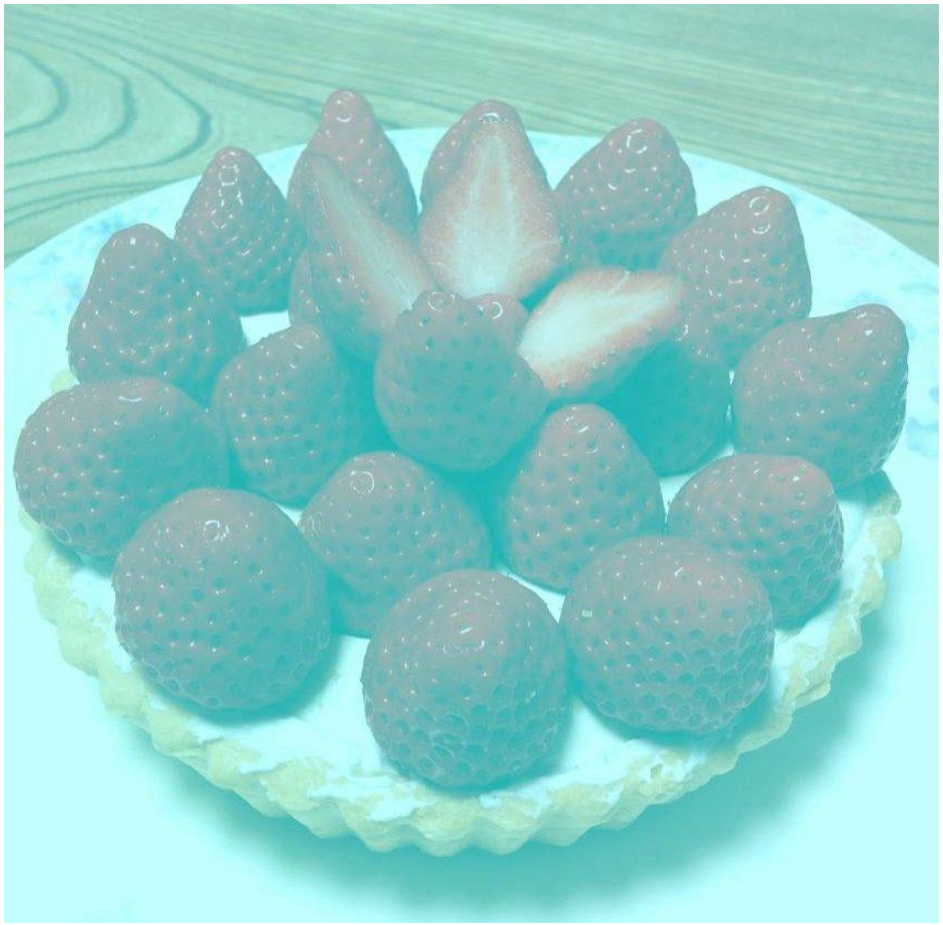
White-Golden

or Blue-Black?

https://www.wired.com/2015/02/science-one-agrees-color-dress/

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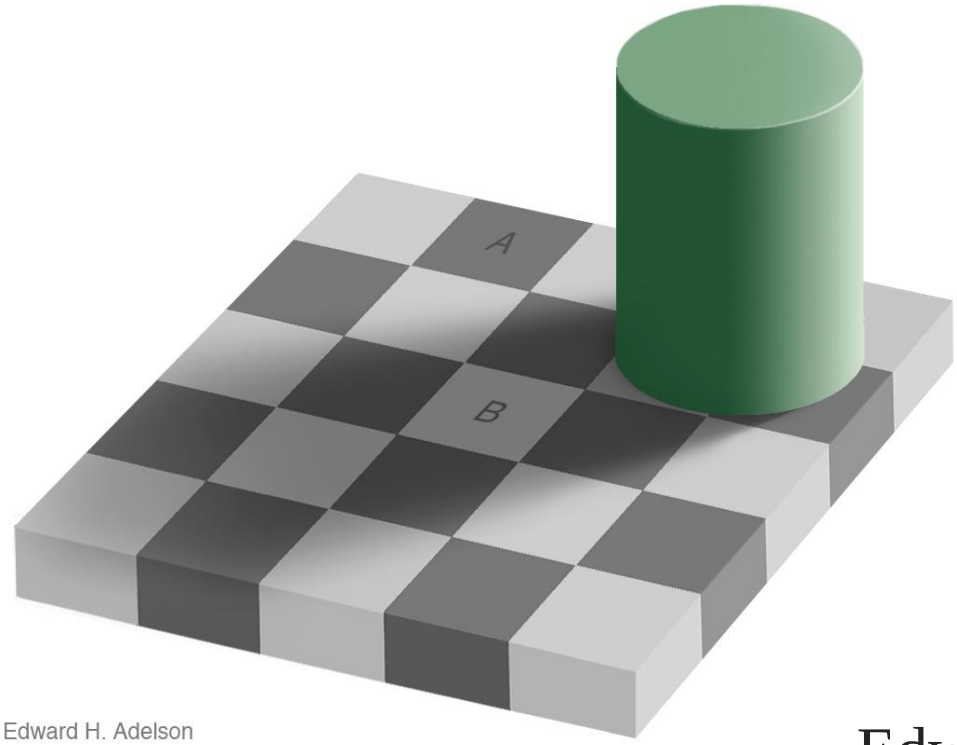
This strawberry cake has no red pixels!



https://www.digitaltrends.com/photography/non-red-strawberries/

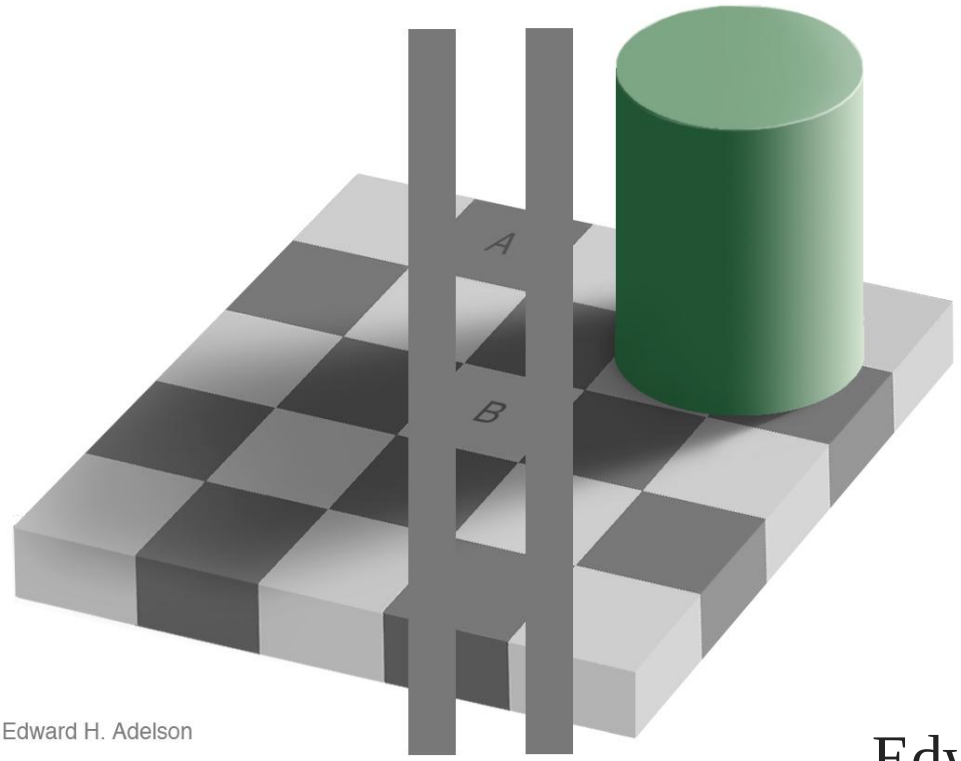
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Brightness perception



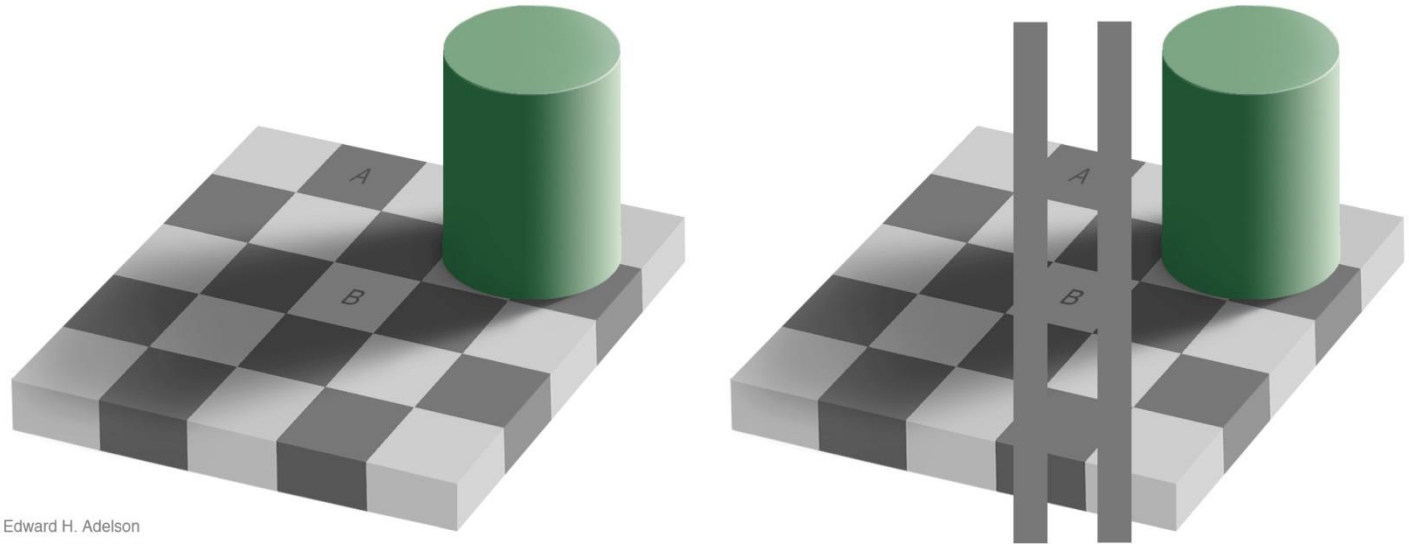
Edward Adelson

http://web.mit.edu/persci/people/adelson/illusions_demos.h



Edward Adelson

http://web.mit.edu/persci/people/adelson/illusions_demos.h



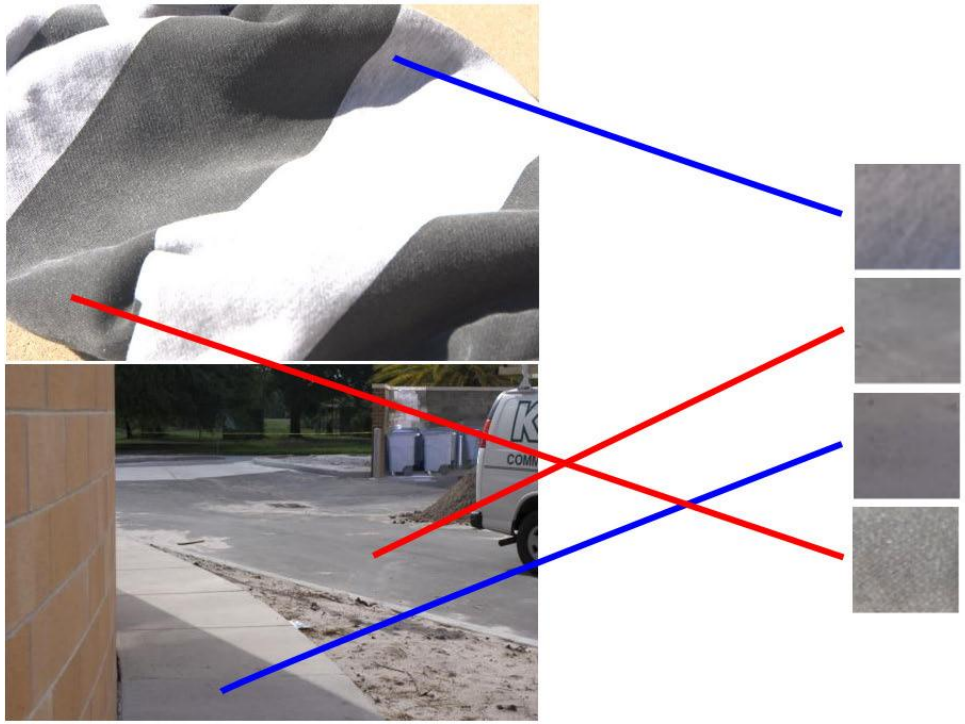
Edward Adelson

http://web.mit.edu/persci/people/adelson/illusions_demos.h

Color constancy

Interpret surface in terms of albedo or “true color”, rather than observed intensity

- Humans are good at it
- Computers are not nearly as good



White balance

Analogous to color constancy mechanismsin human vision, cameras have mechanisms to adapt to the illumination in the environment so that neutral (white or gray) objects look neutral

Incorrect white balance



Correct white balance



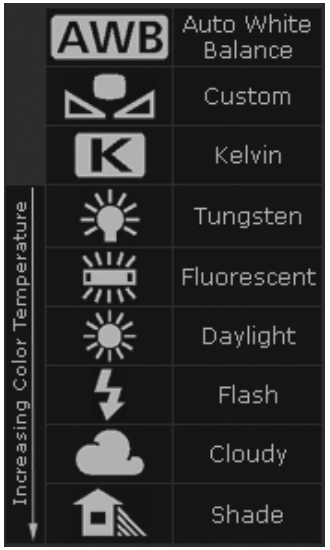
White balance

Film cameras:

- Different types of film or different filters for different illumination conditions

Digital cameras:

- Automatic white balance
- White balance settings corresponding to several common illuminants
- Custom white balance using a reference object



White balance

Von Kries adaptation: Multiply each channel by a *gain factor*

Best way: gray card

Take a picture of a neutral object (white or gray)

If the object is recorded as r_w, g_w, b_w use weights $1/r_w$, $1/g_w$, $1/b_w$



White balance

Without gray cards: we need to "guess" which pixels correspond to white objects

Gray world assumption

- The image average r_{ave}, g_{ave}, b_{ave} is gray
- Use weights $1/r_{ave}$, $1/g_{ave}$, $1/b_{ave}$

Brightest pixel assumption

- Highlights usually have the color of the light source
- Use weights inversely proportional to the values of the brightest pixels

Gamut mapping

- Gamut: convex hull of all pixel colors in an image
- Find the transformation that matches the gamut of the image to the gamut of a "typical" image under white light

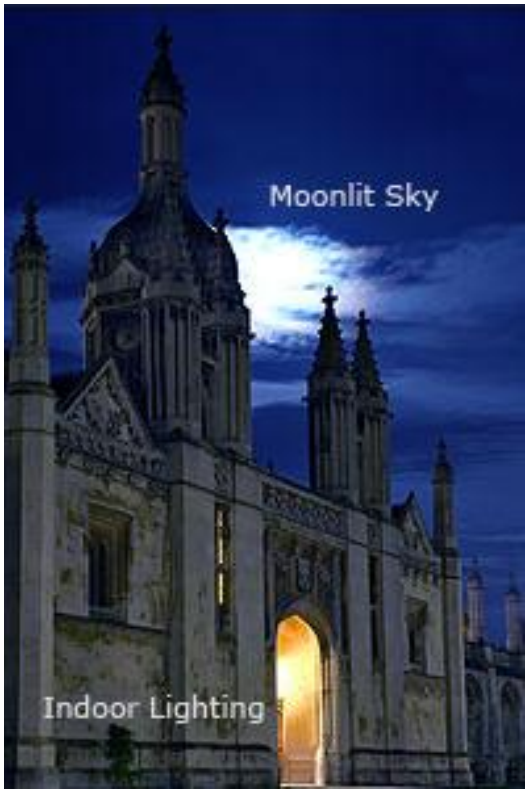
Use image statistics, learning techniques

Mixed illumination

When there are several types of illuminants in the scene, different reference points will yield different results



Reference: moon



Reference: stone

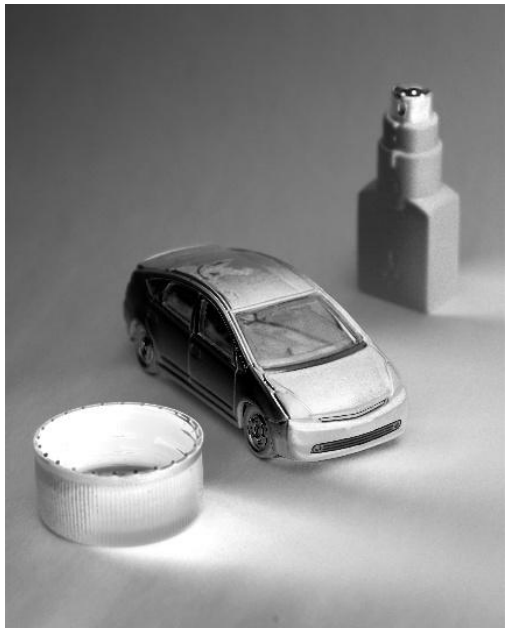
http://www.cambridgeincolour.com/tutorials/white-balance.htm

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Spatially varying white balance



Input



Alpha map



Output

E. Hsu, T. Mertens, S. Paris, S. Avidan, and F. Durand, Light Mixture Estimation for Spatially Varying White Balance, SIGGRAPH 2008

Input Image



Our Result



New Lighting



Our Result



Dim Interior

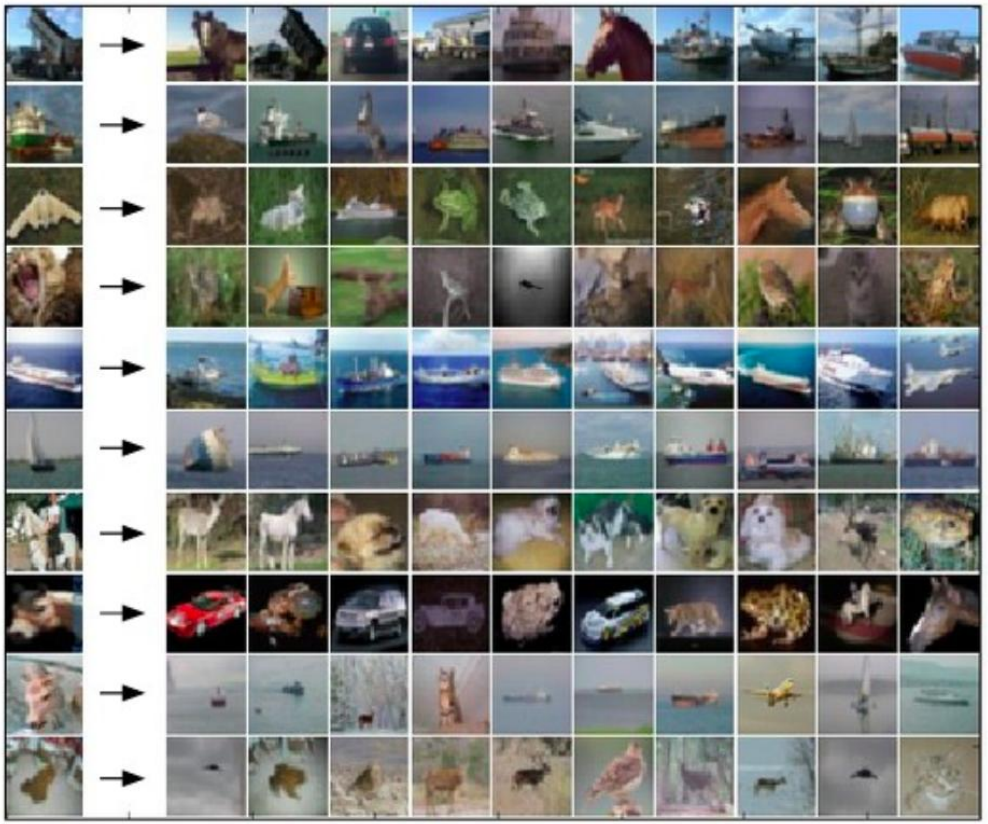


Dim Exterior



Color cues for image understanding

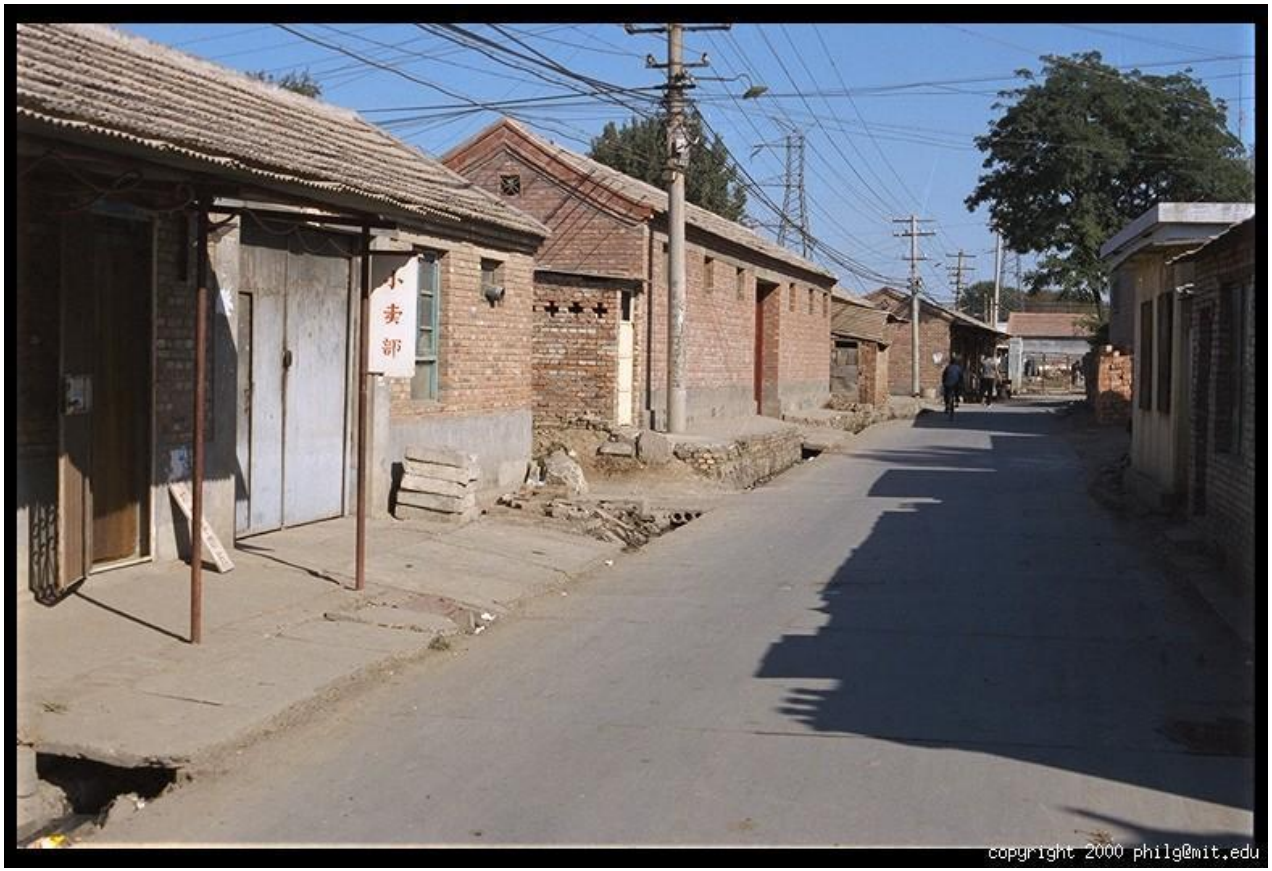
Recognition



Segmentation



Most information in intensity



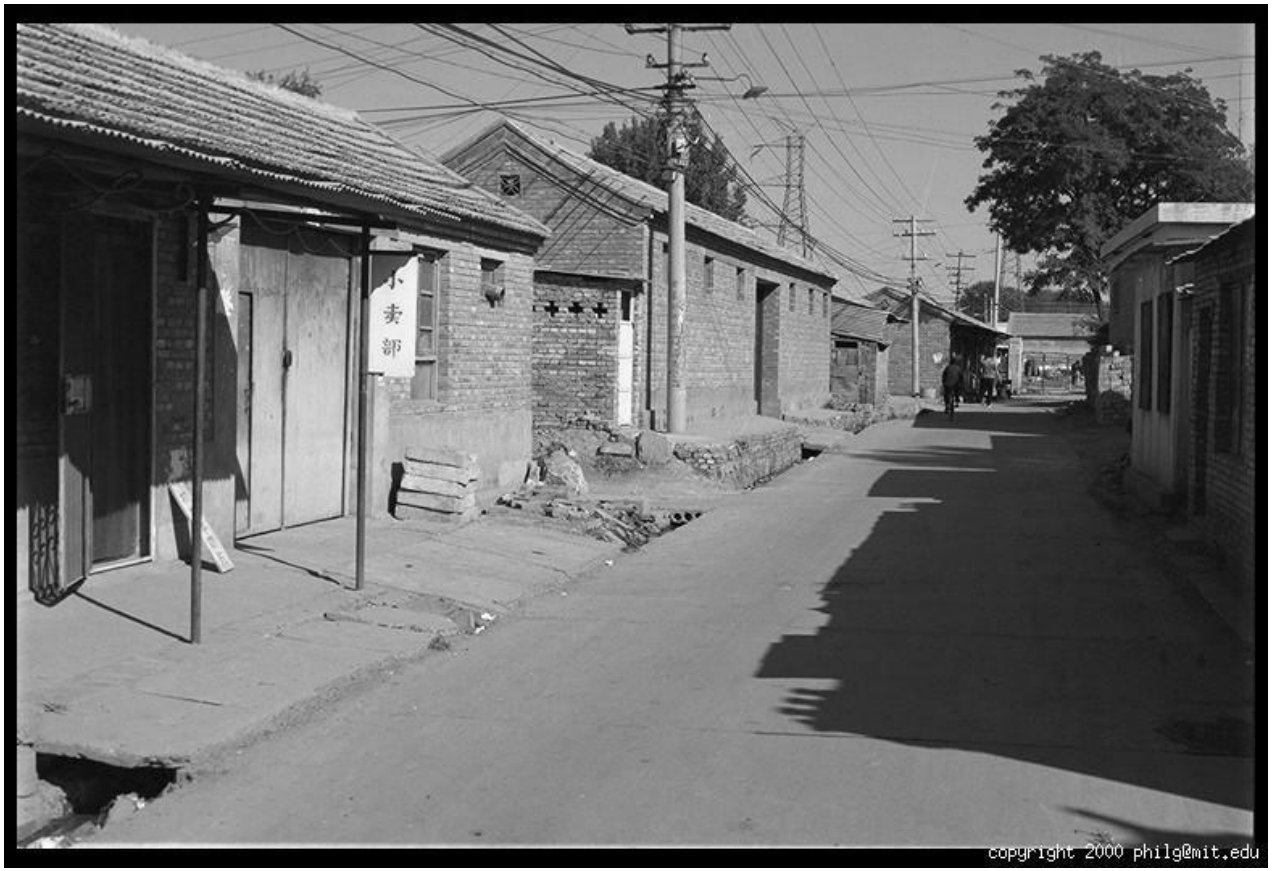
Original image

Most information in intensity



Only color shown – constant intensity

Most information in intensity



Only intensity shown – constant color